

RISK AND DECISION BASELINE

Risk and Mitigation Register

Natural-Language Treatment Planning System (NL-TPS)

Source baseline: NL-TPS controlled documentation suite v0.1

Version: 0.1 - Preliminary engineering baseline

Date: 18 August 2026

Status: Discussion Draft - Not for clinical use

Coverage: 1,948 entities: 1,776 requirements and 172 component responsibilities

STATUS AND SAFETY NOTICE

This register is a preliminary engineering risk screen. It does not assign clinical FMEA scores, establish risk acceptability, claim mitigation effectiveness, replace multidisciplinary hazard analysis, or authorize patient use. All numerical scores and residual-risk decisions require approved evidence and governance.

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8.7	Plan review, selection, QA, and transfer (REV) - 81 records	428
8.8	Data, interoperability, provenance, and audit (DAT) - 90 records	454
8.9	AI and inference-model lifecycle (AIM) - 72 records	485
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8.11 Privacy, cybersecurity, and secure development (SEC) - 72 records	526
8.12 Clinical operations, monitoring, and resilience (OPS) - 63 records	550
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Document control

Field	Value
Document owner	NL-TPS Program - proposed; final owner requires governance approval
Source baseline	NL-TPS controlled documentation suite, Version 0.1
Coverage	1,948 entities: 1,776 requirements and 172 component responsibilities
Risk status	Preliminary failure-mode and control screen; not an approved FMEA, hazard analysis, residual-risk decision, or clinical risk acceptance
Change authority	Clinical governance, risk management, quality system, and configuration control
Supersession	None; source requirements and component documents remain authoritative

Executive summary

This register documents a primary failure condition, potential consequence, mitigation strategy, assurance evidence, accountable owner, and preliminary attention band for every unique controlled requirement and component in the NL-TPS suite. Functional/non-functional/operational requirement IDs are retained as aliases to their 111 source HLRs and 333 source detailed requirements, avoiding duplicate risk records while preserving trace.

The register covers 111 system HLRs, 333 detailed sub-requirements, 333 high-level interface requirements, 999 sub-interface requirements, 41 functional components, 57 interface components, and 74 category-scoped components. The source requirement or component responsibility remains normative; this register describes the risk of failure and the minimum control intent associated with it.

PRELIMINARY RISK STATUS

Attention bands prioritize multidisciplinary review; they are not risk acceptability classifications. Severity, occurrence, detectability, risk-priority number, benefit-risk determination, residual risk, and control-effectiveness conclusions remain TBD until established by the approved team using site, workflow, modality, disease-site, vendor, model, commissioning, and production evidence.

1 Purpose, scope, and interpretation

The register supports process mapping, hazard analysis, FMEA, fault trees, architecture decisions, interface controls, verification planning, commissioning, cybersecurity, human-factors engineering, monitoring, incident response, change control, and release review. A row does not replace the full source obligation, linked hazards, or required evidence.

Risk is evaluated for the reasonably foreseeable failure of the requirement or component responsibility. Mitigation follows a control hierarchy: inherent safety and authority boundaries; deterministic prevention and fail-closed behavior; protected state and data integrity; detection, warning, quarantine, and reconciliation; professional review and independent QA; recovery; and governed

information, training, and procedures.

2 Risk method and record fields

Field	Meaning
Entity and trace	Immutable source ID; F/NF/O alias where applicable; parent, allocation, family, package, or component class.
Failure / risk	Primary foreseeable failure or hazardous condition if the obligation is absent, wrong, stale, bypassed, incompatible, or incomplete.
Potential effect	Plausible patient, user, clinical workflow, data, interoperability, security, operational, or assurance consequence; not a claim of observed harm.
Mitigation	Minimum control strategy. The normative source text, all linked children, interfaces, components, tests, and release controls remain required.
Evidence / owner	Accountable team and expected verification, monitoring, anomaly, release, or lifecycle evidence.
Attention band	Gate: direct release, patient, dose, geometry, QA, or external-TPS gate. High: safety-significant clinical, model, evidence, security, interoperability, or assurance control. Moderate: foundational governance, data, operations, usability, or support control.
FMEA fields	Severity, occurrence, detectability, RPN, residual risk, control effectiveness, and acceptability are explicitly TBD pending approved multidisciplinary analysis.

3 Standards, literature, and evidence constraints

Reference	Use in this baseline	Controlled source
AAPM TG-100 / Report 283	Process mapping, failure modes, effects, causes, controls, FMEA, fault-tree analysis, and risk-informed radiotherapy QM.	AAPM official report page
ISO 14971:2019	Lifecycle risk management, risk analysis and evaluation, risk controls, residual risk, production and post-production information.	ISO official standard page
FDA benefit-risk guidance	Transparent consideration of benefit, risk, uncertainty, controls, evidence, and total-product-lifecycle information.	FDA official guidance page
DICOM PS3 and IHE-RO	Paired conformance, semantic, role, version, security, workflow, and site-specific interoperability risk at external TPS boundaries.	DICOM official source ; IHE-RO official source
Approved clinical evidence	AAPM task-group reports, QUANTEC/HyTEC, peer-reviewed disease-site guidance, prescriptions, and institutional policy remain governed inputs, not universal thresholds in this document.	ConOps EVID and source HLR EVD

4 Coverage and preliminary attention summary

Entity set	Count	Gate	High	Moderate	Status
System high-level requirements	111	31	79	1	Complete
Detailed sub-requirements	333	47	229	57	Complete
High-level interface requirements	333	50	223	60	Complete
Sub-interface requirements	999	150	669	180	Complete
Functional components	41	11	28	2	Complete
Interface components	57	17	34	6	Complete
F/NF/O category components	74	11	55	8	Complete
TOTAL	1,948	317	1317	314	Complete

5 System high-level requirement risk records

5.1 Governance, intended use, and scope (GOV) - 7 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
GOV-001	Category alias HOR-GOV-001; source/hazard SCP; PRN; AUTH; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: operate as a human-supervised treatment-planning support system and shall not be represented as a substitute for the professional....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-GOV; V&V I/ A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
GOV-002	Category alias HOR- GOV-002; source/hazard SCP; MODE; ROAD; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: restrict clinical operation to the institution-approved intended use, patient population, disease sites, modalities, machines, techniques, and tasks defined....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-GOV; V&V I/ T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
GOV-003	Category alias HOR- GOV-003; source/hazard SCP; HAZ; ROAD; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: block clinical execution for an unsupported or out-of-scope workflow until that workflow has completed separate risk assessment, commissioning....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-GOV; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
GOV-004	Category alias HOR- GOV-004; source/hazard MODE; ARCH; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide segregated training/research, commissioning, shadow, clinical-draft, clinical-approved, and de-graded/read-only operating modes with mode-specific permissions and destinations.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-GOV; V&V T/D; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
GOV-005	Category alias HOR-GOV-005; source/hazard AUTH; MODE; QMS; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: enforce institution-approved role authority and separation-of-duties rules for radiation oncologists, dosimetrists, medical physicists, therapists, informatics staff, quality staff....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-GOV; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
GOV-006	Category alias HOR-GOV-006; source/hazard SCP; QMS; ROAD; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain the intended-use statement, approved scope, risk controls, release status, configuration baseline, and applicable regulatory and quality-system records....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-GOV; V&V I/ A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
GOV-007	Category alias HOR-GOV-007; source/hazard AUTH; QMS; ROAD; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require clinical-governance approval for changes to deployment scope, accepted residual risk, evidence policy, model release, monitoring thresholds, or....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-GOV; V&V I/ A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.2 Safety boundaries and human authority (SAF) - 10 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-001	Category alias HNFR-SAF-001; source/hazard PRN; AUTH; MODE; TLR; H-04; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent any AI or automated service from prescribing treatment, signing or approving a prescription or plan, waiving required....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SYS; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
SAF-002	Category alias HNFR-SAF-002; source/hazard WF; ARCH; TLR; H-01; PAT	Wrong, stale, or incomplete patient, course, study, plan, or action context is accepted. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: bind every clinical action to an authenticated user and role and to a confirmed patient, course, study, frame....	Work may be applied to the wrong patient or object, or valid downstream approvals may become unsafe.	Bind authenticated actor and immutable case context, confirm two patient identifiers for consequential actions, reject mismatches, and invalidate stale approvals. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SYS; V&V T/D; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
SAF-003	Category alias HNFR-SAF-003; source/hazard HAZ; H-01; PAT	Wrong, stale, or incomplete patient, course, study, plan, or action context is accepted. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require confirmation of at least two patient identifiers before a state-changing clinical action is executed.	Work may be applied to the wrong patient or object, or valid downstream approvals may become unsafe.	Bind authenticated actor and immutable case context, confirm two patient identifiers for consequential actions, reject mismatches, and invalidate stale approvals. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SYS; V&V T/HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-004	Category alias HNFR-SAF-004; source/hazard PRN; WF; TLR; H-01,H-03,H-08; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: fail closed when a safety-critical field, permission, dependency, context assertion, evidence rule, or commissioned-use condition is missing, ambiguous....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SYS; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
SAF-005	Category alias HNFR-SAF-005; source/hazard MODE; WF; TLR; H-16; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: keep automated contours, objectives, transformations, calculations, and candidate plans in a visibly unsigned draft or sandbox state until....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SYS; V&V T/D; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
SAF-006	Category alias HNFR-SAF-006; source/hazard MODE; WF; PRN; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide a structured preview, explicit confirmation, cancellation, and rollback for reversible state-changing actions before those actions can affect....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SYS; V&V T/HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-007	Category alias HNFR-SAF-007; source/hazard INFO; TLR; H-17; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: invalidate affected downstream approvals, QA evidence, and transfer status whenever an approved dependency changes.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SYS; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
SAF-008	Category alias HNFR-SAF-008; source/hazard ARCH; PRN; HAZ; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement role authorization, state transitions, context assertions, hard stops, approval invalidation, and export policy in a deterministic safety....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SYS; V&V I/ T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
SAF-009	Category alias HNFR-SAF-009; source/hazard PRN; AUTH; VV; TLR; H-13; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: preserve institutionally required independent review, dose or MU verification, data-transfer verification, patient-specific QA, and physics checks without allowing....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SYS; V&V I/ T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-010	Category alias HNFR-SAF-010; source/hazard PRN; OPS; H-16; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify degraded operation and failed dependencies explicitly and shall not use an unannounced fallback that changes clinical behavior....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SYS; V&V T/D; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.3 Natural-language and voice interaction (NLI) - 8 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-001	Category alias HFR-NLI-001; source/hazard WF; NLI; H-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: accept typed text and user-initiated push-to-talk speech while prohibiting background listening and voice-only clinical signatures.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-NLI; V&V T/HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-002	Category alias HFR-NLI-002; source/hazard WF; NLI; TLR; H-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: compile each actionable natural-language request into a versioned, inspectable, typed intent before executing a state-changing action.	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-NLI; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
NLI-003	Category alias HFR-NLI-003; source/hazard WF; NLI; ARCH; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the clinical objects, requested action, parameters, units, priorities, evidence references, preconditions, expected outputs, and affected approval state.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-NLI; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
NLI-004	Category alias HFR-NLI-004; source/hazard ARCH; INFO; HFE; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: distinguish verified patient facts, signed clinical intent, user preferences, evidence-derived rules, system assumptions, and unresolved questions in the....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-NLI; V&V T/HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-005	Category HFR-NLI-005; source/hazard NLI; HAZ; H-02,H-07; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use deterministic validation for safety-critical numbers, units, dose basis, fractionation, negation, laterality, structure identity, and clinical-object references.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-NLI; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
NLI-006	Category alias HFR-NLI-006; source/hazard WF; NLI; TLR; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: display a structured read-back of what will change, the affected patient and object, assumptions, evidence, expected effect, expected....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-NLI; V&V T/HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
NLI-007	Category alias HFR-NLI-007; source/hazard NLI; TLR; H-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require explicit confirmation of speech transcripts containing safety-critical numbers, units, negation, laterality, prescription elements, or clinical-object names.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-NLI; V&V T/HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-008	Category alias HFR-NLI-008; source/hazard WF; INFO; TLR; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record the original user input or confirmed transcript, compiled intent, clarifications, assumptions, pre-view, confirmations, execution result, and user....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, pre-view, explicit confirmation, cancellation, and full interaction audit. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-NLI; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.4 Evidence and literature governance (EVD) - 8 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-001	Category alias HFR-EVD-001; source/hazard EVD; ARCH; TLR; H-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use only an institution-approved, closed evidence service for patient-specific clinical rules and shall not derive such rules from....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-EVD; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-002	Category alias HFR-EVD-002; source/hazard EVD; TLR; H-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: attach the source title, version or date, exact location, population, applicability, limitations, institutional approval status, effective date, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-EVD; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
EVD-003	Category alias HFR-EVD-003; source/hazard EVD; H-03,H-04; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: apply the approved authority order of patient-specific signed intent, applicable trial protocol, institutional standard, professional guidance, and supporting....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-EVD; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
EVD-004	Category alias HFR-EVD-004; source/hazard EVD; H-04; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: treat the signed prescription and physician-approved case intent as the highest patient-specific authority and shall stop when a....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-EVD; V&V T/D; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-005	Category alias HFR-EVD-005; source/hazard EVD; TLR; H-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: block or escalate conflicting, expired, retired, superseded, or in-applicable evidence rather than silently selecting or blending a clinical....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-EVD; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
EVD-006	Category alias HFR-EVD-006; source/hazard EVD; INFO; H-07; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: represent each computable rule with explicit structure definition, units, comparator, dose quantity and basis, population, exclusions, rule type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-EVD; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
EVD-007	Category alias HFR-EVD-007; source/hazard EVD; QMS; OPS; H-15; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: support controlled evidence proposal, independent clinical review, approval, publication, periodic review, retirement, emergency suspension, and rollback.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-EVD; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-008	Category alias HFR-EVD-008; source/hazard EVD; PRN; H-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide a no-source or no-applicable-rule response and require human resolution when a verified clinical rule cannot be produced.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-EVD; V&V T/HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.5 Clinical context, imaging, registration, and contours (CLN) - 9 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-001	Category alias HFR-CLN-001; source/hazard UC; MODEL; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: map verified case facts to candidate workflows or templates, identify missing material information, and require clinician confirmation without....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-IMG; V&V T/HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-002	Category alias HFR-CLN-002; source/hazard WF; INFO; H-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: acquire and bind the approved image studies, frame of reference, registrations, structure sets, prior-treatment information, and signed clinical....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-IMG; V&V T/ I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
CLN-003	Category alias HFR-CLN-003; source/hazard UC; VV; H-06; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: permit rigid or deformable registration and derived dose accumulation only through commissioned methods with registration-specific visualization, quality review....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-IMG; V&V T/ A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
CLN-004	Category alias HFR-CLN-004; source/hazard UC; MODEL; H-05; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: invoke contouring models only for approved anatomy, imaging conditions, structures, populations, and use cases and shall retain all....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-IMG; V&V T/ I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-005	Category alias HFR-CLN-005; source/hazard MODEL; VV; H-05; H-15; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: display contour-model uncertainty, input anomalies, and out-of-distribution status and shall abstain or route to manual contouring when the....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-IMG; V&V T/ HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
CLN-006	Category alias HFR-CLN-006; source/hazard UC; INFO; H-05; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate structure identity, nomenclature mapping, laterality, presence, completeness, topology, image coverage, and geometric plausibility before a contour set....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-IMG; V&V T/ A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
CLN-007	Category alias HFR-CLN-007; source/hazard INFO; TLR; H-05; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: preserve each contour's input study, frame of reference, model and version, configuration, naming map, confidence information, edits, reviewer....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-IMG; V&V I/ T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-008	Category alias HFR-CLN-008; source/hazard UC; VV; HFE; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide slice, surface, difference, and clinically relevant error-review views and shall capture the location and extent of user....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-IMG; V&V D/HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
CLN-009	Category alias HFR-CLN-009; source/hazard AUTH; UC; TLR; H-05; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require radiation-oncologist approval for clinical target contours and shall enforce institution-approved approval roles for organs at risk and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-IMG; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.6 Treatment-plan generation and calculation (PLN) - 12 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-001	Category alias HFR-PLN-001; source/hazard WF; UC; INFO; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create a versioned structured planning intent from the signed prescription, approved anatomy, patient-specific instructions, applicable protocol, institutional planning....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
PLN-002	Category alias HFR-PLN-002; source/hazard UC; ARCH; TLR; H-08; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: generate or modify candidate plans only through commissioned planning, optimization, robustness, and dose-calculation services operating within the validated....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
PLN-003	Category alias HFR-PLN-003; source/hazard PRN; MODEL; H-08; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent language, speech, disease-site, or generative models from directly calculating clinical dose or substituting for a commissioned dose....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-004	Category alias HFR-PLN-004; source/hazard UC; EVD; TLR; H-11; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: distinguish hard constraints, planning goals, optimization objectives, preferences, and reporting-only metrics.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
PLN-005	Category alias HFR-PLN-005; source/hazard UC; TLR; H-11; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent silent relaxation of a hard constraint and shall require an authorized explicit action, documented rationale, and pre/post....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
PLN-006	Category alias HFR-PLN-006; source/hazard UC; MOE; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: generate the number of candidate plans requested by an authorized user when feasible and shall identify when the....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V T/D; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-007	Category alias HFR-PLN-007; source/hazard UC; TLR; H-10; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: document the planning strategy, parameter variation, objective tradeoff, beam or modality variation, or other method used to create....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V I/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
PLN-008	Category alias HFR-PLN-008; source/hazard UC; TLR; H-10; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify infeasible candidates and candidates dominated by another candidate under the declared evaluation criteria.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
PLN-009	Category alias HFR-PLN-009; source/hazard INFO; MODEL; H-08; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: attest at runtime to the active machine model, energy or beam type, technique, dose algorithm, grid, optimizer, biological....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-010	Category alias HFR-PLN-010; source/hazard EVD; INFO; H-07,H-09; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: represent physical dose, RBE-weighted dose, fractionation, normalization, BED or EQD2 when approved, and proton-specific assumptions explicitly and without....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
PLN-011	Category alias HFR-PLN-011; source/hazard UC; VV; H-09; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: apply and record the institution-approved setup, range, motion, interplay, and other robustness scenarios required for the active technique....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
PLN-012	Category alias HFR-PLN-012; source/hazard INFO; ARCH; H-17; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: preserve candidate-plan identity, parent intent, parameters, calculation outputs, warnings, configuration, evaluation results, and lineage through iterative refinement.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-PLAN; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.7 Plan review, selection, QA, and transfer (REV) - 9 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-001	Category alias HFR-REV-001; source/hazard UC; HFE; MOE; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: present candidate plans in a consistent comparison workspace showing absolute values and differences without changing metric definitions or....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-REV; V&V D/HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
REV-002	Category alias HFR-REV-002; source/hazard UC; TLR; H-10; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: display target coverage, organ-at-risk results, constraint status, spatial dose, hotspots and coldspots, robustness, uncertainty, complexity, deliverability, and provenance....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-REV; V&V D/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-003	Category alias HFR-REV-003; source/hazard UC; PRN; H-10,H-18; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: expose the declared tradeoffs and shall not select or recommend a winning plan solely through an opaque composite....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-REV; V&V I/ HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
REV-004	Category alias HFR-REV-004; source/hazard INFO; MOE; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record the selected plan, rejected alternatives, review comments, deviations, tradeoff rationale, and role-specific decisions.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-REV; V&V I/ T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
REV-005	Category alias HFR-REV-005; source/hazard AUTH; MODE; HAZ; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: enforce radiation-oncologist clinical plan approval and qualified-medical-physicist technical review before a candidate can enter the approved state.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-REV; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-006	Category alias HFR-REV-006; source/hazard VV; TLR; H-13; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: route the approved candidate through the institutionally required independent dose or MU verification, patient-specific QA, transfer QA, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-REV; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
REV-007	Category alias HFR-REV-007; source/hazard WF; INFO; H-12,H-17; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: authorize export only after all required approvals and QA states are current and shall verify source-to-destination identity, content....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-REV; V&V T/D; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
REV-008	Category alias HFR-REV-008; source/hazard MODE; INFO; H-17; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: make approved clinical objects immutable or version-controlled and shall require re-review of every affected downstream object after a....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-REV; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-009	Category alias HFR-REV-009; source/hazard PRN; VV; TLR; H-13; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: document the independence assumptions, shared inputs, shared dependencies, ownership boundaries, and common-cause failure risks of each automated verification....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-REV; V&V I/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.8 Data, interoperability, provenance, and audit (DAT) - 10 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-001	Category alias HFR-DAT-001; source/hazard INFO; ARCH; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: assign a unique identity, version, lifecycle state, owner, and source linkage to each clinical context, prescription intent, anatomy....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-DATA; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-002	Category alias HFR-DAT-002; source/hazard INFO; TLR; H-17; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a dependency and provenance graph that traces every derived object to its inputs, transformations, software, model, evidence....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-DATA; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
DAT-003	Category alias HFR-DAT-003; source/hazard INFO; H-07,H-12; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate and preserve explicit units, coordinate systems, dose quantity and basis, structure definitions, and rounding behavior at storage....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-DATA; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
DAT-004	Category alias HFR-DAT-004; source/hazard ARCH; TLR; H-01,H-12; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate DICOM patient and study identities, SOP Instance UIDs, references, frame-of-reference UIDs, required attributes, coordinate transformations, units, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-DATA; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-005	Category alias HFR-DAT-005; source/hazard ARCH; QMS; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: support the approved DICOM objects and transactions required for images, registrations, segmentations or structures, RT Plan, RT Ion....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-DATA; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
DAT-006	Category alias HFR-DAT-006; source/hazard ARCH; QMS; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use approved IHE-RO profiles and FHIR or CodeX radiation-therapy exchanges where implemented and shall document conformance and known....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-DATA; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
DAT-007	Category alias HFR-DAT-007; source/hazard INFO; H-12,H-16; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reject incomplete, ambiguous, inconsistent, orphaned, or partially transferred clinical data rather than promoting it to a valid state.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-DATA; V&V T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-008	Category alias HFR-DAT-008; source/hazard INFO; TLR; H-14; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a tamper-evident, time-synchronized audit ledger of commands, context, inputs, versions, evidence, outputs, warnings, edits, approvals, QA, exports....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-DATA; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
DAT-009	Category alias HFR-DAT-009; source/hazard INFO; SEC; H-12,H-14; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use integrity hashes or equivalent controls to detect unauthorized modification of clinical objects, evidence packages, software artifacts, model....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-DATA; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
DAT-010	Category alias HFR-DAT-010; source/hazard INFO; DEC; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the authoritative system of record and retention, legal-record, export, archival, and reconciliation policy for each clinical object....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-DATA; V&V I/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.9 AI and inference-model lifecycle (AIM) - 8 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
AIM-001	Category alias HFR-AIM-001; source/hazard MODEL; TLR; H-15; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: invoke only institution-approved, locked model versions registered for the active task and clinical operating mode.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-AI; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
AIM-002	Category alias HFR-AIM-002; source/hazard ARCH; MODEL; TLR; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: verify that the active case satisfies the model's intended use, input contract, population, imaging or data conditions, required....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-AI; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
AIM-003	Category alias HFR-AIM-003; source/hazard MODEL; TLR; H-05,H-15; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement a defined abstention, rejection, or manual-review response for unsupported, anomalous, incomplete, or out-of-distribution input.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-AI; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
AIM-004	Category alias HFR-AIM-004; source/hazard MODEL; TLR; H-15; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prohibit production model-weight updates and unapproved behavioral learning from routine patient cases, user edits, prompts, or outcomes.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-AI; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
AIM-005	Category alias HFR-AIM-005; source/hazard MODEL; SEC; QMS; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a model and dependency registry containing version, intended use, owner, supplier, training and evaluation provenance, limitations, approved....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-AI; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
AIM-006	Category alias HFR-AIM-006; source/hazard MODEL; VV; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use separated development, tuning, internal test, external test, and site-acceptance data with documented overlap controls.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-AI; V&V I/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
AIM-007	Category alias HFR- AIM-007; source/hazard MODEL; VV; MOE; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: evaluate and display clinically relevant overall and subgroup performance, calibration, failure detection, uncertainty, edit burden, and known limitations....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-AI; V&V A/ HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
AIM-008	Category alias HFR- AIM-008; source/hazard MODEL; QMS; ROAD; H-15; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: undergo impact assessment, regression testing, approval, versioning, and roll-back planning.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-AI; V&V I/ T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.10 Human factors, usability, and competency (HFE) - 6 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
HFE-001	Category alias HNFR- HFE-001; source/hazard HFE; H-01,H-17; PAT	Wrong, stale, or incomplete patient, course, study, plan, or action context is accepted. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: continuously display the active patient, course, study, plan, operating mode, approval state, and unsaved or unverified changes during....	Work may be applied to the wrong patient or object, or valid downstream approvals may become unsafe.	Bind authenticated actor and immutable case context, confirm two patient identifiers for consequential actions, reject mismatches, and invalidate stale approvals. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-UX; V&V HFE/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
HFE-002	Category alias HNFR- HFE-002; source/hazard NLI; HFE; H-18; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: visually distinguish verified patient facts, signed intent, approved evidence, model suggestions, user preferences, assumptions, warnings, and unresolved questions.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-UX; V&V HFE/I; 3 SUB + 3 HLIR; FMEA S/O/D/ RPN TBD	High
HFE-003	Category alias HNFR- HFE-003; source/hazard HFE; H-18; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the hazard, affected object, required action, and escalation path and shall not rely on color alone.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-UX; V&V HFE/I; 3 SUB + 3 HLIR; FMEA S/O/D/ RPN TBD	Moderate
HFE-004	Category alias HNFR- HFE-004; source/hazard WF; HFE; OPS; H-18; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be proportional to clinical risk, and the NL-TPS shall monitor alert burden, overrides, workarounds, and critical-warning comprehension.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-UX; V&V HFE/A; 3 SUB + 3 HLIR; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
HFE-005	Category alias HNFR-HFE-005; source/hazard VV; HFE; TLR; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: complete formative and summative human-factors validation with representative intended users, critical tasks, environments, interruptions, handoffs, and recovery scenarios....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-UX; V&V HFE/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
HFE-006	Category alias HNFR-HFE-006; source/hazard AUTH; HFE; ROAD; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require role-specific training, demonstrated competency, and current authorization before a user can perform clinical functions assigned to that....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-UX; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.11 Privacy, cybersecurity, and secure development (SEC) - 8 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SEC-001	Category alias HNFR-SEC-001; source/hazard SEC; TLR; H-14; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: enforce unique user identity, multi-factor authentication, least privilege, role and context authorization, session controls, and periodic access review.	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SEC; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
SEC-002	Category alias HNFR-SEC-002; source/hazard SEC; TLR; H-14; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: protect data and services through approved encryption, network segmentation, managed secrets, secure configuration, endpoint controls, and least-privilege service....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SEC; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
SEC-003	Category alias HNFR-SEC-003; source/hazard SEC; H-14; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent protected health information, images, prompts, transcripts, or outputs from being sent to an unapproved connector, model service....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SEC; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SEC-004	Category alias HNFR-SEC-004; source/hazard SEC; H-14; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: treat retrieved text, uploaded documents, DICOM metadata, user content, and model output as untrusted input and shall isolate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SEC; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
SEC-005	Category alias HNFR-SEC-005; source/hazard MODEL; SEC; H-14,H-15; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: verify the identity, integrity, signature, provenance, and approved status of software, model, evidence, configuration, and dependency artifacts before....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SEC; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
SEC-006	Category alias HNFR-SEC-006; source/hazard SEC; QMS; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: include threat modeling, secure coding, code review, software bill of materials, vulnerability scanning, penetration testing, patch governance, and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SEC; V&V I/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SEC-007	Category alias HNFR- SEC-007; source/hazard SEC; OPS; H-14; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: centrally record and detect unauthorized access, anomalous export, integrity failure, malicious input, model or evidence compromise, secrets exposure....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SEC; V&V T/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
SEC-008	Category alias HNFR- SEC-008; source/hazard SEC; OPS; H-14; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: support tested clinical and cybersecurity incident containment, evidence preservation, patient-impact assessment, communication, reporting, recovery, and corrective and preventive....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-SEC; V&V D/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.12 Clinical operations, monitoring, and resilience (OPS) - 7 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OPS-001	Category alias HOR-OPS-001; source/hazard OPS; MOE; TLR; H-15; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: monitor predefined safety, model, evidence, workflow, technical, subgroup, reliability, and security signals against approved action thresholds.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-OPS; V&V T/A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
OPS-002	Category alias HOR-OPS-002; source/hazard OPS; H-16; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use atomic job states and shall prevent a timed-out, failed, canceled, partial, or unreconciled result from appearing complete....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-OPS; V&V T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
OPS-003	Category alias HOR-OPS-003; source/hazard OPS; TLR; H-15; H-16; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: support rapid suspension of a model, evidence rule, connector, workflow, disease site, or complete clinical service and rollback....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-OPS; V&V D/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OPS-004	Category alias HOR-OPS-004; source/hazard MODE; OPS; TLR; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide an explicit degraded or read-only mode and an institution-approved downtime procedure that prohibits new clinical execution while....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-OPS; V&V D/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
OPS-005	Category alias HOR-OPS-005; source/hazard SEC; OPS; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain validated backups and shall perform staged restoration, object and destination reconciliation, approval-state verification, regression testing, and documented....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-OPS; V&V D/I; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
OPS-006	Category alias HOR-OPS-006; source/hazard OPS; QMS; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: support reporting, triage, investigation, patient-impact assessment, evidence preservation, trend analysis, and corrective and preventive action for incidents, near....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-OPS; V&V I/ D; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OPS-007	Category alias HOR-OPS-007; source/hazard OPS; MOE; DEC; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define, approve, monitor, and periodically review clinical service levels and action thresholds for availability, latency, recovery, data retention....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-OPS; V&V I/ A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

5.13 Verification, commissioning, release, and change control (VAL) - 9 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-001	Category alias HOR-VAL-001; source/hazard HAZ; VV; QMS; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: trace each stakeholder need, hazard, risk control, and high-level requirement to allocated design, implementation, verification method, test result....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-VV; V&V I/ A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-002	Category alias HOR-VAL-002; source/hazard VV; ROAD; TLR; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: pass site-specific acceptance, commissioning, end-to-end, regression, shadow-mode, and governance release criteria before clinical use.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-VV; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
VAL-003	Category alias HOR-VAL-003; source/hazard VV; HAZ; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: include locked reference cases and challenge cases for language and voice, evidence computation, segmentation, registration, planning, dose, robustness....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-VV; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
VAL-004	Category alias HOR-VAL-004; source/hazard HAZ; VV; OPS; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: exercise normal, boundary, rare but foreseeable, wrong-context, partial-failure, timeout, rollback, post-approval change, and recovery scenarios through the official....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-VV; V&V I/T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-005	Category alias HOR- VAL-005; source/hazard MODEL; VV; MOE; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: include task-specific technical performance, clinically significant error, dose impact where relevant, edit burden, failure detection, subgroup performance, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-VV; V&V A/ HFE; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	Gate
VAL-006	Category alias HOR- VAL-006; source/hazard VV; ROAD; DEC; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: progress through retrospective, shadow, and controlled limited-clinical stages with predefined endpoints, independent adjudication, and applicable research, ethics, quality....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-VV; V&V I/ A; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
VAL-007	Category alias HOR- VAL-007; source/hazard ROAD; MOE; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent progression through each release gate until its predefined safety, quality, subgroup, usability, reliability, cybersecurity, and governance evidence....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-VV; V&V I/ T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-008	Category alias HOR- VAL-008; source/hazard QMS; ROAD; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: receive documented change-impact assessment, commissioning, human-factors assessment, regression testing, and monitoring approval.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-VV; V&V I/ T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High
VAL-009	Category alias HOR- VAL-009; source/hazard HAZ; VV; QMS; ROAD; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: not enter or remain in clinical service with an unresolved critical anomaly, unacceptable residual risk, failed release threshold....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three detailed children and three HLIRs; link hazards, components, tests, results, monitoring, and release decision.	T-VV; V&V I/ T; 3 SUB + 3 HLIR; FMEA S/O/D/RPN TBD	High

6 Detailed sub-requirement risk records

6.1 Governance, intended use, and scope (GOV) - 21 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
GOV-001-01	Category alias OSR-GOV-001-01; parent GOV-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify AI-produced content as decision support or draft output and shall identify the authorized human roles retaining clinical....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation UI/QMS; preserve the parent obligation; execute I/HFE verification and negative or failure-path tests.	T-GOV; UI/QMS; V&V I/HFE; FMEA S/O/D/RPN TBD	High
GOV-001-02	Category alias OSR-GOV-001-02; parent GOV-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reject any state transition in which model output, an automated service identity, or an unauthenticated actor is presented....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-GOV; SK; V&V T/I; FMEA S/O/D/RPN TBD	High
GOV-001-03	Category alias OSR-GOV-001-03; parent GOV-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: review the intended-use statement, labeling, training content, and release communications for consistency with the human-supervised operating model before....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-GOV; QMS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
GOV-002-01	Category alias OSR- GOV-002-01; parent GOV- 002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned approved-use matrix containing the permitted institution, patient population, disease site, modality, machine, technique, task, operating....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-GOV; CFG; V&V I/T; FMEA S/ O/D/RPN TBD	High
GOV-002-02	Category alias OSR- GOV-002-02; parent GOV- 002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: compare the active case and requested workflow with every applicable field in the approved-use matrix before enabling a....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/CXT; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-GOV; SK/ CXT; V&V T; FMEA S/O/D/ RPN TBD	High
GOV-002-03	Category alias OSR- GOV-002-03; parent GOV- 002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record the evaluated scope attributes, approved-use matrix version, decision, and denial reason for each clinical workflow initiation.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-GOV; AUD; V&V I/T; FMEA S/ O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
GOV-003-01	Category alias OSR-GOV-003-01; parent GOV-003; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: classify any unlisted scope value or workflow combination as unsupported and shall block state change, candidate promotion, and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement allocation SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-GOV; SK; V&V T; FMEA S/O/D/RPN TBD	Moderate
GOV-003-02	Category alias OSR-GOV-003-02; parent GOV-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require linked risk assessment, commissioning, validation, and governance-approval records for each proposed scope expansion.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/VVP; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-GOV; QMS/VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
GOV-003-03	Category alias OSR-GOV-003-03; parent GOV-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent activation of a scope expansion until all required records are complete, approved, effective, and bound to the....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation CFG; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-GOV; CFG; V&V T/I; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
GOV-004-01	Category alias OSR-GOV-004-01; parent GOV-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define, for each operating mode, the permitted actions, data classes, model and evidence versions, approval states, interfaces, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-GOV; CFG; V&V I/T; FMEA S/O/D/RPN TBD	High
GOV-004-02	Category alias OSR-GOV-004-02; parent GOV-004; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: enforce the active mode's permissions and shall display the active mode continuously during user interaction.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation IAM/SK/UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-GOV; IAM/SK/UI; V&V T/HFE; FMEA S/O/D/RPN TBD	High
GOV-004-03	Category alias OSR-GOV-004-03; parent GOV-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent promotion or transfer of objects between operating modes except through an approved, versioned, and audited transition workflow.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation DATA/INT; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-GOV; DATA/INT; V&V T/I; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
GOV-005-01	Category alias OSR-GOV-005-01; parent GOV-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an institution-approved role-permission matrix that identifies allowed functions and incompatible responsibilities for each authorized clinical and technical....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation IAM/QMS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-GOV; IAM/QMS; V&V I/T; FMEA S/O/D/RPN TBD	High
GOV-005-02	Category alias OSR-GOV-005-02; parent GOV-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent one actor or service identity from satisfying approval steps that the active separation-of-duties policy defines as incompatible.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-GOV; SK; V&V T; FMEA S/O/D/RPN TBD	High
GOV-005-03	Category alias OSR-GOV-005-03; parent GOV-005; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: perform and retain periodic access, role, exception, and separation-of-duties reviews, including disposition of identified conflicts.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation QMS/AUD; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-GOV; QMS/AUD; V&V I/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
GOV-006-01	Category alias OSR-GOV-006-01; parent GOV-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: assign an identifier, owner, version, approval state, effective date, and supersession status to each intended-use, scope, risk, release....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS; preserve the parent obligation; execute I verification and negative or failure-path tests.	T-GOV; QMS; V&V I; FMEA S/O/D/RPN TBD	High
GOV-006-02	Category alias OSR-GOV-006-02; parent GOV-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reference the exact approved versions of all required controlled artifacts and shall not activate with an incomplete or....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation CFG; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-GOV; CFG; V&V T/I; FMEA S/O/D/RPN TBD	High
GOV-006-03	Category alias OSR-GOV-006-03; parent GOV-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: generate a reproducible release-baseline report listing each controlled artifact, approval, effective status, and integrity value.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD/QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-GOV; AUD/QMS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
GOV-007-01	Category alias OSR- GOV-007-01; parent GOV- 007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: classify changes to deployment scope, residual risk, evidence policy, model release, monitoring thresholds, and workflow authority as governance-controlled....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-GOV; QMS; V&V I/A; FMEA S/O/D/RPN TBD	High
GOV-007-02	Category alias OSR- GOV-007-02; parent GOV- 007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require the designated clinical-governance approvals before a governance-controlled change can be deployed or made effective.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation CFG; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-GOV; CFG; V&V T/I; FMEA S/O/D/RPN TBD	High
GOV-007-03	Category alias OSR- GOV-007-03; parent GOV- 007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain the prior and new values, rationale, impact assessment, approvers, approval times, effective date, and rollback reference.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD; preserve the parent obligation; execute I verification and negative or failure-path tests.	T-GOV; AUD; V&V I; FMEA S/O/D/RPN TBD	High

6.2 Safety boundaries and human authority (SAF) - 30 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-001-01	Category alias NFSR-SAF-001-01; parent SAF-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: expose no AI-callable capability that can prescribe treatment, provide a clinical signature, waive QA, release a plan, or....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/ARCH; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SYS; SK/ARCH; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SAF-001-02	Category alias NFSR-SAF-001-02; parent SAF-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require an authenticated authorized human acting through the designated clinical workflow and shall reject automated or service-account approval.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation IAM/SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-SYS; IAM/SK; V&V T; FMEA S/O/D/RPN TBD	High
SAF-001-03	Category alias NFSR-SAF-001-03; parent SAF-001; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: demonstrate that prompts, tool calls, model output, malformed requests, and privilege escalation attempts cannot invoke a restricted transition.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation VVP; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-SYS; VVP; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-002-01	Category alias NFSR-SAF-002-01; parent SAF-002; PAT	Wrong, stale, or incomplete patient, course, study, plan, or action context is accepted. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create an immutable action context containing the authenticated user, role, patient, course, study, frame of reference, plan version....	Work may be applied to the wrong patient or object, or valid downstream approvals may become unsafe.	Bind authenticated actor and immutable case context, confirm two patient identifiers for consequential actions, reject mismatches, and invalidate stale approvals. Implement allocation CXT/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SYS; CXT/DATA; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SAF-002-02	Category alias NFSR-SAF-002-02; parent SAF-002; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: block a clinical action when any required context element is absent or inconsistent, and the interface shall present....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation SK/UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-SYS; SK/UI; V&V T/HFE; FMEA S/O/D/RPN TBD	High
SAF-002-03	Category alias NFSR-SAF-002-03; parent SAF-002; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: store the complete action context and the identifiers of every object read, changed, created, or transferred by the....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SYS; AUD; V&V I/T; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-003-01	Category alias NFSR-SAF-003-01; parent SAF-003; PAT	Wrong, stale, or incomplete patient, course, study, plan, or action context is accepted. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: present at least two independently sourced patient identifiers and shall require an active user confirmation that is not....	Work may be applied to the wrong patient or object, or valid downstream approvals may become unsafe.	Bind authenticated actor and immutable case context, confirm two patient identifiers for consequential actions, reject mismatches, and invalidate stale approvals. Implement allocation UI/CXT; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-SYS; UI/CXT; V&V T/HFE; FMEA S/O/D/RPN TBD	Gate
SAF-003-02	Category alias NFSR-SAF-003-02; parent SAF-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reject state-changing execution when the identifier confirmation is absent, stale, associated with another patient, or invalidated by a....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-SYS; SK; V&V T; FMEA S/O/D/RPN TBD	Gate
SAF-003-03	Category alias NFSR-SAF-003-03; parent SAF-003; PAT	Wrong, stale, or incomplete patient, course, study, plan, or action context is accepted. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record identifier-confirmation events and shall include wrong-patient, interruption, and context-switch scenarios in human-factors validation.	Work may be applied to the wrong patient or object, or valid downstream approvals may become unsafe.	Bind authenticated actor and immutable case context, confirm two patient identifiers for consequential actions, reject mismatches, and invalidate stale approvals. Implement allocation AUD/HFE; preserve the parent obligation; execute I/HFE verification and negative or failure-path tests.	T-SYS; AUD/HFE; V&V I/HFE; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-004-01	Category alias NFSR-SAF-004-01; parent SAF-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: represent each safety-critical precondition with explicit satisfied, unsatisfied, or unknown status and shall permit execution only when every....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SYS; SK; V&V I/T; FMEA S/O/D/RPN TBD	High
SAF-004-02	Category alias NFSR-SAF-004-02; parent SAF-004; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify each blocking precondition, the affected action, and the authorized resolution path without offering an ungoverned bypass.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-SYS; UI; V&V T/HFE; FMEA S/O/D/RPN TBD	High
SAF-004-03	Category alias NFSR-SAF-004-03; parent SAF-004; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record every fail-closed decision and shall verify missing, ambiguous, inconsistent, expired, and unsupported preconditions with controlled negative tests.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation AUD/VVP; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SYS; AUD/VVP; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-005-01	Category alias NFSR-SAF-005-01; parent SAF-005; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: assign every automated contour, objective, transformation, calculation, and candidate plan an unsigned draft or sandbox state at creation.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation DATA/SK; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SYS; DATA/SK; V&V I/T; FMEA S/O/D/ RPN TBD	High
SAF-005-02	Category alias NFSR-SAF-005-02; parent SAF-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: distinguish draft and sandbox objects from reviewed and approved clinical objects using persistent text and status indicators that....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation UI; preserve the parent obligation; execute D/HFE verification and negative or failure-path tests.	T-SYS; UI; V&V D/HFE; FMEA S/O/D/ RPN TBD	High
SAF-005-03	Category alias NFSR-SAF-005-03; parent SAF-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reject an automated object until all configured human reviews and approvals are current.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/INT; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-SYS; SK/INT; V&V T; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-006-01	Category alias NFSR- SAF-006-01; parent SAF- 006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the affected objects, current and proposed values, expected consequences, required approvals, cancellation path, and rollback eligibility.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation UI/ DATA; preserve the parent obligation; execute I/HFE verification and negative or failure-path tests.	T-SYS; UI/ DATA; V&V I/ HFE; FMEA S/O/D/RPN TBD	High
SAF-006-02	Category alias NFSR- SAF-006-02; parent SAF- 006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require a distinct confirmation action after preview, and the confirmation shall expire when the patient context, object version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/ UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-SYS; SK/UI; V&V T/HFE; FMEA S/O/D/ RPN TBD	High
SAF-006-03	Category alias NFSR- SAF-006-03; parent SAF- 006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute as recoverable transactions with tested cancellation and rollback behavior that preserves object identity, provenance, and prior approved....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation DATA/ VVP; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-SYS; DATA/ VVP; V&V T; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-007-01	Category alias NFSR- SAF-007-01; parent SAF- 007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify every downstream approval, QA result, transfer status, and derived object affected by a change to an approved....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SYS; DATA; V&V I/T; FMEA S/ O/D/RPN TBD	High
SAF-007-02	Category alias NFSR- SAF-007-02; parent SAF- 007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: atomically invalidate affected approvals, QA evidence, and export eligibility before the changed dependency becomes the active version.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-SYS; SK; V&V T/A; FMEA S/O/D/ RPN TBD	Gate
SAF-007-03	Category alias NFSR- SAF-007-03; parent SAF- 007; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: present the required re-review set, and the audit ledger shall record the invalidation cause, affected objects, prior states....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation UI/AUD; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-SYS; UI/ AUD; V&V D/I; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-008-01	Category alias NFSR- SAF-008-01; parent SAF- 008; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be a versioned deterministic component separated from generative-model prompts, weights, sampling behavior, and model-provided policy text.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation SK/ARCH; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-SYS; SK/ ARCH; V&V I/ A; FMEA S/O/ D/RPN TBD	High
SAF-008-02	Category alias NFSR- SAF-008-02; parent SAF- 008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement explicit allowlisted rules for authorization, state transitions, context assertions, hard stops, approval invalidation, and export eligibility.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SYS; SK; V&V I/T; FMEA S/ O/D/RPN TBD	Gate
SAF-008-03	Category alias NFSR- SAF-008-03; parent SAF- 008; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide an independent test interface and evidence demonstrating repeatable decisions for equivalent inputs across supported model configurations.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation VVP; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-SYS; VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-009-01	Category alias NFSR- SAF-009-01; parent SAF- 009; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify every required independent review, dose or MU verification, transfer verification, patient-specific QA activity, physics check, and responsible....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation CFG/QMS; preserve the parent obligation; execute I verification and negative or failure-path tests.	T-SYS; CFG/ QMS; V&V I; FMEA S/O/D/ RPN TBD	Gate
SAF-009-02	Category alias NFSR- SAF-009-02; parent SAF- 009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent AI output, shared automated evidence, or one incompatible actor from satisfying an independent approval or verification step.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-SYS; SK; V&V T; FMEA S/O/D/RPN TBD	High
SAF-009-03	Category alias NFSR- SAF-009-03; parent SAF- 009; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: remain blocked until current results and authorized signoffs exist for every required independent check, with all evidence linked....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation REVW/AUD; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-SYS; REVV/ AUD; V&V T/I; FMEA S/ O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SAF-010-01	Category alias NFSR-SAF-010-01; parent SAF-010; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: assign an explicit normal, degraded, failed, or unavailable state to each safety-relevant dependency and shall map each state....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation OPS/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SYS; OPS/CFG; V&V I/T; FMEA S/O/D/RPN TBD	High
SAF-010-02	Category alias NFSR-SAF-010-02; parent SAF-010; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: announce degraded operation and the safety kernel shall prohibit any fallback that is not listed and approved for....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation UI/SK; preserve the parent obligation; execute D/T verification and negative or failure-path tests.	T-SYS; UI/SK; V&V D/T; FMEA S/O/D/RPN TBD	High
SAF-010-03	Category alias NFSR-SAF-010-03; parent SAF-010; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record the failed dependency, selected response, affected clinical behavior, user notifications, confirmations, and restoration or escalation outcome.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SYS; AUD; V&V I/T; FMEA S/O/D/RPN TBD	Moderate

6.3 Natural-language and voice interaction (NLI) - 24 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-001-01	Category alias FSR-NLI-001-01; parent NLI-001; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: accept typed input and shall activate speech capture only while an authenticated user deliberately invokes the push-to-talk control.	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement allocation NLU/UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-NLI; NLU/UI; V&V T/HFE; FMEA S/O/D/RPN TBD	High
NLI-001-02	Category alias FSR-NLI-001-02; parent NLI-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be disabled by default, shall terminate when push-to-talk is released or canceled, and shall not persist background audio....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation NLU/SEC; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-NLI; NLU/SEC; V&V I/T; FMEA S/O/D/RPN TBD	High
NLI-001-03	Category alias FSR-NLI-001-03; parent NLI-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prohibit voice-only clinical signatures and shall route all approval actions through an authenticated non-voice confirmation control.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-NLI; SK/UI; V&V T/HFE; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-002-01	Category alias FSR-NLI-002-01; parent NLI-002; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: transform each actionable request into a schema-valid typed-intent object with a unique identity, schema version, content version, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation NLU/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-NLI; NLU/DATA; V&V I/T; FMEA S/O/D/RPN TBD	High
NLI-002-02	Category alias FSR-NLI-002-02; parent NLI-002; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent a state-changing action until the typed intent has passed deterministic validation and the required preview and confirmation....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement allocation SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-NLI; SK; V&V T; FMEA S/O/D/RPN TBD	Gate
NLI-002-03	Category alias FSR-NLI-002-03; parent NLI-002; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain every typed-intent version and its relationship to the original input, clarifications, validation result, confirmation, and execution result.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-NLI; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-003-01	Category alias FSR-NLI-003-01; parent NLI-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide explicit fields for clinical-object identifiers, action, parameters, units, priorities, evidence references, preconditions, expected outputs, and approval-state effects.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation NLU/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-NLI; NLU/DATA; V&V I/T; FMEA S/O/D/RPN TBD	High
NLI-003-02	Category alias FSR-NLI-003-02; parent NLI-003; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reject a required field that is missing, untyped, unresolved, internally inconsistent, or not bound to a current clinical....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation NLU/SK; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-NLI; NLU/SK; V&V T/A; FMEA S/O/D/RPN TBD	High
NLI-003-03	Category alias FSR-NLI-003-03; parent NLI-003; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: render the same serialized typed-intent content used for deterministic execution.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation UI/AUD; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-NLI; UI/AUD; V&V D/I; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-004-01	Category alias FSR-NLI-004-01; parent NLI-004; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: carry a source classification of verified fact, signed clinical intent, user preference, evidence-derived rule, system assumption, or unresolved....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation DATA/NLU; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-NLI; DATA/NLU; V&V I/T; FMEA S/O/D/RPN TBD	High
NLI-004-02	Category alias FSR-NLI-004-02; parent NLI-004; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: distinguish source classifications with persistent text labels and a visible legend that does not rely on color alone.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation UI; preserve the parent obligation; execute HFE/I verification and negative or failure-path tests.	T-NLI; UI; V&V HFE/I; FMEA S/O/D/RPN TBD	High
NLI-004-03	Category alias FSR-NLI-004-03; parent NLI-004; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent a model suggestion, preference, assumption, or unresolved question from overwriting or being promoted as a verified fact....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-NLI; SK; V&V T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-005-01	Category alias FSR-NLI-005-01; parent NLI-005; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: parse and normalize safety-critical numbers, units, dose basis, fractionation, negation, laterality, structure identity, and clinical-object references.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation NLU/SK; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-NLI; NLU/SK; V&V I/T; FMEA S/O/D/RPN TBD	Gate
NLI-005-02	Category alias FSR-NLI-005-02; parent NLI-005; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reject or request clarification for ambiguous, incompatible, implausible, or unresolved safety-critical values before intent confirmation.	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement allocation NLU/UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-NLI; NLU/UI; V&V T/HFE; FMEA S/O/D/RPN TBD	High
NLI-005-03	Category alias FSR-NLI-005-03; parent NLI-005; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: include accepted, rejected, boundary, negated, unit-conversion, laterality, alias, dictation-error, and adversarial examples for every supported safety-critical field.	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement allocation VVP; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-NLI; VVP; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-006-01	Category alias FSR-NLI-006-01; parent NLI-006; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the action, affected patient and objects, current and proposed values, assumptions, evidence, expected effect, expected run-time, and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement allocation UI; preserve the parent obligation; execute I/HFE verification and negative or failure-path tests.	T-NLI; UI; V&V I/HFE; FMEA S/O/D/RPN TBD	High
NLI-006-02	Category alias FSR-NLI-006-02; parent NLI-006; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require the read-back to be generated from the validated typed intent and shall regenerate it whenever the intent....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement allocation SK/UI; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-NLI; SK/UI; V&V T; FMEA S/O/D/RPN TBD	High
NLI-006-03	Category alias FSR-NLI-006-03; parent NLI-006; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be retained in the audit ledger.	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-NLI; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-007-01	Category alias FSR-NLI-007-01; parent NLI-007; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: flag utterances containing safety-critical numbers, units, negation, laterality, prescription elements, or clinical-object names.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation NLU; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-NLI; NLU; V&V I/T; FMEA S/O/D/RPN TBD	Gate
NLI-007-02	Category alias FSR-NLI-007-02; parent NLI-007; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be displayed for active user confirmation or correction, and the safety kernel shall block execution until confirmation is....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement allocation UI/SK; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-NLI; UI/SK; V&V T/HFE; FMEA S/O/D/RPN TBD	High
NLI-007-03	Category alias FSR-NLI-007-03; parent NLI-007; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain the captured transcript, flagged tokens, user corrections, confirmed transcript, and confirmation identity and time.	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-NLI; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NLI-008-01	Category alias FSR- NLI-008-01; parent NLI- 008; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: link the original text or confirmed transcript, typed-intent versions, clarification turns, assumptions, previews, confirmations, execution outputs, and user....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement allocation AUD/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-NLI; AUD/DATA; V&V I/T; FMEA S/O/D/RPN TBD	High
NLI-008-02	Category alias FSR- NLI-008-02; parent NLI- 008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be append-only, integrity protected, access controlled, and time synchronized according to the approved retention policy.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD/SEC; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-NLI; AUD/SEC; V&V T/I; FMEA S/O/D/RPN TBD	High
NLI-008-03	Category alias FSR- NLI-008-03; parent NLI- 008; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be able to reconstruct the ordered interaction and identify the exact inputs, versions, decisions, and outputs used for....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement allocation AUD/VVP; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-NLI; AUD/VVP; V&V T; FMEA S/O/D/RPN TBD	High

6.4 Evidence and literature governance (EVD) - 24 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-001-01	Category alias FSR-EVD-001-01; parent EVD-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use an allowlisted evidence repository and retrieval configuration containing only institution-approved sources, rules, versions, and effective states.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation EVID/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-EVD; EVID/CFG; V&V I/T; FMEA S/O/D/RPN TBD	High
EVD-001-02	Category alias FSR-EVD-001-02; parent EVD-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: block unrestricted internet retrieval and unapproved external connectors for patient-specific clinical-rule generation in clinical mode.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SEC/EVID; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-EVD; SEC/EVID; V&V T; FMEA S/O/D/RPN TBD	High
EVD-001-03	Category alias FSR-EVD-001-03; parent EVD-001; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record the repository, retrieval configuration, source set, operating mode, result, and any blocked retrieval attempt.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-EVD; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-002-01	Category alias FSR-EVD-002-01; parent EVD-002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require source title, version or date, exact location, population, applicability, limitations, approval status, effective date, and review date.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation EVID/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-EVD; EVID/DATA; V&V I/T; FMEA S/O/D/RPN TBD	High
EVD-002-02	Category alias FSR-EVD-002-02; parent EVD-002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reject an evidence-derived rule when any required provenance field is missing, invalid, expired, or inconsistent with the approved....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation EVID/QMS; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-EVD; EVID/QMS; V&V T/I; FMEA S/O/D/RPN TBD	High
EVD-002-03	Category alias FSR-EVD-002-03; parent EVD-002; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: make the complete provenance and applicability record available from every displayed or applied evidence-derived rule.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation UI; preserve the parent obligation; execute D/HFE verification and negative or failure-path tests.	T-EVD; UI; V&V D/HFE; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-003-01	Category alias FSR-EVD-003-01; parent EVD-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: encode the approved authority tiers and ordering for signed patient intent, trial protocol, institutional standard, professional guidance, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation EVID/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-EVD; EVID/CFG; V&V I/T; FMEA S/O/D/RPN TBD	High
EVD-003-02	Category alias FSR-EVD-003-02; parent EVD-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: apply the highest applicable authority and shall identify unresolved conflicts rather than combining rules across authority tiers without....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation EVID/SK; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-EVD; EVID/SK; V&V T/A; FMEA S/O/D/RPN TBD	High
EVD-003-03	Category alias FSR-EVD-003-03; parent EVD-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: display and record the authority tier, applicability determination, selected rule, excluded rules, and conflict-resolution rationale.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation UI/AUD; preserve the parent obligation; execute I/D verification and negative or failure-path tests.	T-EVD; UI/AUD; V&V I/D; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-004-01	Category alias FSR-EVD-004-01; parent EVD-004; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: bind the current signed prescription and physician-approved case intent as the highest patient-specific authority for the active case.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation CXT/EVID; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-EVD; CXT/EVID; V&V I/T; FMEA S/O/D/RPN TBD	Gate
EVD-004-02	Category alias FSR-EVD-004-02; parent EVD-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: issue a hard stop when a lower-tier rule conflicts with signed patient-specific authority and shall prohibit silent substitution....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-EVD; SK; V&V T; FMEA S/O/D/RPN TBD	Gate
EVD-004-03	Category alias FSR-EVD-004-03; parent EVD-004; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the conflicting fields, source versions, affected objects, and required physician resolution.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation UI/AUD; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-EVD; UI/AUD; V&V D/I; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-005-01	Category alias FSR-EVD-005-01; parent EVD-005; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: have a controlled lifecycle state, effective interval, supersession link, applicability predicate, and conflict status.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation EVID/QMS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-EVD; EVID/QMS; V&V I/T; FMEA S/O/D/RPN TBD	High
EVD-005-02	Category alias FSR-EVD-005-02; parent EVD-005; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: exclude expired, retired, suspended, superseded, and inapplicable rules and shall block or escalate unresolved conflicts.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation EVID/SK; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-EVD; EVID/SK; V&V T/A; FMEA S/O/D/RPN TBD	High
EVD-005-03	Category alias FSR-EVD-005-03; parent EVD-005; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain all candidate rules, exclusion reasons, lifecycle states, conflict results, and the human disposition when escalation occurs.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-EVD; AUD; V&V I/T; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-006-01	Category alias FSR- EVD-006-01; parent EVD- 006; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: contain a controlled structure definition, units, comparator, dose quantity and basis, population, exclusions, rule type, tolerance, and conflict....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation EVID/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-EVD; EVID/ DATA; V&V I/ T; FMEA S/O/ D/RPN TBD	Gate
EVD-006-02	Category alias FSR- EVD-006-02; parent EVD- 006; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: verify dimensional consistency, structure mapping, comparator behavior, boundary behavior, exclusions, and conflict-policy execution.	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement allocation EVID; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-EVD; EVID; V&V T/A; FMEA S/O/D/ RPN TBD	Moderate
EVD-006-03	Category alias FSR- EVD-006-03; parent EVD- 006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: include versioned positive, negative, boundary, and conflict test cases with approved expected results.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation VVP/QMS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-EVD; VVP/ QMS; V&V I/T; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-007-01	Category alias FSR-EVD-007-01; parent EVD-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement proposal, independent clinical review, approval, publication, periodic review, re-tirement, emergency suspension, and rollback states with authorized roles.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/EVID; preserve the parent obligation; execute I verification and negative or failure-path tests.	T-EVD; QMS/EVID; V&V I; FMEA S/O/D/RPN TBD	High
EVD-007-02	Category alias FSR-EVD-007-02; parent EVD-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be available in clinical mode, and rollback shall restore a previously approved version without altering historical case records.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation EVID/CFG; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-EVD; EVID/CFG; V&V T/I; FMEA S/O/D/RPN TBD	High
EVD-007-03	Category alias FSR-EVD-007-03; parent EVD-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain the actor, role, rationale, source change, impact assessment, approval, effective time, and affected release configurations.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-EVD; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
EVD-008-01	Category alias FSR-EVD-008-01; parent EVD-008; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: return an explicit no-source or no-applicable-rule status when it cannot produce a verified rule for the active context.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation EVID; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-EVD; EVID; V&V T; FMEA S/O/D/RPN TBD	High
EVD-008-02	Category alias FSR-EVD-008-02; parent EVD-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: block dependent automated reasoning, identify the missing evidence need, and route the case to an authorized human for....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-EVD; SK/UI; V&V T/HFE; FMEA S/O/D/RPN TBD	High
EVD-008-03	Category alias FSR-EVD-008-03; parent EVD-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record the query, evaluated sources, exclusion reasons, no-rule status, human resolution, and any approved case-specific decision.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-EVD; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

6.5 Clinical context, imaging, registration, and contours (CLN) - 27 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-001-01	Category alias FSR-CLN-001-01; parent CLN-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: derive candidate workflow matches only from verified case facts and approved workflow definitions.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation CXT; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; CXT; V&V I/T; FMEA S/O/D/RPN TBD	High
CLN-001-02	Category alias FSR-CLN-001-02; parent CLN-001; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: display candidate rationale and missing material information, shall require clinician confirmation, and shall not assign an autonomous diagnosis....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation UI/SK; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-IMG; UI/SK; V&V T/HFE; FMEA S/O/D/RPN TBD	Moderate
CLN-001-03	Category alias FSR-CLN-001-03; parent CLN-001; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record the verified inputs, workflow candidates, missing-data findings, clinician selection or rejection, and corrections.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; AUD; V&V I/T; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-002-01	Category alias FSR-CLN-002-01; parent CLN-002; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the approved image studies, frames of reference, registrations, structure sets, prior-treatment information, and signed clinical intent by....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation CXT/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; CXT/DATA; V&V I/T; FMEA S/O/D/RPN TBD	High
CLN-002-02	Category alias FSR-CLN-002-02; parent CLN-002; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reject a planning task with a missing, conflicting, stale, or incorrectly referenced required input.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation CXT/SK; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-IMG; CXT/SK; V&V T/A; FMEA S/O/D/RPN TBD	High
CLN-002-03	Category alias FSR-CLN-002-03; parent CLN-002; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: preserve the validated planning-context manifest and bind it to every derived anatomy, intent, and candidate-plan version.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-003-01	Category alias FSR- CLN-003-01; parent CLN- 003; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: allow only commissioned rigid-registration, deformable-registration, and dose-accumulation methods for the active anatomy, image conditions, and clinical purpose.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation REG/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; REG/CFG; V&V I/T; FMEA S/O/D/RPN TBD	Gate
CLN-003-02	Category alias FSR- CLN-003-02; parent CLN- 003; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide registration-specific visualization, quality measures, uncertainty documentation, and an authorized approval step before derived dose is used.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation REG/UI; preserve the parent obligation; execute D/HFE verification and negative or failure-path tests.	T-IMG; REG/UI; V&V D/HFE; FMEA S/O/D/RPN TBD	Gate
CLN-003-03	Category alias FSR- CLN-003-03; parent CLN- 003; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain the source and target images, transform, method and version, settings, quality results, uncertainty, reviewer, approval, and accumulated-dose....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation REG/AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; REG/AUD; V&V I/T; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-004-01	Category alias FSR-CLN-004-01; parent CLN-004; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: verify the approved anatomy, imaging conditions, structure list, patient population, use case, model version, and operating mode before....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation MGW/SEG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; MGW/SEG; V&V I/T; FMEA S/O/D/RPN TBD	High
CLN-004-02	Category alias FSR-CLN-004-02; parent CLN-004; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be created in a draft state and shall retain its model, input, configuration, and applicability decision.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation SEG/DATA; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-IMG; SEG/DATA; V&V T/I; FMEA S/O/D/RPN TBD	High
CLN-004-03	Category alias FSR-CLN-004-03; parent CLN-004; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prohibit a model or automated service from approving a generated contour or promoting it directly to an approved....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-IMG; SK; V&V T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-005-01	Category alias FSR-CLN-005-01; parent CLN-005; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: calculate the commissioned uncertainty, input-quality, anomaly, and out-of-distribution indicators defined for the active model use.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation SEG/MGW; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; SEG/MGW; V&V I/T; FMEA S/O/D/RPN TBD	High
CLN-005-02	Category alias FSR-CLN-005-02; parent CLN-005; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: abstain or route to manual contouring when an approved applicability threshold is not met and shall display the....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation SK/UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-IMG; SK/UI; V&V T/HFE; FMEA S/O/D/RPN TBD	High
CLN-005-03	Category alias FSR-CLN-005-03; parent CLN-005; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record all applicability indicators, thresholds, decisions, user responses, and performance against commissioned abstinence and failure-detection cases.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation AUD/VVP; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; AUD/VVP; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-006-01	Category alias FSR-CLN-006-01; parent CLN-006; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: evaluate structure identity, approved nomenclature mapping, laterality, required presence, completeness, topology, image coverage, and configured geometric plausibility rules.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation SEG/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; SEG/DATA; V&V I/T; FMEA S/O/D/RPN TBD	High
CLN-006-02	Category alias FSR-CLN-006-02; parent CLN-006; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: block promotion to clinical-candidate status while any required contour validation is failed, unresolved, or stale.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation SK; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-IMG; SK; V&V T/A; FMEA S/O/D/RPN TBD	High
CLN-006-03	Category alias FSR-CLN-006-03; parent CLN-006; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide and retain a validation report identifying each check, result, affected structure, evidence, disposition, and reviewer.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation UI/AUD; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-IMG; UI/AUD; V&V D/I; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-007-01	Category alias FSR-CLN-007-01; parent CLN-007; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reference its input study, frame of reference, model and version, configuration, naming map, confidence information, and parent contour....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation DATA/SEG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; DATA/SEG; V&V I/T; FMEA S/O/D/RPN TBD	Gate
CLN-007-02	Category alias FSR-CLN-007-02; parent CLN-007; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create a new contour version and shall retain the changed region, editor, time, reason when required, and pre-edit....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation DATA/UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-IMG; DATA/UI; V&V T/HFE; FMEA S/O/D/RPN TBD	High
CLN-007-03	Category alias FSR-CLN-007-03; parent CLN-007; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the exact contour version, lineage, reviewer, approval state, and integrity value transferred downstream.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation AUD/INT; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; AUD/INT; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-008-01	Category alias FSR-CLN-008-01; parent CLN-008; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide synchronized slice, surface, difference, and configured clinically relevant error views for the selected contour versions.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation UI/SEG; preserve the parent obligation; execute D/HFE verification and negative or failure-path tests.	T-IMG; UI/SEG; V&V D/HFE; FMEA S/O/D/RPN TBD	High
CLN-008-02	Category alias FSR-CLN-008-02; parent CLN-008; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: capture the location, spatial extent, operation type, and before-and-after geometry for each user edit.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation UI/DATA; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-IMG; UI/DATA; V&V T/I; FMEA S/O/D/RPN TBD	High
CLN-008-03	Category alias FSR-CLN-008-03; parent CLN-008; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: verify that representative users can find, interpret, edit, and confirm clinically important contour errors using the provided views.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation HFE/VVP; preserve the parent obligation; execute HFE/A verification and negative or failure-path tests.	T-IMG; HFE/VVP; V&V HFE/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
CLN-009-01	Category alias FSR-CLN-009-01; parent CLN-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the authorized approval role for targets, organs at risk, derived structures, and each supported clinical workflow.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation IAM/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; IAM/CFG; V&V I/T; FMEA S/O/D/RPN TBD	High
CLN-009-02	Category alias FSR-CLN-009-02; parent CLN-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require radiation-oncologist approval for clinical targets and shall enforce the configured authorized role before any structure becomes approved.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-IMG; SK; V&V T; FMEA S/O/D/RPN TBD	High
CLN-009-03	Category alias FSR-CLN-009-03; parent CLN-009; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: bind the approver, role, contour version, review time, approval decision, and any conditions, and later edits shall invalidate....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation DATA/AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-IMG; DATA/AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

6.6 Treatment-plan generation and calculation (PLN) - 36 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-001-01	Category alias FSR-PLN-001-01; parent PLN-001; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reference the signed prescription, approved anatomy, patient-specific instructions, applicable protocol, institutional planning standard, and confirmed user priorities by....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation PLAN/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; PLAN/DATA; V&V I/T; FMEA S/O/D/RPN TBD	Gate
PLN-001-02	Category alias FSR-PLN-001-02; parent PLN-001; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify missing inputs and conflicts among authoritative sources and shall block candidate generation until required conflicts are resolved.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation PLAN/SK; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-PLAN; PLAN/SK; V&V T/A; FMEA S/O/D/RPN TBD	Gate
PLN-001-03	Category alias FSR-PLN-001-03; parent PLN-001; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain its source versions, normalized content, assumptions, clarifications, author or confirmer, integrity value, and supersedion relationship.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; AUD; V&V I/T; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-002-01	Category alias FSR-PLN-002-01; parent PLN-002; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: list each commissioned planning, optimization, robustness, and dose-calculation service with its validated machine, technique, algorithm, and use envelope.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation CFG/PLAN; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; CFG/PLAN; V&V I/T; FMEA S/O/D/RPN TBD	Gate
PLN-002-02	Category alias FSR-PLN-002-02; parent PLN-002; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: verify the active request and configuration against the commissioned-use envelope before invoking a clinical planning or dose service.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation SK/PLAN; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-PLAN; SK/PLAN; V&V T; FMEA S/O/D/RPN TBD	Gate
PLN-002-03	Category alias FSR-PLN-002-03; parent PLN-002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record the service identity, version, approved configuration, envelope decision, input identity, and output identity for every candidate calculation.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-003-01	Category alias FSR-PLN-003-01; parent PLN-003; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: separate language, speech, disease-site, and generative-model services from commissioned numerical dose-calculation components and credentials.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation ARCH/SK; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-PLAN; ARCH/SK; V&V I/A; FMEA S/O/D/RPN TBD	Gate
PLN-003-02	Category alias FSR-PLN-003-02; parent PLN-003; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: accept clinical dose only from an approved commissioned dose service and shall reject model-supplied dose arrays, dose metrics....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation PLAN/DATA; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-PLAN; PLAN/DATA; V&V T; FMEA S/O/D/RPN TBD	Gate
PLN-003-03	Category alias FSR-PLN-003-03; parent PLN-003; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: demonstrate rejection of spoofed, mislabeled, malformed, or model-generated dose results at every supported interface boundary.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation VVP/SEC; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-PLAN; VVP/SEC; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-004-01	Category alias FSR-PLN-004-01; parent PLN-004; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: have an explicit type of hard constraint, planning goal, optimization objective, preference, or reporting-only metric.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation PLAN/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; PLAN/DATA; V&V I/T; FMEA S/O/D/RPN TBD	Gate
PLN-004-02	Category alias FSR-PLN-004-02; parent PLN-004; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: display criterion type, priority, units, source, and status without presenting a preference or reporting metric as a hard....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation UI; preserve the parent obligation; execute D/HFE verification and negative or failure-path tests.	T-PLAN; UI; V&V D/HFE; FMEA S/O/D/RPN TBD	Moderate
PLN-004-03	Category alias FSR-PLN-004-03; parent PLN-004; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: apply behavior consistent with the declared criterion type and shall prevent implicit type conversion.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation PLAN/SK; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-PLAN; PLAN/SK; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-005-01	Category alias FSR-PLN-005-01; parent PLN-005; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prohibit modification or relaxation of a hard constraint unless the active policy permits it and the user holds....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement allocation SK/IAM; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-PLAN; SK/IAM; V&V T/I; FMEA S/O/D/RPN TBD	Moderate
PLN-005-02	Category alias FSR-PLN-005-02; parent PLN-005; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require an explicit action, structured rationale, and a pre-change and post-change difference review before recalculation or promotion.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation UI/PLAN; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-PLAN; UI/PLAN; V&V T/HFE; FMEA S/O/D/RPN TBD	Gate
PLN-005-03	Category alias FSR-PLN-005-03; parent PLN-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create a new planning-intent version and shall retain the actor, authority, rationale, old and new values, affected candidates....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation DATA/AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; DATA/AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-006-01	Category alias FSR-PLN-006-01; parent PLN-006; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: contain the authorized requested candidate count and shall assign a unique identity to each attempted candidate.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation PLAN; preserve the parent obligation; execute T/D verification and negative or failure-path tests.	T-PLAN; PLAN; V&V T/D; FMEA S/O/D/RPN TBD	Gate
PLN-006-02	Category alias FSR-PLN-006-02; parent PLN-006; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: report generated, failed, infeasible, canceled, and pending candidates and shall state why the requested set was not completed.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation PLAN/UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-PLAN; PLAN/UI; V&V T/HFE; FMEA S/O/D/RPN TBD	Gate
PLN-006-03	Category alias FSR-PLN-006-03; parent PLN-006; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain the requested count, generation attempts, parameters, job states, results, and user decision to continue, retry, or accept....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; AUD; V&V I/T; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-007-01	Category alias FSR-PLN-007-01; parent PLN-007; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the planning strategy and the parameter, objective, beam, modality, robustness, or other controlled variation used to distinguish....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation PLAN/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; PLAN/DATA; V&V I/T; FMEA S/O/D/RPN TBD	Gate
PLN-007-02	Category alias FSR-PLN-007-02; parent PLN-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: calculate and present configured diversity descriptors against the parent candidate set without claiming clinically meaningful diversity from an....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation PLAN/REVIEW; preserve the parent obligation; execute A/D verification and negative or failure-path tests.	T-PLAN; PLAN/REVIEW; V&V A/D; FMEA S/O/D/RPN TBD	High
PLN-007-03	Category alias FSR-PLN-007-03; parent PLN-007; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain the diversity method, configuration, descriptors, related candidates, and any user-directed variation.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation AUD; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-PLAN; AUD; V&V I/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-008-01	Category alias FSR-PLN-008-01; parent PLN-008; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use versioned feasibility rules and declared dominance criteria appropriate to the active modality, technique, and evaluation set.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation PLAN/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; PLAN/CFG; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
PLN-008-02	Category alias FSR-PLN-008-02; parent PLN-008; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: label infeasible candidates and candidates dominated under the declared criteria and shall expose the specific failed conditions or....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation PLAN/REVIEW; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-PLAN; PLAN/REVIEW; V&V T/A; FMEA S/O/D/RPN TBD	Gate
PLN-008-03	Category alias FSR-PLN-008-03; parent PLN-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain rule versions, input metrics, comparator results, excluded candidates, and any authorized human disposition.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-009-01	Category alias FSR- PLN-009-01; parent PLN- 009; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: expose a signed runtime manifest containing machine model, energy or beam type, technique, dose algorithm, grid, optimizer, biological....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation DOSE/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; DOSE/ CFG; V&V I/T; FMEA S/ O/D/RPN TBD	Gate
PLN-009-02	Category alias FSR- PLN-009-02; parent PLN- 009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: compare the runtime manifest with the approved release configuration and shall block calculation on a missing or mismatched....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/PLAN; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-PLAN; SK/ PLAN; V&V T; FMEA S/O/D/ RPN TBD	High
PLN-009-03	Category alias FSR- PLN-009-03; parent PLN- 009; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be bound to each calculation output and candidate-plan version.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation DATA/AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; DATA/AUD; V&V I/T; FMEA S/ O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-010-01	Category alias FSR- PLN-010-01; parent PLN- 010; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify numeric units, physical or RBE-weighted basis, fractionation, normalization, biological transformation when approved, and applicable proton-specific assumptions.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation DATA/PLAN; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; DATA/PLAN; V&V I/T; FMEA S/ O/D/RPN TBD	Gate
PLN-010-02	Category alias FSR- PLN-010-02; parent PLN- 010; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent comparison, summation, or conversion of incompatible dose quantities unless an approved explicit transformation and its assumptions are....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation SK/REVIEW; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-PLAN; SK/ REVIEW; V&V T/A; FMEA S/ O/D/RPN TBD	Gate
PLN-010-03	Category alias FSR- PLN-010-03; parent PLN- 010; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: present the dose basis and assumptions adjacent to each value and shall retain every transformation and source quantity.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation UI/AUD; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-PLAN; UI/ AUD; V&V D/I; FMEA S/O/D/ RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-011-01	Category alias FSR-PLN-011-01; parent PLN-011; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the required setup, range, motion, interplay, and other robustness scenarios, including parameter values and evaluation rules.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation CFG/PLAN; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; CFG/PLAN; V&V I/T; FMEA S/O/D/RPN TBD	Gate
PLN-011-02	Category alias FSR-PLN-011-02; parent PLN-011; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute every required scenario and shall block candidate eligibility when a scenario is missing, failed, stale, or outside....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation PLAN/SK; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-PLAN; PLAN/SK; V&V T/A; FMEA S/O/D/RPN TBD	High
PLN-011-03	Category alias FSR-PLN-011-03; parent PLN-011; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain and present each scenario definition, result, aggregate, warning, and eligibility decision.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation DATA/REVIEW; preserve the parent obligation; execute I/D verification and negative or failure-path tests.	T-PLAN; DATA/REVIEW; V&V I/D; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
PLN-012-01	Category alias FSR-PLN-012-01; parent PLN-012; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: have a unique immutable identity and shall reference its parent intent, predecessor candidate, parameters, configuration, and calculation outputs.	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement allocation DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-PLAN; DATA; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
PLN-012-02	Category alias FSR-PLN-012-02; parent PLN-012; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create a new candidate version without overwriting prior parameters, results, warnings, evaluation states, or lineage.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation PLAN; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-PLAN; PLAN; V&V T; FMEA S/O/D/RPN TBD	Moderate
PLN-012-03	Category alias FSR-PLN-012-03; parent PLN-012; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be able to compare any retained candidate versions and reconstruct the ordered refinement history and decision trail.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation REVW/AUD; preserve the parent obligation; execute I/D verification and negative or failure-path tests.	T-PLAN; REVW/AUD; V&V I/D; FMEA S/O/D/RPN TBD	Moderate

6.7 Plan review, selection, QA, and transfer (REV) - 27 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-001-01	Category alias FSR-REV-001-01; parent REV-001; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use one versioned metric definition, unit convention, calculation method, and default display scale for all candidates in a....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation REVW/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-REV; REVW/CFG; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
REV-001-02	Category alias FSR-REV-001-02; parent REV-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: present absolute values and candidate-to-reference differences and shall visibly identify any authorized scale or reference change.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation REVW/UI; preserve the parent obligation; execute D/HFE verification and negative or failure-path tests.	T-REV; REVW/UI; V&V D/HFE; FMEA S/O/D/RPN TBD	High
REV-001-03	Category alias FSR-REV-001-03; parent REV-001; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain the candidate set, metric-definition version, scales, reference candidate, filters, and user changes.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-REV; AUD; V&V I/T; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-002-01	Category alias FSR- REV-002-01; parent REV- 002; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide target, organ-at-risk, constraint, spatial dose, hotspot, coldspot, robustness, uncertainty, complexity, deliverability, and provenance views for every candidate.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation REVW; preserve the parent obligation; execute I/D verification and negative or failure-path tests.	T-REV; REVVW; V&V I/ D; FMEA S/O/ D/RPN TBD	Gate
REV-002-02	Category alias FSR- REV-002-02; parent REV- 002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify missing or stale required review content and shall block promotion until the configured review package is complete.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/REVVW; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-REV; SK/ REVVW; V&V T; FMEA S/O/ D/RPN TBD	High
REV-002-03	Category alias FSR- REV-002-03; parent REV- 002; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: link to the exact candidate, calculation, definition, scenario, and provenance record from which it was derived.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation DATA/AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-REV; DATA/ AUD; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-003-01	Category alias FSR-REV-003-01; parent REV-003; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: display each declared tradeoff criterion, value, priority or weight, and candidate-specific contribution used in an ordering or recommendation.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation REVW; preserve the parent obligation; execute I/HFE verification and negative or failure-path tests.	T-REV; REVW; V&V I/HFE; FMEA S/O/D/RPN TBD	Moderate
REV-003-02	Category alias FSR-REV-003-02; parent REV-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: not automatically select a final candidate and shall not present an unexplained composite score as sufficient clinical justification.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/RE VW; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-REV; SK/REVW; V&V T/A; FMEA S/O/D/RPN TBD	High
REV-003-03	Category alias FSR-REV-003-03; parent REV-003; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: include an inspectable rationale, uncertainty and limitations, and the audit record shall identify the authorized user who made....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation UI/AUD; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-REV; UI/AUD; V&V D/I; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-004-01	Category alias FSR-REV-004-01; parent REV-004; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the selected candidate, rejected alternatives, reviewer roles, comments, deviations, tradeoff rationale, and decision state.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation DATA/REVW; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-REV; DATA/REVW; V&V I/T; FMEA S/O/D/RPN TBD	High
REV-004-02	Category alias FSR-REV-004-02; parent REV-004; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require completion of configured decision fields and role-specific attestations before the review can be finalized.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation REVW/UI; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-REV; REVW/UI; V&V T/HFE; FMEA S/O/D/RPN TBD	High
REV-004-03	Category alias FSR-REV-004-03; parent REV-004; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be integrity protected, version linked, time stamped, and retained with all subsequent amendments or superseding decisions.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-REV; AUD; V&V I/T; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-005-01	Category alias FSR-REV-005-01; parent REV-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: assign clinical plan approval to an authorized radiation oncologist and technical review to an authorized qualified medical physicist.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation IAM/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-REV; IAM/CFG; V&V I/T; FMEA S/O/D/RPN TBD	High
REV-005-02	Category alias FSR-REV-005-02; parent REV-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require both current role-specific decisions and shall enforce configured ordering and separation-of-duties constraints before approved status.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-REV; SK; V&V T; FMEA S/O/D/RPN TBD	High
REV-005-03	Category alias FSR-REV-005-03; parent REV-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: bind the actor, role, candidate version, reviewed evidence, decision, conditions, time, and authentication event.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-REV; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-006-01	Category alias FSR-REV-006-01; parent REV-006; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define required independent dose or MU verification, patient-specific QA, transfer QA, and technique-specific checks for each supported workflow.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation CFG/QMS; preserve the parent obligation; execute I verification and negative or failure-path tests.	T-REV; CFG/QMS; V&V I; FMEA S/O/D/RPN TBD	Gate
REV-006-02	Category alias FSR-REV-006-02; parent REV-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create and track every required QA task and shall block release while a required task is incomplete, failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation REVW/SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-REV; REVW/SK; V&V T; FMEA S/O/D/RPN TBD	High
REV-006-03	Category alias FSR-REV-006-03; parent REV-006; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain each QA input, method, result, tolerance, reviewer, disposition, linked candidate version, and supersession state.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation DATA/AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-REV; DATA/AUD; V&V I/T; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-007-01	Category alias FSR-REV-007-01; parent REV-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: verify that all required approvals and QA states are current for the exact source-object versions before initiating transfer.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/INT; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-REV; SK/INT; V&V T; FMEA S/O/D/RPN TBD	High
REV-007-02	Category alias FSR-REV-007-02; parent REV-007; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: compare patient and object identity, content, units, geometry, references, integrity values, and destination acknowledgment with the source package.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation INT; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-REV; INT; V&V T/A; FMEA S/O/D/RPN TBD	High
REV-007-03	Category alias FSR-REV-007-03; parent REV-007; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain the source package, destination, transmitted objects, validation results, acknowledgments, failures, retries, and reconciliation outcome.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-REV; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-008-01	Category alias FSR-REV-008-01; parent REV-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be immutable; any authorized change shall create a new version linked to the approved predecessor.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-REV; DATA; V&V I/T; FMEA S/O/D/RPN TBD	High
REV-008-02	Category alias FSR-REV-008-02; parent REV-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify and invalidate every downstream approval, QA result, transfer status, and derived object affected by the new version.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation DATA/SK; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-REV; DATA/SK; V&V T/A; FMEA S/O/D/RPN TBD	High
REV-008-03	Category alias FSR-REV-008-03; parent REV-008; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: present the required re-review set and shall retain each completed re-review and resulting restoration of eligibility.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation REVW/AUD; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-REV; REVW/AUD; V&V D/I; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
REV-009-01	Category alias FSR-REV-009-01; parent REV-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: have a controlled independence record identifying its owner, purpose, inputs, algorithms, dependencies, data sources, and approval boundaries.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/VVP; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-REV; QMS/VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
REV-009-02	Category alias FSR-REV-009-02; parent REV-009; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: evaluate shared inputs, common software, models, configurations, infrastructure, personnel, and failure modes and shall define compensating controls.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation VVP; preserve the parent obligation; execute A/T verification and negative or failure-path tests.	T-REV; VVP; V&V A/T; FMEA S/O/D/RPN TBD	High
REV-009-03	Category alias FSR-REV-009-03; parent REV-009; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: expose material independence limitations, and the system shall retain the assessment version used for each verification result.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation REVW/AUD; preserve the parent obligation; execute I/D verification and negative or failure-path tests.	T-REV; REVW/AUD; V&V I/D; FMEA S/O/D/RPN TBD	High

6.8 Data, interoperability, provenance, and audit (DAT) - 30 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-001-01	Category alias FSR-DAT-001-01; parent DAT-001; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require a globally unique identity, version, lifecycle state, owner, creation source, and source linkage for every controlled object....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; DATA; V&V I/T; FMEA S/O/D/RPN TBD	High
DAT-001-02	Category alias FSR-DAT-001-02; parent DAT-001; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reject identifier collision, missing ownership, invalid lifecycle transition, or creation of a derived object without a valid source....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation DATA; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-DATA; DATA; V&V T; FMEA S/O/D/RPN TBD	Moderate
DAT-001-03	Category alias FSR-DAT-001-03; parent DAT-001; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: expose the identity, version, state, owner, and source linkage for each clinical object.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation UI/AUD; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-DATA; UI/AUD; V&V D/I; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-002-01	Category alias FSR-DAT-002-01; parent DAT-002; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: represent typed edges from every derived object to its input objects, transformations, software, models, evidence, configuration, user actions....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; DATA; V&V I/T; FMEA S/O/D/RPN TBD	High
DAT-002-02	Category alias FSR-DAT-002-02; parent DAT-002; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent promotion of an object with a missing mandatory provenance edge, orphaned dependency, or unresolved version reference.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation DATA/SK; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-DATA; DATA/SK; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
DAT-002-03	Category alias FSR-DAT-002-03; parent DAT-002; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be able to query transitive upstream and downstream provenance and reproduce the dependency set for a selected object....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation UI/AUD; preserve the parent obligation; execute D/T verification and negative or failure-path tests.	T-DATA; UI/AUD; V&V D/T; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-003-01	Category alias FSR-DAT-003-01; parent DAT-003; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define canonical units, coordinate systems, dose quantities and bases, structure definitions, and rounding rules for each stored and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation DATA/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; DATA/CFG; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
DAT-003-02	Category alias FSR-DAT-003-02; parent DAT-003; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be explicit, versioned, dimensionally validated, and rejected when the source or destination representation is ambiguous or unsupported.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation DATA/INT; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-DATA; DATA/INT; V&V T/A; FMEA S/O/D/ RPN TBD	Moderate
DAT-003-03	Category alias FSR-DAT-003-03; parent DAT-003; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: cover storage, calculation, display, export, round-trip, precision, rounding, coordinate-transform, and incompatible-unit cases.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation VVP; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-DATA; VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-004-01	Category alias FSR-DAT-004-01; parent DAT-004; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate patient and study identity, SOP Instance UIDs, object references, frame-of-reference UIDs, required attributes, transformations, and units for....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation INT; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; INT; V&V I/T; FMEA S/O/D/RPN TBD	High
DAT-004-02	Category alias FSR-DAT-004-02; parent DAT-004; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: reject or quarantine a transfer containing a missing, duplicate, inconsistent, orphaned, or destination-altered required element.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation INT/SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-DATA; INT/SK; V&V T; FMEA S/O/D/RPN TBD	Moderate
DAT-004-03	Category alias FSR-DAT-004-03; parent DAT-004; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: compare source and destination identities, references, content, geometry, units, integrity values, and acknowledgment status.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation INT/AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; INT/AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-005-01	Category alias FSR-DAT-005-01; parent DAT-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: list the supported DICOM SOP classes, roles, transfer syntaxes, transactions, required attributes, and known limitations for each approved....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation INT/CFG; preserve the parent obligation; execute I verification and negative or failure-path tests.	T-DATA; INT/CFG; V&V I; FMEA S/O/D/RPN TBD	High
DAT-005-02	Category alias FSR-DAT-005-02; parent DAT-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: import, validate, create, and export only the DICOM objects and transactions approved for the active deployment and operating....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation INT; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-DATA; INT; V&V T; FMEA S/O/D/RPN TBD	High
DAT-005-03	Category alias FSR-DAT-005-03; parent DAT-005; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use representative and challenge objects for images, registrations, segmentations or structures, plans, ion plans, dose, treatment records, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation VVP/QMS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; VVP/QMS; V&V I/T; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-006-01	Category alias FSR-DAT-006-01; parent DAT-006; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the profile or guide version, actor or resource role, supported transactions, terminology, and optional features.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation INT/CFG; preserve the parent obligation; execute I verification and negative or failure-path tests.	T-DATA; INT/CFG; V&V I; FMEA S/O/D/RPN TBD	High
DAT-006-02	Category alias FSR-DAT-006-02; parent DAT-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate conformance to the approved profile or guide and shall prevent use of an unsupported exchange as though....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation INT; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-DATA; INT; V&V T/A; FMEA S/O/D/RPN TBD	High
DAT-006-03	Category alias FSR-DAT-006-03; parent DAT-006; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain and make available a conformance statement and known-limitations record for every implemented exchange.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation QMS/UI; preserve the parent obligation; execute I/D verification and negative or failure-path tests.	T-DATA; QMS/UI; V&V I/D; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-007-01	Category alias FSR-DAT-007-01; parent DAT-007; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: classify missing, ambiguous, inconsistent, orphaned, duplicate, and partial-transfer conditions with explicit invalid or quarantine states.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation DATA/INT; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; DATA/INT; V&V I/T; FMEA S/O/D/RPN TBD	High
DAT-007-02	Category alias FSR-DAT-007-02; parent DAT-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent invalid or quarantined data from becoming active, approved, or eligible as input to a clinical calculation or...	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-DATA; SK; V&V T; FMEA S/O/D/RPN TBD	High
DAT-007-03	Category alias FSR-DAT-007-03; parent DAT-007; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify each detected condition, affected object, source, disposition, correction, and reconciliation result.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation UI/AUD; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-DATA; UI/AUD; V&V D/I; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-008-01	Category alias FSR-DAT-008-01; parent DAT-008; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use append-only records, trusted time, ordered event identity, actor identity, object versions, action type, outcome, and integrity protection.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; AUD; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
DAT-008-02	Category alias FSR-DAT-008-02; parent DAT-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: capture commands, context, inputs, versions, evidence, outputs, warnings, edits, approvals, QA, exports, failures, overrides, and reconciliation events.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-DATA; AUD; V&V T; FMEA S/O/D/RPN TBD	High
DAT-008-03	Category alias FSR-DAT-008-03; parent DAT-008; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: preserve event order and integrity while enforcing access control, retention, legal hold, and privacy policy.	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement allocation AUD/SEC; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; AUD/SEC; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-009-01	Category alias FSR-DAT-009-01; parent DAT-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create and retain an integrity hash, digital signature, or approved equivalent for each protected clinical, evidence, software, model....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation DATA/SEC; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; DATA/SEC; V&V I/T; FMEA S/O/D/RPN TBD	High
DAT-009-02	Category alias FSR-DAT-009-02; parent DAT-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be verified at approved creation, load, use, transfer, restoration, and reconciliation boundaries, with failure preventing clinical promotion or....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation DATA/INT; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-DATA; DATA/INT; V&V T; FMEA S/O/D/RPN TBD	High
DAT-009-03	Category alias FSR-DAT-009-03; parent DAT-009; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create a security and clinical-safety event containing the artifact, expected and observed values, context, containment action, and disposition.	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement allocation AUD/OPS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; AUD/OPS; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
DAT-010-01	Category alias FSR-DAT-010-01; parent DAT-010; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify the authoritative system of record, retention period, legal-record status, export policy, archive, and reconciliation method for each....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation QMS/DATA; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-DATA; QMS/DATA; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
DAT-010-02	Category alias FSR-DAT-010-02; parent DAT-010; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: enforce the active information-governance matrix and applicable legal holds.	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement allocation DATA/INT; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-DATA; DATA/INT; V&V T; FMEA S/O/D/RPN TBD	Moderate
DAT-010-03	Category alias FSR-DAT-010-03; parent DAT-010; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: compare authoritative and dependent records and shall retain discrepancies, corrective actions, and closure evidence.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation OPS/AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-DATA; OPS/AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

6.9 AI and inference-model lifecycle (AIM) - 24 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
AIM-001-01	Category alias FSR-AIM-001-01; parent AIM-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify each approved model version, task, intended use, operating modes, clinical scope, artifact signature, and effective release state.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation MGW/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-AI; MGW/CFG; V&V I/T; FMEA S/O/D/RPN TBD	High
AIM-001-02	Category alias FSR-AIM-001-02; parent AIM-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: invoke only the exact locked artifact approved for the active task, scope, and operating mode and shall reject....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation MGW; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-AI; MGW; V&V T; FMEA S/O/D/RPN TBD	High
AIM-001-03	Category alias FSR-AIM-001-03; parent AIM-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain the requested and resolved model identities, registry version, signature result, approval status, and invocation outcome.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-AI; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
AIM-002-01	Category alias FSR-AIM-002-01; parent AIM-002; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: contain a machine-readable input contract for intended use, population, data or imaging conditions, required fields, preprocessing, and resource....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation MGW/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-AI; MGW/DATA; V&V I/T; FMEA S/O/D/RPN TBD	High
AIM-002-02	Category alias FSR-AIM-002-02; parent AIM-002; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate the active case and prepared inputs against every applicable input-contract condition before inference and shall reject an....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation MGW; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-AI; MGW; V&V T/A; FMEA S/O/D/RPN TBD	High
AIM-002-03	Category alias FSR-AIM-002-03; parent AIM-002; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: expose and retain the input-contract results, preprocessing version, resource checks, rejected conditions, and routing decision.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation UI/AUD; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-AI; UI/AUD; V&V D/I; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
AIM-003-01	Category alias FSR-AIM-003-01; parent AIM-003; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define versioned abstention, rejection, and manual-review triggers for unsupported, anomalous, incomplete, and out-of-distribution inputs.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation MGW/MLOPS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-AI; MGW/MLOPS; V&V I/T; FMEA S/O/D/RPN TBD	High
AIM-003-02	Category alias FSR-AIM-003-02; parent AIM-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: produce no promotable clinical output and shall route the case to the configured manual or alternate approved workflow.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation MGW/SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-AI; MGW/SK; V&V T; FMEA S/O/D/RPN TBD	High
AIM-003-03	Category alias FSR-AIM-003-03; parent AIM-003; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: measure and retain trigger performance, false acceptance, false rejection, routing outcome, and user disposition.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation VVP/AUD; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-AI; VVP/AUD; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
AIM-004-01	Category alias FSR-AIM-004-01; parent AIM-004; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be read-only to clinical workflows and service identities.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation MLOPS/SEC; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-AI; MLOPS/SEC; V&V I/T; FMEA S/O/D/RPN TBD	High
AIM-004-02	Category alias FSR-AIM-004-02; parent AIM-004; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: not write to or update a production model or retrieval behavior through the clinical execution path.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation MGW/DATA; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-AI; MGW/DATA; V&V T; FMEA S/O/D/RPN TBD	High
AIM-004-03	Category alias FSR-AIM-004-03; parent AIM-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: occur in a segregated governed lifecycle with approved data use, evaluation, change control, and release authorization.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/MLOPS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-AI; QMS/MLOPS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
AIM-005-01	Category alias FSR-AIM-005-01; parent AIM-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require model version, intended use, owner, supplier, training and evaluation provenance, limitations, approved configuration, signed artifacts, dependencies, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation MLOPS/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-AI; MLOPS/DATA; V&V I/T; FMEA S/O/D/RPN TBD	High
AIM-005-02	Category alias FSR-AIM-005-02; parent AIM-005; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: verify completeness, signatures, dependency identities, vulnerability status, and compatibility before the model becomes approved.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation MLOPS/SEC; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-AI; MLOPS/SEC; V&V T/I; FMEA S/O/D/RPN TBD	High
AIM-005-03	Category alias FSR-AIM-005-03; parent AIM-005; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be queryable by authorized users and linked to the inference and resulting clinical object.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation AUD/UI; preserve the parent obligation; execute I/D verification and negative or failure-path tests.	T-AI; AUD/UI; V&V I/D; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
AIM-006-01	Category alias FSR-AIM-006-01; parent AIM-006; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: assign immutable identities and intended purposes to development, tuning, internal-test, external-test, and site-acceptance datasets.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation MLOPS/VVP; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-AI; MLOPS/VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
AIM-006-02	Category alias FSR-AIM-006-02; parent AIM-006; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: perform patient, study, source, temporal, and derived-data overlap checks and shall report all identified overlap and leakage risks.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation MLOPS; preserve the parent obligation; execute A/T verification and negative or failure-path tests.	T-AI; MLOPS; V&V A/T; FMEA S/O/D/RPN TBD	Moderate
AIM-006-03	Category alias FSR-AIM-006-03; parent AIM-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: receive documented approval before model release.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/VVP; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-AI; QMS/VVP; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
AIM-007-01	Category alias FSR- AIM-007-01; parent AIM- 007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define overall and subgroup performance, calibration, failure detection, uncertainty, edit burden, and known-limitation measures for each approved use.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation VVP/MLOPS; preserve the parent obligation; execute A/T verification and negative or failure-path tests.	T-AI; VVP/ MLOPS; V&V A/T; FMEA S/ O/D/RPN TBD	High
AIM-007-02	Category alias FSR- AIM-007-02; parent AIM- 007; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: display the applicable model performance summary, limitations, uncertainty meaning, and commissioned action thresholds to authorized users.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation UI/MGW; preserve the parent obligation; execute D/HFE verification and negative or failure-path tests.	T-AI; UI/MGW; V&V D/HFE; FMEA S/O/D/ RPN TBD	High
AIM-007-03	Category alias FSR- AIM-007-03; parent AIM- 007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: compare observed model and subgroup signals with approved thresholds and shall document resulting action.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation OPS/QMS; preserve the parent obligation; execute A/I verification and negative or failure-path tests.	T-AI; OPS/ QMS; V&V A/ I; FMEA S/O/ D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
AIM-008-01	Category alias FSR-AIM-008-01; parent AIM-008; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: classify changes to weights, behavior-altering prompts, retrieval configuration, clinical rules, preprocessing, dependencies, and runtime infrastructure and shall identify....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation QMS/MLOPS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-AI; QMS/MLOPS; V&V I/A; FMEA S/O/D/RPN TBD	High
AIM-008-02	Category alias FSR-AIM-008-02; parent AIM-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: receive documented impact assessment and the required regression, performance, safety, cybersecurity, and compatibility testing before approval.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation VVP/MLOPS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-AI; VVP/MLOPS; V&V I/T; FMEA S/O/D/RPN TBD	High
AIM-008-03	Category alias FSR-AIM-008-03; parent AIM-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create a new versioned release with an approved rollback plan and shall preserve the prior known-good configuration until....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation CFG/OPS; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-AI; CFG/OPS; V&V T/I; FMEA S/O/D/RPN TBD	High

6.10 Human factors, usability, and competency (HFE) - 18 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
HFE-001-01	Category alias NFSR-HFE-001-01; parent HFE-001; PAT	Wrong, stale, or incomplete patient, course, study, plan, or action context is accepted. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: display the active patient, course, study, plan, operating mode, approval state, and unsaved or unverified-change status throughout clinical....	Work may be applied to the wrong patient or object, or valid downstream approvals may become unsafe.	Bind authenticated actor and immutable case context, confirm two patient identifiers for consequential actions, reject mismatches, and invalidate stale approvals. Implement allocation UI; preserve the parent obligation; execute HFE/T verification and negative or failure-path tests.	T-UX; UI; V&V HFE/T; FMEA S/O/D/RPN TBD	Gate
HFE-001-02	Category alias NFSR-HFE-001-02; parent HFE-001; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require explicit transition handling and shall invalidate stale confirmations, previews, and unsaved actions associated with the prior context.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation UI/SK; preserve the parent obligation; execute T/HFE verification and negative or failure-path tests.	T-UX; UI/SK; V&V T/HFE; FMEA S/O/D/RPN TBD	Moderate
HFE-001-03	Category alias NFSR-HFE-001-03; parent HFE-001; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: verify context visibility and correct user response during interruptions, hand-offs, multi-patient work, and version changes.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation HFE/AUD; preserve the parent obligation; execute HFE/I verification and negative or failure-path tests.	T-UX; HFE/AUD; V&V HFE/I; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
HFE-002-01	Category alias NFSR- HFE-002-01; parent HFE- 002; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: carry an explicit semantic source label.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation UI/DATA; preserve the parent obligation; execute I/HFE verification and negative or failure-path tests.	T-UX; UI/ DATA; V&V I/ HFE; FMEA S/O/D/RPN TBD	High
HFE-002-02	Category alias NFSR- HFE-002-02; parent HFE- 002; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use text, shape, iconography, or grouping in addition to color and shall remain consistent across supported workflows.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation UI; preserve the parent obligation; execute HFE/I verification and negative or failure-path tests.	T-UX; UI; V&V HFE/I; FMEA S/O/D/RPN TBD	Moderate
HFE-002-03	Category alias NFSR- HFE-002-03; parent HFE- 002; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: verify correct classification and interpretation of source types in normal, ambiguous, and time-pressured scenarios.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation HFE/VVP; preserve the parent obligation; execute HFE/A verification and negative or failure-path tests.	T-UX; HFE/ VVP; V&V HFE/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
HFE-003-01	Category alias NFSR- HFE-003-01; parent HFE- 003; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: state the hazard or condition, affected patient or object, consequence when material, required action, and escalation path.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation UI; preserve the parent obligation; execute I/HFE verification and negative or failure-path tests.	T-UX; UI; V&V I/HFE; FMEA S/O/D/RPN TBD	Moderate
HFE-003-02	Category alias NFSR- HFE-003-02; parent HFE- 003; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use accessible multimodal indicators, shall not rely on color alone, and shall require the configured acknowledgment or corrective....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation UI; preserve the parent obligation; execute HFE/T verification and negative or failure-path tests.	T-UX; UI; V&V HFE/T; FMEA S/O/D/RPN TBD	High
HFE-003-03	Category alias NFSR- HFE-003-03; parent HFE- 003; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: record warning presentation and response, and usability review shall assess warning recognition, comprehension, and appropriate action.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation AUD/HFE; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-UX; AUD/ HFE; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
HFE-004-01	Category alias NFSR- HFE-004-01; parent HFE- 004; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: assign a risk tier, presentation method, acknowledgment requirement, escalation, suppression rule, and owner to each warning or confirmation....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation HFE/CFG; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-UX; HFE/CFG; V&V I/A; FMEA S/O/D/RPN TBD	High
HFE-004-02	Category alias NFSR- HFE-004-02; parent HFE- 004; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: collect alert frequency, repeated exposure, acknowledgment, override, cancellation, workaround, and response-time signals without weakening a required hard stop.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation UI/OPS; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-UX; UI/OPS; V&V T/A; FMEA S/O/D/RPN TBD	Gate
HFE-004-03	Category alias NFSR- HFE-004-03; parent HFE- 004; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: periodically review alert burden and critical-warning comprehension and shall control corrective changes through risk and human-factors assessment.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement allocation QMS/HFE; preserve the parent obligation; execute A/I verification and negative or failure-path tests.	T-UX; QMS/HFE; V&V A/I; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
HFE-005-01	Category alias NFSR- HFE-005-01; parent HFE- 005; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify intended users, use environments, critical tasks, hazards, foreseeable misuse, interruptions, hand-offs, recovery scenarios, and acceptance criteria.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement allocation HFE/VVP; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-UX; HFE/VVP; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
HFE-005-02	Category alias NFSR- HFE-005-02; parent HFE- 005; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: document findings, risk links, design changes, unresolved issues, and verification of corrective actions.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation HFE/QMS; preserve the parent obligation; execute HFE/I verification and negative or failure-path tests.	T-UX; HFE/QMS; V&V HFE/I; FMEA S/O/D/RPN TBD	High
HFE-005-03	Category alias NFSR- HFE-005-03; parent HFE- 005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use representative users and conditions and shall demonstrate successful completion of critical tasks without unacceptable use-related risk before....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation HFE/VVP; preserve the parent obligation; execute HFE/A verification and negative or failure-path tests.	T-UX; HFE/VVP; V&V HFE/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
HFE-006-01	Category alias NFSR-HFE-006-01; parent HFE-006; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define role-specific curricula, prerequisites, supervised practice, competency assessments, renewal intervals, and remediation for each clinical function.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation HFE/QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-UX; HFE/QMS; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
HFE-006-02	Category alias NFSR-HFE-006-02; parent HFE-006; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: enable a role-specific clinical function only when the user's training, competency, and authorization records are current.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement allocation IAM; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-UX; IAM; V&V T/I; FMEA S/O/D/RPN TBD	Moderate
HFE-006-03	Category alias NFSR-HFE-006-03; parent HFE-006; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain training and competency evidence, authorization decisions, expirations, revocations, exceptions, and periodic review results.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation AUD/QMS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-UX; AUD/QMS; V&V I/T; FMEA S/O/D/RPN TBD	High

6.11 Privacy, cybersecurity, and secure development (SEC) - 24 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SEC-001-01	Category alias NFSR-SEC-001-01; parent SEC-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide unique user identity, approved multi-factor authentication, role and context authorization, least privilege, and configurable session controls for....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation IAM; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SEC; IAM; V&V I/T; FMEA S/O/D/RPN TBD	High
SEC-001-02	Category alias NFSR-SEC-001-02; parent SEC-001; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: evaluate current identity, role, context, session, and object authorization and shall reject absent, expired, or insufficient authorization.	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement allocation IAM/SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-SEC; IAM/SK; V&V T; FMEA S/O/D/RPN TBD	High
SEC-001-03	Category alias NFSR-SEC-001-03; parent SEC-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: perform periodic access and privilege reviews and shall retain approvals, changes, dormant accounts, exceptions, and remediation.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/AUD; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-SEC; QMS/AUD; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SEC-002-01	Category alias NFSR- SEC-002-01; parent SEC- 002; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use approved encryption in transit and at rest, network segmentation, managed secrets, end-point controls, secure configuration, and least-privilege....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement allocation SEC/ARCH; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SEC; SEC/ARCH; V&V I/T; FMEA S/O/D/RPN TBD	High
SEC-002-02	Category alias NFSR- SEC-002-02; parent SEC- 002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be continuously or periodically assessed against the approved baseline, with material deviation generating an actionable event.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SEC/OPS; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-SEC; SEC/OPS; V&V T/I; FMEA S/O/D/RPN TBD	High
SEC-002-03	Category alias NFSR- SEC-002-03; parent SEC- 002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain security-baseline versions, key and certificate status, configuration assessments, exceptions, approvals, and remediation evidence.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/AUD; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-SEC; QMS/AUD; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SEC-003-01	Category alias NFSR- SEC-003-01; parent SEC- 003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: allowlist approved connectors, model services, tenants, destinations, data classes, purposes, and required protection controls.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SEC/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SEC; SEC/CFG; V&V I/T; FMEA S/O/D/RPN TBD	High
SEC-003-02	Category alias NFSR- SEC-003-02; parent SEC- 003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: block protected health information, images, prompts, transcripts, and outputs from an unapproved egress path or incompatible tenant.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SEC/SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-SEC; SEC/SK; V&V T; FMEA S/O/D/RPN TBD	High
SEC-003-03	Category alias NFSR- SEC-003-03; parent SEC- 003; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain the source, destination, data classification, policy decision, actor, purpose, and incident escalation when applicable.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement allocation AUD/OPS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SEC; AUD/OPS; V&V I/T; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SEC-004-01	Category alias NFSR-SEC-004-01; parent SEC-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: enter the system through an untrusted-data boundary separate from system policy and tool authority.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SEC/ARCH; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-SEC; SEC/ARCH; V&V I/A; FMEA S/O/D/RPN TBD	High
SEC-004-02	Category alias NFSR-SEC-004-02; parent SEC-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: sanitize, constrain, or reject embedded instructions that attempt to alter system authority, safety policy, credentials, tool permissions, or...	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SEC/NLU; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-SEC; SEC/NLU; V&V T; FMEA S/O/D/RPN TBD	High
SEC-004-03	Category alias NFSR-SEC-004-03; parent SEC-004; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: cover prompt injection, indirect instruction, malformed metadata, encoded content, tool redirection, data exfiltration, and privilege escalation.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation VVP/SEC; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-SEC; VVP/SEC; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SEC-005-01	Category alias NFSR- SEC-005-01; parent SEC- 005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: have an approved manifest containing identity, version, provenance, integrity value, signature, and release status.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SEC/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SEC; SEC/CFG; V&V I/T; FMEA S/O/D/RPN TBD	High
SEC-005-02	Category alias NFSR- SEC-005-02; parent SEC- 005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: occur before load or clinical use and shall block a missing, unsigned, altered, revoked, incompatible, or unapproved artifact.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SEC/SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-SEC; SEC/SK; V&V T; FMEA S/O/D/RPN TBD	High
SEC-005-03	Category alias NFSR- SEC-005-03; parent SEC- 005; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be retained for each artifact load.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation AUD/OPS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SEC; AUD/OPS; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SEC-006-01	Category alias NFSR- SEC-006-01; parent SEC- 006; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require threat modeling, secure coding standards, code review, software bill of materials, vulnerability scanning, penetration testing, patch governance....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement allocation QMS/ SEC; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-SEC; QMS/ SEC; V&V I/A; FMEA S/O/D/ RPN TBD	High
SEC-006-02	Category alias NFSR- SEC-006-02; parent SEC- 006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: include objective evidence that required security activities are complete and that identified vulnerabilities are remediated or formally risk....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SEC/ QMS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SEC; SEC/ QMS; V&V I/T; FMEA S/O/D/ RPN TBD	High
SEC-006-03	Category alias NFSR- SEC-006-03; parent SEC- 006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: undergo provenance, vulnerability, license, support, incident, and continuity assessment before approval.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-SEC; QMS; V&V I/A; FMEA S/ O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SEC-007-01	Category alias NFSR-SEC-007-01; parent SEC-007; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: collect identity, access, export, integrity, connector, model, evidence, secret, and data-loss events with synchronized time and protected provenance.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation SEC/AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-SEC; SEC/AUD; V&V I/T; FMEA S/O/D/ RPN TBD	High
SEC-007-02	Category alias NFSR-SEC-007-02; parent SEC-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify unauthorized access, anomalous export, integrity failure, malicious input, artifact compromise, secrets exposure, and suspected protected-information leakage.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SEC/OPS; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-SEC; SEC/OPS; V&V T; FMEA S/O/D/ RPN TBD	High
SEC-007-03	Category alias NFSR-SEC-007-03; parent SEC-007; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create a triage record containing severity, affected users and objects, clinical impact, containment, evidence, disposition, and escalation.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation OPS/AUD; preserve the parent obligation; execute I/D verification and negative or failure-path tests.	T-SEC; OPS/AUD; V&V I/D; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SEC-008-01	Category alias NFSR- SEC-008-01; parent SEC- 008; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define clinical and cybersecurity containment, evidence preservation, patient-impact assessment, communication, reporting, recovery, and corrective-action responsibilities.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation OPS/SEC; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-SEC; OPS/SEC; V&V I/A; FMEA S/O/D/RPN TBD	High
SEC-008-02	Category alias NFSR- SEC-008-02; parent SEC- 008; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: perform documented exercises covering representative compromise, outage, data-integrity, privacy, model, evidence, and interface incidents.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation OPS; preserve the parent obligation; execute D/T verification and negative or failure-path tests.	T-SEC; OPS; V&V D/T; FMEA S/O/D/RPN TBD	High
SEC-008-03	Category alias NFSR- SEC-008-03; parent SEC- 008; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain chronology, decisions, affected cases, notifications, evidence, recovery validation, lessons learned, and corrective and preventive actions.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation QMS/AUD; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-SEC; QMS/AUD; V&V I/A; FMEA S/O/D/RPN TBD	High

6.12 Clinical operations, monitoring, and resilience (OPS) - 21 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OPS-001-01	Category alias OSR-OPS-001-01; parent OPS-001; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define each safety, model, evidence, workflow, technical, subgroup, reliability, and security signal with data source, owner, threshold, action....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation OPS/CFG; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-OPS; OPS/CFG; V&V I/A; FMEA S/O/D/ RPN TBD	High
OPS-001-02	Category alias OSR-OPS-001-02; parent OPS-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: calculate signals against the approved thresholds and shall trigger the configured notification, restriction, suspension, or investigation action.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation OPS; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-OPS; OPS; V&V T/A; FMEA S/O/D/ RPN TBD	High
OPS-001-03	Category alias OSR-OPS-001-03; parent OPS-001; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain signal values, cohort or subgroup definition, threshold version, alert, action, acknowledgment, disposition, and periodic review.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement allocation AUD/QMS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-OPS; AUD/QMS; V&V I/T; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OPS-002-01	Category alias OSR- OPS-002-01; parent OPS- 002; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use a versioned state machine with pending, running, completed, failed, canceled, timed-out, partial, and reconciliation states as applicable.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation OPS/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-OPS; OPS/ DATA; V&V I/ T; FMEA S/O/ D/RPN TBD	Moderate
OPS-002-02	Category alias OSR- OPS-002-02; parent OPS- 002; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: be eligible for promotion; every other terminal or uncertain state shall remain ineligible and visibly noncomplete.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement allocation SK/DATA; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-OPS; SK/ DATA; V&V T; FMEA S/O/D/ RPN TBD	Moderate
OPS-002-03	Category alias OSR- OPS-002-03; parent OPS- 002; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain all job transitions, checkpoints, partial artifacts, cleanup actions, reconciliation decisions, and final eligibility.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation OPS/AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-OPS; OPS/ AUD; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OPS-003-01	Category alias OSR-OPS-003-01; parent OPS-003; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: provide authorized suspension controls for a model, evidence rule, connector, workflow, disease site, and the complete clinical service.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation OPS/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-OPS; OPS/CFG; V&V I/T; FMEA S/O/D/RPN TBD	High
OPS-003-02	Category alias OSR-OPS-003-02; parent OPS-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent new affected execution, identify in-progress and downstream affected work, and transition the service to the approved safe....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/OPS; preserve the parent obligation; execute D/T verification and negative or failure-path tests.	T-OPS; SK/OPS; V&V D/T; FMEA S/O/D/RPN TBD	High
OPS-003-03	Category alias OSR-OPS-003-03; parent OPS-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: restore a signed known-good configuration and shall require documented validation, reconciliation, approval, and communication before resumption.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation OPS/QMS; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-OPS; OPS/QMS; V&V D/I; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OPS-004-01	Category alias OSR-OPS-004-01; parent OPS-004; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define allowed read functions, prohibited executions, authorized users, data availability, indicators, entry criteria, and exit criteria.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement allocation OPS/CFG; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-OPS; OPS/CFG; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
OPS-004-02	Category alias OSR-OPS-004-02; parent OPS-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or pre-serves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: display the condition and shall block new clinical execution, modification, approval, and export except as explicitly authorized.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/UI; preserve the parent obligation; execute D/T verification and negative or failure-path tests.	T-OPS; SK/UI; V&V D/T; FMEA S/O/D/RPN TBD	High
OPS-004-03	Category alias OSR-OPS-004-03; parent OPS-004; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define access to completed work and audit records, external work practices, reconciliation, communication, and controlled restoration.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement allocation QMS/OPS; preserve the parent obligation; execute I/D verification and negative or failure-path tests.	T-OPS; QMS/OPS; V&V I/D; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OPS-005-01	Category alias OSR- OPS-005-01; parent OPS- 005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify protected objects, versions, encryption, integrity checks, retention, recovery objectives, storage locations, and restoration authority.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation OPS/SEC; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-OPS; OPS/SEC; V&V I/T; FMEA S/O/D/RPN TBD	High
OPS-005-02	Category alias OSR- OPS-005-02; parent OPS- 005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: occur in a staged environment and shall verify object identity, dependencies, destination reconciliation, approval states, audit continuity, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation OPS/DATA; preserve the parent obligation; execute D/T verification and negative or failure-path tests.	T-OPS; OPS/DATA; V&V D/T; FMEA S/O/D/RPN TBD	High
OPS-005-03	Category alias OSR- OPS-005-03; parent OPS- 005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: conduct and retain periodic recovery exercises, observed results, deviations, corrective actions, and approval for continued adequacy.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/AUD; preserve the parent obligation; execute I/D verification and negative or failure-path tests.	T-OPS; QMS/AUD; V&V I/D; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OPS-006-01	Category alias OSR-OPS-006-01; parent OPS-006; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: support reports for incidents, near misses, unsafe workarounds, and QA discrepancies with severity, triage, ownership, and escalation.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement allocation OPS/QMS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-OPS; OPS/QMS; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
OPS-006-02	Category alias OSR-OPS-006-02; parent OPS-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: link affected patients, cases, objects, releases, models, evidence, interfaces, and prior related events while preserving required privacy.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation OPS/DATA; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-OPS; OPS/DATA; V&V I/A; FMEA S/O/D/RPN TBD	High
OPS-006-03	Category alias OSR-OPS-006-03; parent OPS-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: document patient impact, evidence preservation, root or contributing causes, trends, corrective and preventive action, effectiveness checks, and closure....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-OPS; QMS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OPS-007-01	Category alias OSR- OPS-007-01; parent OPS- 007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define approved measures, thresholds, owners, escalation, and review intervals for availability, latency, recovery, retention, reliability, safety, model performance....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation OPS/QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-OPS; OPS/QMS; V&V I/A; FMEA S/O/D/RPN TBD	High
OPS-007-02	Category alias OSR- OPS-007-02; parent OPS- 007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: measure each service-level indicator using approved definitions and shall notify the responsible owner when an action threshold is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation OPS; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-OPS; OPS; V&V T/A; FMEA S/O/D/RPN TBD	High
OPS-007-03	Category alias OSR- OPS-007-03; parent OPS- 007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain performance trends, threshold breaches, actions, exceptions, proposed threshold changes, and governance approval.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/AUD; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-OPS; QMS/AUD; V&V I/A; FMEA S/O/D/RPN TBD	High

6.13 Verification, commissioning, release, and change control (VAL) - 27 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-001-01	Category alias OSR-VAL-001-01; parent VAL-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: assign stable identities and typed links among stakeholder needs, hazards, risk controls, high-level requirements, child requirements, design elements....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation VVP/QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-VV; VVP/QMS; V&V I/A; FMEA S/O/D/RPN TBD	High
VAL-001-02	Category alias OSR-VAL-001-02; parent VAL-001; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify missing, orphaned, duplicate, circular, stale, or failed links and shall prevent baseline acceptance with an unresolved required....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement allocation VVP; preserve the parent obligation; execute A/T verification and negative or failure-path tests.	T-VV; VVP; V&V A/T; FMEA S/O/D/RPN TBD	High
VAL-001-03	Category alias OSR-VAL-001-03; parent VAL-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain an approved traceability report, anomaly dispositions, coverage results, and the configuration of all traced artifacts.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/AUD; preserve the parent obligation; execute I verification and negative or failure-path tests.	T-VV; QMS/AUD; V&V I; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-002-01	Category alias OSR-VAL-002-01; parent VAL-002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define acceptance, commissioning, end-to-end, regression, shadow-mode, and governance criteria for the approved site configuration and intended use.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation VVP/QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-VV; VVP/QMS; V&V I/A; FMEA S/O/D/RPN TBD	High
VAL-002-02	Category alias OSR-VAL-002-02; parent VAL-002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute the approved site test protocols using controlled configurations, expected results, tolerances, independent reviewers, and recorded deviations.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation VVP; preserve the parent obligation; execute T/A verification and negative or failure-path tests.	T-VV; VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
VAL-002-03	Category alias OSR-VAL-002-03; parent VAL-002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: remain disabled until all required site evidence is complete, anomalies are acceptably dispositioned, and designated authorities approve release.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation CFG/QMS; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-VV; CFG/QMS; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-003-01	Category alias OSR-VAL-003-01; parent VAL-003; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: contain locked reference and challenge cases for language, voice, evidence, segmentation, registration, planning, dose, robustness, interoperability, security, recovery....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation VVP; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-VV; VVP; V&V I/A; FMEA S/O/D/RPN TBD	Gate
VAL-003-02	Category alias OSR-VAL-003-02; parent VAL-003; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: include immutable inputs, expected results or adjudication rules, configuration, applicability, provenance, and acceptance criteria.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement allocation VVP/DATA; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-VV; VVP/DATA; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
VAL-003-03	Category alias OSR-VAL-003-03; parent VAL-003; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: map cases to requirements, hazards, boundaries, failure modes, supported configurations, and known residual gaps.	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement allocation VVP/QMS; preserve the parent obligation; execute A/I verification and negative or failure-path tests.	T-VV; VVP/QMS; V&V A/I; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-004-01	Category alias OSR-VAL-004-01; parent VAL-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: include normal, boundary, rare but foreseeable, wrong-context, partial-failure, time-out, rollback, post-approval change, and recovery scenarios.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation VVP; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-VV; VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
VAL-004-02	Category alias OSR-VAL-004-02; parent VAL-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through the official user, model, clinical-service, and interface paths using the release-candidate configuration.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation VVP; preserve the parent obligation; execute T/D verification and negative or failure-path tests.	T-VV; VVP; V&V T/D; FMEA S/O/D/RPN TBD	High
VAL-004-03	Category alias OSR-VAL-004-03; parent VAL-004; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain ordered evidence, expected and observed behavior, data lineage, screenshots or logs when applicable, anomalies, dispositions, and regression....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation VVP/QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-VV; VVP/QMS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-005-01	Category alias OSR-VAL-005-01; parent VAL-005; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define task-specific technical performance, clinically significant error, dose impact where relevant, edit burden, failure detection, subgroup performance, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement allocation VVP/MLOPS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-VV; VVP/MLOPS; V&V I/A; FMEA S/O/D/RPN TBD	Gate
VAL-005-02	Category alias OSR-VAL-005-02; parent VAL-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use representative and challenge data, prespecified analyses, independent adjudication, confidence intervals or uncertainty characterization, and approved acceptance criteria.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation VVP; preserve the parent obligation; execute A/T verification and negative or failure-path tests.	T-VV; VVP; V&V A/T; FMEA S/O/D/RPN TBD	High
VAL-005-03	Category alias OSR-VAL-005-03; parent VAL-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: document limitations, subgroup findings, user interaction effects, residual risk, monitoring needs, and approval for the bounded intended use.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/HFE; preserve the parent obligation; execute I/HFE verification and negative or failure-path tests.	T-VV; QMS/HFE; V&V I/HFE; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-006-01	Category alias OSR-VAL-006-01; parent VAL-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: have an approved protocol defining scope, endpoints, stop rules, sample rationale, adjudication, monitoring, and progression criteria.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation VVP/QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-VV; VVP/QMS; V&V I/A; FMEA S/O/D/RPN TBD	High
VAL-006-02	Category alias OSR-VAL-006-02; parent VAL-006; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: obtain and retain all applicable research, ethics, privacy, quality, institutional, and regulatory authorizations before initiating a clinical-evaluation stage.	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement allocation QMS; preserve the parent obligation; execute I verification and negative or failure-path tests.	T-VV; QMS; V&V I; FMEA S/O/D/RPN TBD	High
VAL-006-03	Category alias OSR-VAL-006-03; parent VAL-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require independent endpoint adjudication, anomaly and patient-impact review, residual-risk assessment, and documented approval to progress.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation VVP/QMS; preserve the parent obligation; execute A/I verification and negative or failure-path tests.	T-VV; VVP/QMS; V&V A/I; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-007-01	Category alias OSR-VAL-007-01; parent VAL-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: have a versioned checklist of required safety, quality, subgroup, usability, reliability, cybersecurity, clinical, and governance evidence and authorized....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/CFG; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-VV; QMS/CFG; V&V I/A; FMEA S/O/D/RPN TBD	High
VAL-007-02	Category alias OSR-VAL-007-02; parent VAL-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prevent progression or clinical activation when required evidence is missing, failed, expired, inconsistent, or not approved.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation SK/CFG; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-VV; SK/CFG; V&V T; FMEA S/O/D/RPN TBD	High
VAL-007-03	Category alias OSR-VAL-007-03; parent VAL-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: retain submitted evidence versions, automated checks, reviewer decisions, exceptions, residual risks, approval times, and resulting release state.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation AUD; preserve the parent obligation; execute I/T verification and negative or failure-path tests.	T-VV; AUD; V&V I/T; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-008-01	Category alias OSR-VAL-008-01; parent VAL-008; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: classify expansions by disease site, model, modality, machine, technique, institution, adaptive workflow, interface, and evidence package and shall....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement allocation QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-VV; QMS; V&V I/A; FMEA S/O/D/RPN TBD	High
VAL-008-02	Category alias OSR-VAL-008-02; parent VAL-008; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: prescribe the required commissioning, regression, human-factors, interoperability, cybersecurity, clinical, and monitoring activities for the expansion.	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement allocation VVP/QMS; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-VV; VVP/QMS; V&V I/A; FMEA S/O/D/RPN TBD	High
VAL-008-03	Category alias OSR-VAL-008-03; parent VAL-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: remain unavailable in clinical mode until all prescribed activities and approvals are complete and bound to a new....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation CFG; preserve the parent obligation; execute T/I verification and negative or failure-path tests.	T-VV; CFG; V&V T/I; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
VAL-009-01	Category alias OSR-VAL-009-01; parent VAL-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: identify severity, clinical impact, affected scope, risk-control status, release-threshold effect, owner, disposition, and closure evidence.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation QMS/VVP; preserve the parent obligation; execute I/A verification and negative or failure-path tests.	T-VV; QMS/VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
VAL-009-02	Category alias OSR-VAL-009-02; parent VAL-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: block entry to or continued operation in clinical mode when a critical anomaly, unacceptable residual risk, failed threshold....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation CFG/SK; preserve the parent obligation; execute T verification and negative or failure-path tests.	T-VV; CFG/SK; V&V T; FMEA S/O/D/RPN TBD	High
VAL-009-03	Category alias OSR-VAL-009-03; parent VAL-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: require containment, affected-case assessment, corrective action, regression evidence, residual-risk approval, and documented authorization.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement allocation OPS/QMS; preserve the parent obligation; execute D/I verification and negative or failure-path tests.	T-VV; OPS/QMS; V&V D/I; FMEA S/O/D/RPN TBD	High

7 High-level interface requirement risk records

7.1 Governance, intended use, and scope (GOV) - 21 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-GOV-001-01	Parent GOV-001; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-001-02	Parent GOV-001; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce deterministic authorization....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-001-03	Parent GOV-001; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from GOV-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-GOV-002-01	Parent GOV-002; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
IF-GOV-002-02	Parent GOV-002; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V T/D; FMEA S/O/D/ RPN TBD	Moderate
IF-GOV-002-03	Parent GOV-002; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from GOV-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V I/T/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-GOV-003-01	Parent GOV-003; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-003-02	Parent GOV-003; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce deterministic authorization....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-003-03	Parent GOV-003; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from GOV-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-GOV-004-01	Parent GOV-004; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DICOM-01; IF-TPS-01; IF-OPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DICOM-01; IF-TPS-01; IF-OPS-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-004-02	Parent GOV-004; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DICOM-01; IF-TPS-01; IF-OPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DICOM-01; IF-TPS-01; IF-OPS-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-004-03	Parent GOV-004; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from GOV-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DICOM-01; IF-TPS-01; IF-OPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DICOM-01; IF-TPS-01; IF-OPS-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-GOV-005-01	Parent GOV-005; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-005-02	Parent GOV-005; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-005-03	Parent GOV-005; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from GOV-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-GOV-006-01	Parent GOV-006; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DATA-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-006-02	Parent GOV-006; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce deterministic authorization....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DATA-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-006-03	Parent GOV-006; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from GOV-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-GOV-007-01	Parent GOV-007; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; IF-AI-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; IF-AI-01; IF-OPS-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-007-02	Parent GOV-007; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; IF-AI-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; IF-AI-01; IF-OPS-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-GOV-007-03	Parent GOV-007; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from GOV-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement all three child SIRs across IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; IF-AI-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-GOV; IF-CFG-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; IF-AI-01; IF-OPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

7.2 Safety boundaries and human authority (SAF) - 30 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SAF-001-01	Parent SAF-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SAF-001-02	Parent SAF-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall preserve independent review....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-SAF-001-03	Parent SAF-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SAF-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SAF-002-01	Parent SAF-002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; IF-DICOM-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; IF-DICOM-01; IF-TPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SAF-002-02	Parent SAF-002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; IF-DICOM-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; IF-DICOM-01; IF-TPS-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-SAF-002-03	Parent SAF-002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SAF-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; IF-DICOM-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; IF-DICOM-01; IF-TPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SAF-003-01	Parent SAF-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SAF-003-02	Parent SAF-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-SAF-003-03	Parent SAF-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SAF-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-CTX-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SAF-004-01	Parent SAF-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SAF-004-02	Parent SAF-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-SAF-004-03	Parent SAF-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SAF-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SAF-005-01	Parent SAF-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SAF-005-02	Parent SAF-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-SAF-005-03	Parent SAF-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SAF-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SAF-006-01	Parent SAF-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-OPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SAF-006-02	Parent SAF-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-OPS-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-SAF-006-03	Parent SAF-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SAF-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-OPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SAF-007-01	Parent SAF-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SAF-007-02	Parent SAF-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-SAF-007-03	Parent SAF-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SAF-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SAF-008-01	Parent SAF-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-AI-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-AI-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SAF-008-02	Parent SAF-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce intended use....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-AI-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-AI-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-SAF-008-03	Parent SAF-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SAF-008 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-AI-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-AI-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SAF-009-01	Parent SAF-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SAF-009-02	Parent SAF-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-SAF-009-03	Parent SAF-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SAF-009 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-CTX-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SAF-010-01	Parent SAF-010; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SAF-010-02	Parent SAF-010; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-SAF-010-03	Parent SAF-010; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SAF-010 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SYS; IF-SAF-01; IF-IAM-01; IF-DATA-01; IF-EVD-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

7.3 Natural-language and voice interaction (NLI) - 24 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-NLI-001-01	Parent NLI-001; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-NLI-001-02	Parent NLI-001; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-NLI-001-03	Parent NLI-001; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from NLI-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-NLI-002-01	Parent NLI-002; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-NLI-002-02	Parent NLI-002; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-NLI-002-03	Parent NLI-002; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from NLI-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-NLI-003-01	Parent NLI-003; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-NLI-003-02	Parent NLI-003; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-NLI-003-03	Parent NLI-003; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from NLI-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IAM-01; IF-EVD-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-NLI-004-01	Parent NLI-004; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-NLI-004-02	Parent NLI-004; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-NLI-004-03	Parent NLI-004; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from NLI-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-NLI-005-01	Parent NLI-005; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-NLI-005-02	Parent NLI-005; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-NLI-005-03	Parent NLI-005; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from NLI-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-NLI-006-01	Parent NLI-006; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-OPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-NLI-006-02	Parent NLI-006; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-OPS-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-NLI-006-03	Parent NLI-006; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from NLI-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-OPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-NLI-007-01	Parent NLI-007; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-CTX-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-NLI-007-02	Parent NLI-007; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-CTX-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-NLI-007-03	Parent NLI-007; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from NLI-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-CTX-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-NLI-008-01	Parent NLI-008; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-REV-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-REV-01; IF-DATA-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-NLI-008-02	Parent NLI-008; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-REV-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-REV-01; IF-DATA-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-NLI-008-03	Parent NLI-008; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from NLI-008 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement all three child SIRs across IF-NLI-01; IF-UI-01; IF-SAF-01; IF-REV-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-NLI; IF-NLI-01; IF-UI-01; IF-SAF-01; IF-REV-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

7.4 Evidence and literature governance (EVD) - 24 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-EVD-001-01	Parent EVD-001; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-EVD-001-02	Parent EVD-001; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-EVD-001-03	Parent EVD-001; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from EVD-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-EVD-002-01	Parent EVD-002; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-EVD-002-02	Parent EVD-002; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-EVD-002-03	Parent EVD-002; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from EVD-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-EVD-003-01	Parent EVD-003; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-EVD-003-02	Parent EVD-003; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-EVD-003-03	Parent EVD-003; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from EVD-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-EVD-004-01	Parent EVD-004; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-EVD-004-02	Parent EVD-004; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-EVD-004-03	Parent EVD-004; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from EVD-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-CTX-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-EVD-005-01	Parent EVD-005; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-EVD-005-02	Parent EVD-005; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-EVD-005-03	Parent EVD-005; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from EVD-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-EVD-006-01	Parent EVD-006; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-EVD-006-02	Parent EVD-006; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-EVD-006-03	Parent EVD-006; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from EVD-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IMG-01; IF-DICOM-01; IF-PLN-01; IF-TPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-EVD-007-01	Parent EVD-007; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; IF-OPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-EVD-007-02	Parent EVD-007; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; IF-OPS-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-EVD-007-03	Parent EVD-007; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from EVD-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; IF-IAM-01; IF-REV-01; IF-OPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-EVD-008-01	Parent EVD-008; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-EVD-008-02	Parent EVD-008; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-EVD-008-03	Parent EVD-008; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from EVD-008 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement all three child SIRs across IF-EVD-01; IF-SAF-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-EVD; IF-EVD-01; IF-SAF-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

7.5 Clinical context, imaging, registration, and contours (CLN) - 27 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-CLN-001-01	Parent CLN-001; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-CLN-001-02	Parent CLN-001; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-CLN-001-03	Parent CLN-001; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from CLN-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-CLN-002-01	Parent CLN-002; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-CLN-002-02	Parent CLN-002; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-CLN-002-03	Parent CLN-002; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from CLN-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-CLN-003-01	Parent CLN-003; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-PLN-01; IF-TPS-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-PLN-01; IF-TPS-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-CLN-003-02	Parent CLN-003; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-PLN-01; IF-TPS-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-PLN-01; IF-TPS-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-CLN-003-03	Parent CLN-003; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from CLN-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-PLN-01; IF-TPS-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-PLN-01; IF-TPS-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-CLN-004-01	Parent CLN-004; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-AI-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-CLN-004-02	Parent CLN-004; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-AI-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-CLN-004-03	Parent CLN-004; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from CLN-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-REV-01; IF-AI-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-CLN-005-01	Parent CLN-005; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-AI-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-AI-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-CLN-005-02	Parent CLN-005; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-AI-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-AI-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-CLN-005-03	Parent CLN-005; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from CLN-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-AI-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-AI-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-CLN-006-01	Parent CLN-006; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-CLN-006-02	Parent CLN-006; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-CLN-006-03	Parent CLN-006; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from CLN-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-CLN-007-01	Parent CLN-007; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; IF-AI-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; IF-AI-01; IF-DATA-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-CLN-007-02	Parent CLN-007; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; IF-AI-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; IF-AI-01; IF-DATA-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-CLN-007-03	Parent CLN-007; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from CLN-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; IF-AI-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; IF-AI-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-CLN-008-01	Parent CLN-008; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-CLN-008-02	Parent CLN-008; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-CLN-008-03	Parent CLN-008; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from CLN-008 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-UI-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-CLN-009-01	Parent CLN-009; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-CLN-009-02	Parent CLN-009; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-CLN-009-03	Parent CLN-009; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from CLN-009 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement all three child SIRs across IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-IMG; IF-CTX-01; IF-IMG-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

7.6 Treatment-plan generation and calculation (PLN) - 36 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-001-01	Parent PLN-001; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-CTX-01; IF-IMG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-CTX-01; IF-IMG-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-PLN-001-02	Parent PLN-001; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-CTX-01; IF-IMG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-CTX-01; IF-IMG-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-PLN-001-03	Parent PLN-001; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-CTX-01; IF-IMG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-CTX-01; IF-IMG-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-002-01	Parent PLN-002; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-PLN-002-02	Parent PLN-002; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-PLN-002-03	Parent PLN-002; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-003-01	Parent PLN-003; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-NLI-01; IF-UI-01; IF-AI-01; IF-VV-01; base-line contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-NLI-01; IF-UI-01; IF-AI-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-PLN-003-02	Parent PLN-003; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-NLI-01; IF-UI-01; IF-AI-01; IF-VV-01; base-line contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-NLI-01; IF-UI-01; IF-AI-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-PLN-003-03	Parent PLN-003; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-NLI-01; IF-UI-01; IF-AI-01; IF-VV-01; base-line contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-NLI-01; IF-UI-01; IF-AI-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-004-01	Parent PLN-004; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IMG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IMG-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-PLN-004-02	Parent PLN-004; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IMG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IMG-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-PLN-004-03	Parent PLN-004; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IMG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IMG-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-005-01	Parent PLN-005; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-PLN-005-02	Parent PLN-005; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-PLN-005-03	Parent PLN-005; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-006-01	Parent PLN-006; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-IAM-01; IF-SAF-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-PLN-006-02	Parent PLN-006; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-IAM-01; IF-SAF-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-PLN-006-03	Parent PLN-006; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-IAM-01; IF-SAF-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-007-01	Parent PLN-007; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-PLN-007-02	Parent PLN-007; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-PLN-007-03	Parent PLN-007; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-008-01	Parent PLN-008; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-PLN-008-02	Parent PLN-008; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-PLN-008-03	Parent PLN-008; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-008 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-009-01	Parent PLN-009; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-CFG-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-CFG-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-PLN-009-02	Parent PLN-009; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-CFG-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-CFG-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-PLN-009-03	Parent PLN-009; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-009 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-CFG-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; IF-CFG-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-010-01	Parent PLN-010; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-PLN-010-02	Parent PLN-010; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-PLN-010-03	Parent PLN-010; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-010 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-011-01	Parent PLN-011; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; V&V I/A; FMEA S/O/D/ RPN TBD	High
IF-PLN-011-02	Parent PLN-011; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; V&V T/D; FMEA S/O/D/ RPN TBD	Gate
IF-PLN-011-03	Parent PLN-011; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-011 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-PLN-012-01	Parent PLN-012; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; IF-CFG-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-PLN-012-02	Parent PLN-012; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; IF-CFG-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-PLN-012-03	Parent PLN-012; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from PLN-012 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-PLAN; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-DATA-01; IF-CFG-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

7.7 Plan review, selection, QA, and transfer (REV) - 27 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-REV-001-01	Parent REV-001; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-REV-001-02	Parent REV-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-REV-001-03	Parent REV-001; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from REV-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-REV-002-01	Parent REV-002; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; IF-DATA-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-REV-002-02	Parent REV-002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; IF-DATA-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-REV-002-03	Parent REV-002; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from REV-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-UI-01; IF-PLN-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-REV-003-01	Parent REV-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-REV-003-02	Parent REV-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-REV-003-03	Parent REV-003; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from REV-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-REV-004-01	Parent REV-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-DATA-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-REV-004-02	Parent REV-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-DATA-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-REV-004-03	Parent REV-004; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from REV-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-REV-005-01	Parent REV-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-REV-005-02	Parent REV-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-REV-005-03	Parent REV-005; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from REV-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-REV-006-01	Parent REV-006; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-CTX-01; IF-PLN-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-CTX-01; IF-PLN-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-REV-006-02	Parent REV-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-CTX-01; IF-PLN-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-CTX-01; IF-PLN-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-REV-006-03	Parent REV-006; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from REV-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-CTX-01; IF-PLN-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-CTX-01; IF-PLN-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-REV-007-01	Parent REV-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-REV-007-02	Parent REV-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-REV-007-03	Parent REV-007; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from REV-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-IAM-01; IF-SAF-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-REV-008-01	Parent REV-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-REV-008-02	Parent REV-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-REV-008-03	Parent REV-008; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from REV-008 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-REV-009-01	Parent REV-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-REV-009-02	Parent REV-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-REV-009-03	Parent REV-009; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from REV-009 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement all three child SIRs across IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-REV; IF-REV-01; IF-TPS-01; IF-DICOM-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

7.8 Data, interoperability, provenance, and audit (DAT) - 30 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-DAT-001-01	Parent DAT-001; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-DAT-001-02	Parent DAT-001; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-DAT-001-03	Parent DAT-001; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from DAT-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-DAT-002-01	Parent DAT-002; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-AI-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-DAT-002-02	Parent DAT-002; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-AI-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-DAT-002-03	Parent DAT-002; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from DAT-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-AI-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-DAT-003-01	Parent DAT-003; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-DAT-003-02	Parent DAT-003; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-DAT-003-03	Parent DAT-003; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from DAT-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-UI-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-DAT-004-01	Parent DAT-004; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-CTX-01; IF-REV-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-CTX-01; IF-REV-01; IF-TPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-DAT-004-02	Parent DAT-004; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-CTX-01; IF-REV-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-CTX-01; IF-REV-01; IF-TPS-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-DAT-004-03	Parent DAT-004; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from DAT-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-CTX-01; IF-REV-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-CTX-01; IF-REV-01; IF-TPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-DAT-005-01	Parent DAT-005; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-DAT-005-02	Parent DAT-005; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-DAT-005-03	Parent DAT-005; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from DAT-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IMG-01; IF-PLN-01; IF-TPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-DAT-006-01	Parent DAT-006; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IHE-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IHE-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-DAT-006-02	Parent DAT-006; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IHE-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IHE-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-DAT-006-03	Parent DAT-006; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from DAT-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IHE-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IHE-01; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-DAT-007-01	Parent DAT-007; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-DAT-007-02	Parent DAT-007; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-DAT-007-03	Parent DAT-007; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from DAT-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-DAT-008-01	Parent DAT-008; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-DAT-008-02	Parent DAT-008; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-DAT-008-03	Parent DAT-008; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from DAT-008 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-DAT-009-01	Parent DAT-009; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; IF-AI-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-DAT-009-02	Parent DAT-009; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; IF-AI-01; V&V T/D; FMEA S/O/D/RPN TBD	Gate
IF-DAT-009-03	Parent DAT-009; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from DAT-009 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-REV-01; IF-AI-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-DAT-010-01	Parent DAT-010; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; V&V I/A; FMEA S/O/D/ RPN TBD	High
IF-DAT-010-02	Parent DAT-010; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; V&V T/D; FMEA S/O/D/ RPN TBD	Gate
IF-DAT-010-03	Parent DAT-010; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from DAT-010 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement all three child SIRs across IF-DATA-01; IF-DICOM-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-DATA; IF-DATA-01; IF-DICOM-01; IF-SEC-01; V&V I/T/A; FMEA S/O/D/ RPN TBD	High

7.9 AI and inference-model lifecycle (AIM) - 24 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-AIM-001-01	Parent AIM-001; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-AIM-001-02	Parent AIM-001; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce intended use....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-AIM-001-03	Parent AIM-001; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from AIM-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-AIM-002-01	Parent AIM-002; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-AIM-002-02	Parent AIM-002; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce intended use....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-AIM-002-03	Parent AIM-002; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from AIM-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-AIM-003-01	Parent AIM-003; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-AIM-003-02	Parent AIM-003; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce intended use....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-AIM-003-03	Parent AIM-003; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from AIM-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-AIM-004-01	Parent AIM-004; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; IF-CTX-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-AIM-004-02	Parent AIM-004; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; IF-CTX-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-AIM-004-03	Parent AIM-004; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from AIM-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; IF-CTX-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-AIM-005-01	Parent AIM-005; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-DATA-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-DATA-01; IF-SEC-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-AIM-005-02	Parent AIM-005; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce intended use....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-DATA-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-DATA-01; IF-SEC-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-AIM-005-03	Parent AIM-005; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from AIM-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-DATA-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-DATA-01; IF-SEC-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-AIM-006-01	Parent AIM-006; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-AIM-006-02	Parent AIM-006; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce intended use....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-AIM-006-03	Parent AIM-006; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from AIM-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-AIM-007-01	Parent AIM-007; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-AIM-007-02	Parent AIM-007; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-AIM-007-03	Parent AIM-007; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from AIM-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-UI-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-AIM-008-01	Parent AIM-008; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; V&V I/A; FMEA S/O/D/ RPN TBD	High
IF-AIM-008-02	Parent AIM-008; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; V&V T/D; FMEA S/O/D/ RPN TBD	High
IF-AIM-008-03	Parent AIM-008; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from AIM-008 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement all three child SIRs across IF-AI-01; IF-CFG-01; IF-OPS-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-AI; IF-AI-01; IF-CFG-01; IF-OPS-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; V&V I/T/A; FMEA S/O/D/ RPN TBD	High

7.10 Human factors, usability, and competency (HFE) - 18 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-HFE-001-01	Parent HFE-001; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; IF-CTX-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-001-02	Parent HFE-001; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; IF-CTX-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-001-03	Parent HFE-001; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from HFE-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; IF-CTX-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; IF-CTX-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-HFE-002-01	Parent HFE-002; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-EVD-01; IF-CTX-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-EVD-01; IF-CTX-01; IF-AI-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-002-02	Parent HFE-002; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-EVD-01; IF-CTX-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-EVD-01; IF-CTX-01; IF-AI-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-002-03	Parent HFE-002; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from HFE-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-EVD-01; IF-CTX-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-EVD-01; IF-CTX-01; IF-AI-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

Continued on next page

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-HFE-003-01	Parent HFE-003; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-003-02	Parent HFE-003; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-003-03	Parent HFE-003; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from HFE-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-HFE-004-01	Parent HFE-004; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-OPS-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-004-02	Parent HFE-004; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-OPS-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-004-03	Parent HFE-004; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from HFE-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-OPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-HFE-005-01	Parent HFE-005; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-CFG-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-CFG-01; IF-OPS-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-005-02	Parent HFE-005; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-CFG-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-CFG-01; IF-OPS-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-005-03	Parent HFE-005; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from HFE-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-CFG-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-CFG-01; IF-OPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-HFE-006-01	Parent HFE-006; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-006-02	Parent HFE-006; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-HFE-006-03	Parent HFE-006; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from HFE-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement all three child SIRs across IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-UX; IF-UI-01; IF-VV-01; IF-IAM-01; IF-SAF-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

7.11 Privacy, cybersecurity, and secure development (SEC) - 24 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SEC-001-01	Parent SEC-001; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-SAF-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-SAF-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SEC-001-02	Parent SEC-001; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-SAF-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-SAF-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-SEC-001-03	Parent SEC-001; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SEC-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-SAF-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-SAF-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SEC-002-01	Parent SEC-002; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-IMG-01; IF-DICOM-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-IMG-01; IF-DICOM-01; IF-CFG-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SEC-002-02	Parent SEC-002; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-IMG-01; IF-DICOM-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-IMG-01; IF-DICOM-01; IF-CFG-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-SEC-002-03	Parent SEC-002; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SEC-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-IMG-01; IF-DICOM-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-IMG-01; IF-DICOM-01; IF-CFG-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SEC-003-01	Parent SEC-003; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-NLI-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-NLI-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-TPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SEC-003-02	Parent SEC-003; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-NLI-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-NLI-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-TPS-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-SEC-003-03	Parent SEC-003; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SEC-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-NLI-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-TPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-NLI-01; IF-UI-01; IF-IMG-01; IF-DICOM-01; IF-TPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SEC-004-01	Parent SEC-004; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-DICOM-01; IF-TPS-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-DICOM-01; IF-TPS-01; IF-AI-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SEC-004-02	Parent SEC-004; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-DICOM-01; IF-TPS-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-DICOM-01; IF-TPS-01; IF-AI-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-SEC-004-03	Parent SEC-004; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SEC-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-DICOM-01; IF-TPS-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-UI-01; IF-DICOM-01; IF-TPS-01; IF-AI-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SEC-005-01	Parent SEC-005; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; IF-CFG-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SEC-005-02	Parent SEC-005; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; IF-CFG-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-SEC-005-03	Parent SEC-005; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SEC-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; IF-CFG-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SEC-006-01	Parent SEC-006; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-REV-01; IF-AI-01; IF-CFG-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-REV-01; IF-AI-01; IF-CFG-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SEC-006-02	Parent SEC-006; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce intended use....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-REV-01; IF-AI-01; IF-CFG-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-REV-01; IF-AI-01; IF-CFG-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-SEC-006-03	Parent SEC-006; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SEC-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-REV-01; IF-AI-01; IF-CFG-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-REV-01; IF-AI-01; IF-CFG-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SEC-007-01	Parent SEC-007; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SEC-007-02	Parent SEC-007; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-SEC-007-03	Parent SEC-007; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SEC-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-SAF-01; IF-EVD-01; IF-AI-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-SEC-008-01	Parent SEC-008; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-EVD-01; IF-CTX-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-EVD-01; IF-CTX-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-SEC-008-02	Parent SEC-008; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-EVD-01; IF-CTX-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-EVD-01; IF-CTX-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-SEC-008-03	Parent SEC-008; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from SEC-008 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement all three child SIRs across IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-EVD-01; IF-CTX-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-SEC; IF-SEC-01; IF-IAM-01; IF-OPS-01; IF-EVD-01; IF-CTX-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

7.12 Clinical operations, monitoring, and resilience (OPS) - 21 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-OPS-001-01	Parent OPS-001; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-001-02	Parent OPS-001; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-001-03	Parent OPS-001; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from OPS-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-OPS-002-01	Parent OPS-002; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-002-02	Parent OPS-002; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall make timeout, retry....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-002-03	Parent OPS-002; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from OPS-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-OPS-003-01	Parent OPS-003; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; IF-CFG-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-003-02	Parent OPS-003; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; IF-CFG-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-003-03	Parent OPS-003; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from OPS-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; IF-CFG-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-AI-01; IF-CFG-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-OPS-004-01	Parent OPS-004; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-004-02	Parent OPS-004; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall make timeout, retry....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-004-03	Parent OPS-004; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from OPS-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-OPS-005-01	Parent OPS-005; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-DICOM-01; IF-TPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-DICOM-01; IF-TPS-01; IF-VV-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-005-02	Parent OPS-005; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-DICOM-01; IF-TPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-DICOM-01; IF-TPS-01; IF-VV-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-005-03	Parent OPS-005; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from OPS-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-DICOM-01; IF-TPS-01; IF-VV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-IAM-01; IF-SAF-01; IF-DICOM-01; IF-TPS-01; IF-VV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-OPS-006-01	Parent OPS-006; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-REV-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-006-02	Parent OPS-006; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-REV-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-006-03	Parent OPS-006; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from OPS-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-REV-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-CTX-01; IF-REV-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-OPS-007-01	Parent OPS-007; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-REV-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-REV-01; IF-AI-01; V&V I/A; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-007-02	Parent OPS-007; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall reject inapplicable, conflicting....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-REV-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-REV-01; IF-AI-01; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
IF-OPS-007-03	Parent OPS-007; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from OPS-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement all three child SIRs across IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-REV-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-OPS; IF-OPS-01; IF-DATA-01; IF-SEC-01; IF-EVD-01; IF-REV-01; IF-AI-01; V&V I/T/A; FMEA S/O/D/RPN TBD	Moderate

7.13 Verification, commissioning, release, and change control (VAL) - 27 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-VAL-001-01	Parent VAL-001; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; V&V I/A; FMEA S/O/D/ RPN TBD	High
IF-VAL-001-02	Parent VAL-001; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce deterministic authorization....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; V&V T/D; FMEA S/O/D/ RPN TBD	High
IF-VAL-001-03	Parent VAL-001; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from VAL-001 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/ RPN TBD	High

Continued on next page

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-VAL-002-01	Parent VAL-002; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-SEC-01; V&V I/A; FMEA S/O/D/ RPN TBD	High
IF-VAL-002-02	Parent VAL-002; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall authenticate and authorize....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-SEC-01; V&V T/D; FMEA S/O/D/ RPN TBD	High
IF-VAL-002-03	Parent VAL-002; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from VAL-002 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-SEC-01; V&V I/T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-VAL-003-01	Parent VAL-003; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-NLI-01; IF-UI-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-NLI-01; IF-UI-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; V&V I/A; FMEA S/O/D/ RPN TBD	High
IF-VAL-003-02	Parent VAL-003; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-NLI-01; IF-UI-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-NLI-01; IF-UI-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; V&V T/D; FMEA S/O/D/ RPN TBD	High
IF-VAL-003-03	Parent VAL-003; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from VAL-003 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-NLI-01; IF-UI-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-NLI-01; IF-UI-01; IF-EVD-01; IF-IMG-01; IF-DICOM-01; V&V I/T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-VAL-004-01	Parent VAL-004; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-OPS-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-VAL-004-02	Parent VAL-004; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-OPS-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-VAL-004-03	Parent VAL-004; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from VAL-004 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-OPS-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-OPS-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-VAL-005-01	Parent VAL-005; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-VAL-005-02	Parent VAL-005; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall validate patient and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-VAL-005-03	Parent VAL-005; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from VAL-005 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-PLN-01; IF-TPS-01; IF-DICOM-01; IF-AI-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-VAL-006-01	Parent VAL-006; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; V&V I/A; FMEA S/O/D/ RPN TBD	High
IF-VAL-006-02	Parent VAL-006; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce deterministic authorization....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; V&V T/D; FMEA S/O/D/ RPN TBD	High
IF-VAL-006-03	Parent VAL-006; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from VAL-006 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-VAL-007-01	Parent VAL-007; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-EVD-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-EVD-01; IF-SEC-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-VAL-007-02	Parent VAL-007; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-EVD-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-EVD-01; IF-SEC-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-VAL-007-03	Parent VAL-007; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from VAL-007 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-EVD-01; IF-SEC-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-EVD-01; IF-SEC-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

Continued on next page

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-VAL-008-01	Parent VAL-008; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-PLN-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-PLN-01; V&V I/A; FMEA S/O/D/RPN TBD	High
IF-VAL-008-02	Parent VAL-008; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall present the active....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-PLN-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-PLN-01; V&V T/D; FMEA S/O/D/RPN TBD	High
IF-VAL-008-03	Parent VAL-008; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from VAL-008 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-PLN-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; IF-UI-01; IF-IAM-01; IF-SAF-01; IF-EVD-01; IF-PLN-01; V&V I/T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IF-VAL-009-01	Parent VAL-009; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional content, terminology, units, version....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; V&V I/A; FMEA S/O/D/ RPN TBD	High
IF-VAL-009-02	Parent VAL-009; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance; shall enforce deterministic authorization....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; V&V T/D; FMEA S/O/D/ RPN TBD	High
IF-VAL-009-03	Parent VAL-009; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain a versioned trace from VAL-009 to interface-control documents, implementations, test cases, results, anomalies, monitoring, and release decisions....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, re-release gates, monitoring, change regression, and revalidation. Implement all three child SIRs across IF-VV-01; IF-CFG-01; IF-DATA-01; baseline contract, runtime validation, evidence, reconciliation, and release gating.	T-VV; IF-VV-01; IF-CFG-01; IF-DATA-01; V&V I/T/A; FMEA S/O/D/ RPN TBD	High

8 Sub-interface requirement risk records

8.1 Governance, intended use, and scope (GOV) - 63 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-001-01-01	Parent IF-GOV-001-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-001-01-02	Parent IF-GOV-001-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-001-01-03	Parent IF-GOV-001-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-001-02-01	Parent IF-GOV-001-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-001-02-02	Parent IF-GOV-001-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-GOV-001-02-03	Parent IF-GOV-001-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-001-03-01	Parent IF-GOV-001-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-001-03-02	Parent IF-GOV-001-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-001-03-03	Parent IF-GOV-001-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-002-01-01	Parent IF-GOV-002-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-002-01-02	Parent IF-GOV-002-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-002-01-03	Parent IF-GOV-002-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-002-02-01	Parent IF-GOV-002-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-002-02-02	Parent IF-GOV-002-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-GOV-002-02-03	Parent IF-GOV-002-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-002-03-01	Parent IF-GOV-002-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-002-03-02	Parent IF-GOV-002-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-GOV-002-03-03	Parent IF-GOV-002-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-003-01-01	Parent IF-GOV-003-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-003-01-02	Parent IF-GOV-003-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-003-01-03	Parent IF-GOV-003-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-003-02-01	Parent IF-GOV-003-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-003-02-02	Parent IF-GOV-003-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-GOV-003-02-03	Parent IF-GOV-003-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-003-03-01	Parent IF-GOV-003-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-003-03-02	Parent IF-GOV-003-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-003-03-03	Parent IF-GOV-003-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-004-01-01	Parent IF-GOV-004-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-004-01-02	Parent IF-GOV-004-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-004-01-03	Parent IF-GOV-004-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-004-02-01	Parent IF-GOV-004-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-004-02-02	Parent IF-GOV-004-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-GOV-004-02-03	Parent IF-GOV-004-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-004-03-01	Parent IF-GOV-004-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-004-03-02	Parent IF-GOV-004-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-GOV-004-03-03	Parent IF-GOV-004-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-005-01-01	Parent IF-GOV-005-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-005-01-02	Parent IF-GOV-005-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-005-01-03	Parent IF-GOV-005-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-005-02-01	Parent IF-GOV-005-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-005-02-02	Parent IF-GOV-005-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-GOV-005-02-03	Parent IF-GOV-005-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-005-03-01	Parent IF-GOV-005-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-005-03-02	Parent IF-GOV-005-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-GOV-005-03-03	Parent IF-GOV-005-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-006-01-01	Parent IF-GOV-006-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-006-01-02	Parent IF-GOV-006-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-006-01-03	Parent IF-GOV-006-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-006-02-01	Parent IF-GOV-006-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-006-02-02	Parent IF-GOV-006-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-GOV-006-02-03	Parent IF-GOV-006-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-006-03-01	Parent IF-GOV-006-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-006-03-02	Parent IF-GOV-006-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-GOV-006-03-03	Parent IF-GOV-006-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-007-01-01	Parent IF-GOV-007-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-007-01-02	Parent IF-GOV-007-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-007-01-03	Parent IF-GOV-007-01; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-007-02-01	Parent IF-GOV-007-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-007-02-02	Parent IF-GOV-007-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-GOV-007-02-03	Parent IF-GOV-007-02; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-GOV-007-03-01	Parent IF-GOV-007-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-007-03-02	Parent IF-GOV-007-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-GOV-007-03-03	Parent IF-GOV-007-03; GOV	Intended use, approved scope, policy, ownership, configuration, change, or residual-risk decision is absent, inconsistent, or bypassed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may operate outside the institution's approved clinical, technical, regulatory, or support envelope.	Control intended use and scope, assign accountable authority, baseline policy and configuration, require impact and risk analysis, document residual-risk decisions, and gate activation and change. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-GOV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

8.2 Safety boundaries and human authority (SAF) - 90 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-001-01-01	Parent IF-SAF-001-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-001-01-02	Parent IF-SAF-001-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SAF-001-01-03	Parent IF-SAF-001-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-001-02-01	Parent IF-SAF-001-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-001-02-02	Parent IF-SAF-001-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-001-02-03	Parent IF-SAF-001-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-001-03-01	Parent IF-SAF-001-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-001-03-02	Parent IF-SAF-001-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-001-03-03	Parent IF-SAF-001-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-002-01-01	Parent IF-SAF-002-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-002-01-02	Parent IF-SAF-002-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SAF-002-01-03	Parent IF-SAF-002-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-002-02-01	Parent IF-SAF-002-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-002-02-02	Parent IF-SAF-002-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-002-02-03	Parent IF-SAF-002-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-002-03-01	Parent IF-SAF-002-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-002-03-02	Parent IF-SAF-002-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-002-03-03	Parent IF-SAF-002-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-003-01-01	Parent IF-SAF-003-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-003-01-02	Parent IF-SAF-003-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SAF-003-01-03	Parent IF-SAF-003-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-003-02-01	Parent IF-SAF-003-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-003-02-02	Parent IF-SAF-003-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-003-02-03	Parent IF-SAF-003-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-003-03-01	Parent IF-SAF-003-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-003-03-02	Parent IF-SAF-003-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-003-03-03	Parent IF-SAF-003-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-004-01-01	Parent IF-SAF-004-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-004-01-02	Parent IF-SAF-004-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SAF-004-01-03	Parent IF-SAF-004-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-004-02-01	Parent IF-SAF-004-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-004-02-02	Parent IF-SAF-004-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-004-02-03	Parent IF-SAF-004-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-004-03-01	Parent IF-SAF-004-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-004-03-02	Parent IF-SAF-004-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-004-03-03	Parent IF-SAF-004-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-005-01-01	Parent IF-SAF-005-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-005-01-02	Parent IF-SAF-005-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SAF-005-01-03	Parent IF-SAF-005-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-005-02-01	Parent IF-SAF-005-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-005-02-02	Parent IF-SAF-005-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-005-02-03	Parent IF-SAF-005-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-005-03-01	Parent IF-SAF-005-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-005-03-02	Parent IF-SAF-005-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-005-03-03	Parent IF-SAF-005-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-006-01-01	Parent IF-SAF-006-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-006-01-02	Parent IF-SAF-006-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-006-01-03	Parent IF-SAF-006-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-006-02-01	Parent IF-SAF-006-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-006-02-02	Parent IF-SAF-006-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-006-02-03	Parent IF-SAF-006-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-006-03-01	Parent IF-SAF-006-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-006-03-02	Parent IF-SAF-006-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-006-03-03	Parent IF-SAF-006-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-007-01-01	Parent IF-SAF-007-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-007-01-02	Parent IF-SAF-007-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SAF-007-01-03	Parent IF-SAF-007-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-007-02-01	Parent IF-SAF-007-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-007-02-02	Parent IF-SAF-007-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-007-02-03	Parent IF-SAF-007-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-007-03-01	Parent IF-SAF-007-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-007-03-02	Parent IF-SAF-007-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-007-03-03	Parent IF-SAF-007-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-008-01-01	Parent IF-SAF-008-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-008-01-02	Parent IF-SAF-008-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SAF-008-01-03	Parent IF-SAF-008-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-008-02-01	Parent IF-SAF-008-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-008-02-02	Parent IF-SAF-008-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-008-02-03	Parent IF-SAF-008-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-008-03-01	Parent IF-SAF-008-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-008-03-02	Parent IF-SAF-008-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-008-03-03	Parent IF-SAF-008-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-009-01-01	Parent IF-SAF-009-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-009-01-02	Parent IF-SAF-009-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SAF-009-01-03	Parent IF-SAF-009-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-009-02-01	Parent IF-SAF-009-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-009-02-02	Parent IF-SAF-009-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-009-02-03	Parent IF-SAF-009-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-009-03-01	Parent IF-SAF-009-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-009-03-02	Parent IF-SAF-009-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-009-03-03	Parent IF-SAF-009-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-010-01-01	Parent IF-SAF-010-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-010-01-02	Parent IF-SAF-010-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-010-01-03	Parent IF-SAF-010-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-010-02-01	Parent IF-SAF-010-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-010-02-02	Parent IF-SAF-010-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-SAF-010-02-03	Parent IF-SAF-010-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SAF-010-03-01	Parent IF-SAF-010-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SAF-010-03-02	Parent IF-SAF-010-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SAF-010-03-03	Parent IF-SAF-010-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SYS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

8.3 Natural-language and voice interaction (NLI) - 72 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-001-01-01	Parent IF-NLI-001-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-001-01-02	Parent IF-NLI-001-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-NLI-001-01-03	Parent IF-NLI-001-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-001-02-01	Parent IF-NLI-001-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-NLI-001-02-02	Parent IF-NLI-001-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-NLI-001-02-03	Parent IF-NLI-001-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-001-03-01	Parent IF-NLI-001-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-001-03-02	Parent IF-NLI-001-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-NLI-001-03-03	Parent IF-NLI-001-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-002-01-01	Parent IF-NLI-002-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-002-01-02	Parent IF-NLI-002-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-002-01-03	Parent IF-NLI-002-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-002-02-01	Parent IF-NLI-002-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-NLI-002-02-02	Parent IF-NLI-002-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-NLI-002-02-03	Parent IF-NLI-002-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-002-03-01	Parent IF-NLI-002-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-002-03-02	Parent IF-NLI-002-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-NLI-002-03-03	Parent IF-NLI-002-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-003-01-01	Parent IF-NLI-003-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-003-01-02	Parent IF-NLI-003-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-003-01-03	Parent IF-NLI-003-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-003-02-01	Parent IF-NLI-003-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-NLI-003-02-02	Parent IF-NLI-003-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-NLI-003-02-03	Parent IF-NLI-003-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-003-03-01	Parent IF-NLI-003-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-003-03-02	Parent IF-NLI-003-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-NLI-003-03-03	Parent IF-NLI-003-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-004-01-01	Parent IF-NLI-004-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-004-01-02	Parent IF-NLI-004-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-004-01-03	Parent IF-NLI-004-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-004-02-01	Parent IF-NLI-004-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-NLI-004-02-02	Parent IF-NLI-004-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-NLI-004-02-03	Parent IF-NLI-004-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-004-03-01	Parent IF-NLI-004-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-004-03-02	Parent IF-NLI-004-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-NLI-004-03-03	Parent IF-NLI-004-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-005-01-01	Parent IF-NLI-005-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-005-01-02	Parent IF-NLI-005-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-005-01-03	Parent IF-NLI-005-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-005-02-01	Parent IF-NLI-005-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-NLI-005-02-02	Parent IF-NLI-005-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-NLI-005-02-03	Parent IF-NLI-005-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-005-03-01	Parent IF-NLI-005-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-005-03-02	Parent IF-NLI-005-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-NLI-005-03-03	Parent IF-NLI-005-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-006-01-01	Parent IF-NLI-006-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-006-01-02	Parent IF-NLI-006-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-006-01-03	Parent IF-NLI-006-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-006-02-01	Parent IF-NLI-006-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-NLI-006-02-02	Parent IF-NLI-006-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-NLI-006-02-03	Parent IF-NLI-006-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-006-03-01	Parent IF-NLI-006-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-006-03-02	Parent IF-NLI-006-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-NLI-006-03-03	Parent IF-NLI-006-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-007-01-01	Parent IF-NLI-007-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-007-01-02	Parent IF-NLI-007-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-007-01-03	Parent IF-NLI-007-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-007-02-01	Parent IF-NLI-007-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-NLI-007-02-02	Parent IF-NLI-007-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-NLI-007-02-03	Parent IF-NLI-007-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-007-03-01	Parent IF-NLI-007-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-007-03-02	Parent IF-NLI-007-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-NLI-007-03-03	Parent IF-NLI-007-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-008-01-01	Parent IF-NLI-008-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-008-01-02	Parent IF-NLI-008-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-008-01-03	Parent IF-NLI-008-01; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-008-02-01	Parent IF-NLI-008-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-NLI-008-02-02	Parent IF-NLI-008-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-NLI-008-02-03	Parent IF-NLI-008-02; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

Continued on next page

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-NLI-008-03-01	Parent IF-NLI-008-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-NLI-008-03-02	Parent IF-NLI-008-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-NLI-008-03-03	Parent IF-NLI-008-03; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-NLI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

8.4 Evidence and literature governance (EVD) - 72 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-001-01-01	Parent IF-EVD-001-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-001-01-02	Parent IF-EVD-001-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-001-01-03	Parent IF-EVD-001-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-001-02-01	Parent IF-EVD-001-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-001-02-02	Parent IF-EVD-001-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-EVD-001-02-03	Parent IF-EVD-001-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-001-03-01	Parent IF-EVD-001-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-001-03-02	Parent IF-EVD-001-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-EVD-001-03-03	Parent IF-EVD-001-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-002-01-01	Parent IF-EVD-002-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-002-01-02	Parent IF-EVD-002-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-002-01-03	Parent IF-EVD-002-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-002-02-01	Parent IF-EVD-002-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-002-02-02	Parent IF-EVD-002-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-EVD-002-02-03	Parent IF-EVD-002-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-002-03-01	Parent IF-EVD-002-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-002-03-02	Parent IF-EVD-002-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-EVD-002-03-03	Parent IF-EVD-002-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-003-01-01	Parent IF-EVD-003-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-003-01-02	Parent IF-EVD-003-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-003-01-03	Parent IF-EVD-003-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-003-02-01	Parent IF-EVD-003-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-003-02-02	Parent IF-EVD-003-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-EVD-003-02-03	Parent IF-EVD-003-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-003-03-01	Parent IF-EVD-003-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-003-03-02	Parent IF-EVD-003-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-EVD-003-03-03	Parent IF-EVD-003-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-004-01-01	Parent IF-EVD-004-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-004-01-02	Parent IF-EVD-004-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-004-01-03	Parent IF-EVD-004-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-004-02-01	Parent IF-EVD-004-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-004-02-02	Parent IF-EVD-004-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-EVD-004-02-03	Parent IF-EVD-004-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-004-03-01	Parent IF-EVD-004-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-004-03-02	Parent IF-EVD-004-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-EVD-004-03-03	Parent IF-EVD-004-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-005-01-01	Parent IF-EVD-005-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-005-01-02	Parent IF-EVD-005-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-005-01-03	Parent IF-EVD-005-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-005-02-01	Parent IF-EVD-005-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-005-02-02	Parent IF-EVD-005-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-EVD-005-02-03	Parent IF-EVD-005-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-005-03-01	Parent IF-EVD-005-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-005-03-02	Parent IF-EVD-005-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-EVD-005-03-03	Parent IF-EVD-005-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-006-01-01	Parent IF-EVD-006-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-006-01-02	Parent IF-EVD-006-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-006-01-03	Parent IF-EVD-006-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-006-02-01	Parent IF-EVD-006-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-006-02-02	Parent IF-EVD-006-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-EVD-006-02-03	Parent IF-EVD-006-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-006-03-01	Parent IF-EVD-006-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-006-03-02	Parent IF-EVD-006-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-EVD-006-03-03	Parent IF-EVD-006-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-007-01-01	Parent IF-EVD-007-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-007-01-02	Parent IF-EVD-007-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-007-01-03	Parent IF-EVD-007-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-007-02-01	Parent IF-EVD-007-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-007-02-02	Parent IF-EVD-007-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-EVD-007-02-03	Parent IF-EVD-007-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-007-03-01	Parent IF-EVD-007-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-007-03-02	Parent IF-EVD-007-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-EVD-007-03-03	Parent IF-EVD-007-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-008-01-01	Parent IF-EVD-008-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-EVD-008-01-02	Parent IF-EVD-008-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-008-01-03	Parent IF-EVD-008-01; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-008-02-01	Parent IF-EVD-008-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-EVD-008-02-02	Parent IF-EVD-008-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-EVD-008-02-03	Parent IF-EVD-008-02; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-EVD-008-03-01	Parent IF-EVD-008-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
SIR-EVD-008-03-02	Parent IF-EVD-008-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-EVD-008-03-03	Parent IF-EVD-008-03; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-EVD; RP-RLS; V&V I/A; FMEA S/O/D/RPN TBD	High

8.5 Clinical context, imaging, registration, and contours (CLN) - 81 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-001-01-01	Parent IF-CLN-001-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-001-01-02	Parent IF-CLN-001-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-CLN-001-01-03	Parent IF-CLN-001-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-001-02-01	Parent IF-CLN-001-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-CLN-001-02-02	Parent IF-CLN-001-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-CLN-001-02-03	Parent IF-CLN-001-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-001-03-01	Parent IF-CLN-001-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-001-03-02	Parent IF-CLN-001-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-CLN-001-03-03	Parent IF-CLN-001-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-002-01-01	Parent IF-CLN-002-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-002-01-02	Parent IF-CLN-002-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-CLN-002-01-03	Parent IF-CLN-002-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-002-02-01	Parent IF-CLN-002-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-CLN-002-02-02	Parent IF-CLN-002-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-CLN-002-02-03	Parent IF-CLN-002-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-002-03-01	Parent IF-CLN-002-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-002-03-02	Parent IF-CLN-002-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-CLN-002-03-03	Parent IF-CLN-002-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-003-01-01	Parent IF-CLN-003-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-003-01-02	Parent IF-CLN-003-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-CLN-003-01-03	Parent IF-CLN-003-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-003-02-01	Parent IF-CLN-003-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-CLN-003-02-02	Parent IF-CLN-003-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-CLN-003-02-03	Parent IF-CLN-003-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-003-03-01	Parent IF-CLN-003-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-003-03-02	Parent IF-CLN-003-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-CLN-003-03-03	Parent IF-CLN-003-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-004-01-01	Parent IF-CLN-004-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-004-01-02	Parent IF-CLN-004-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-CLN-004-01-03	Parent IF-CLN-004-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-004-02-01	Parent IF-CLN-004-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-CLN-004-02-02	Parent IF-CLN-004-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-CLN-004-02-03	Parent IF-CLN-004-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-004-03-01	Parent IF-CLN-004-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-004-03-02	Parent IF-CLN-004-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-CLN-004-03-03	Parent IF-CLN-004-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-005-01-01	Parent IF-CLN-005-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-005-01-02	Parent IF-CLN-005-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-CLN-005-01-03	Parent IF-CLN-005-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-005-02-01	Parent IF-CLN-005-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-CLN-005-02-02	Parent IF-CLN-005-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-CLN-005-02-03	Parent IF-CLN-005-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-005-03-01	Parent IF-CLN-005-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-005-03-02	Parent IF-CLN-005-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-CLN-005-03-03	Parent IF-CLN-005-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-006-01-01	Parent IF-CLN-006-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-006-01-02	Parent IF-CLN-006-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-CLN-006-01-03	Parent IF-CLN-006-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-006-02-01	Parent IF-CLN-006-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-CLN-006-02-02	Parent IF-CLN-006-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-CLN-006-02-03	Parent IF-CLN-006-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-006-03-01	Parent IF-CLN-006-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-006-03-02	Parent IF-CLN-006-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-CLN-006-03-03	Parent IF-CLN-006-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-007-01-01	Parent IF-CLN-007-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-007-01-02	Parent IF-CLN-007-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-CLN-007-01-03	Parent IF-CLN-007-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-007-02-01	Parent IF-CLN-007-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-CLN-007-02-02	Parent IF-CLN-007-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-CLN-007-02-03	Parent IF-CLN-007-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-007-03-01	Parent IF-CLN-007-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-007-03-02	Parent IF-CLN-007-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-CLN-007-03-03	Parent IF-CLN-007-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-008-01-01	Parent IF-CLN-008-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-008-01-02	Parent IF-CLN-008-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-CLN-008-01-03	Parent IF-CLN-008-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-008-02-01	Parent IF-CLN-008-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-CLN-008-02-02	Parent IF-CLN-008-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-CLN-008-02-03	Parent IF-CLN-008-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-008-03-01	Parent IF-CLN-008-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-008-03-02	Parent IF-CLN-008-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-CLN-008-03-03	Parent IF-CLN-008-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-009-01-01	Parent IF-CLN-009-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-009-01-02	Parent IF-CLN-009-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-CLN-009-01-03	Parent IF-CLN-009-01; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-009-02-01	Parent IF-CLN-009-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-CLN-009-02-02	Parent IF-CLN-009-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-CLN-009-02-03	Parent IF-CLN-009-02; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-CLN-009-03-01	Parent IF-CLN-009-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-CLN-009-03-02	Parent IF-CLN-009-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-CLN-009-03-03	Parent IF-CLN-009-03; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-IMG; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

8.6 Treatment-plan generation and calculation (PLN) - 108 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-001-01-01	Parent IF-PLN-001-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-001-01-02	Parent IF-PLN-001-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-001-01-03	Parent IF-PLN-001-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-001-02-01	Parent IF-PLN-001-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-001-02-02	Parent IF-PLN-001-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-001-02-03	Parent IF-PLN-001-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-001-03-01	Parent IF-PLN-001-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-001-03-02	Parent IF-PLN-001-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-001-03-03	Parent IF-PLN-001-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-002-01-01	Parent IF-PLN-002-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-002-01-02	Parent IF-PLN-002-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-002-01-03	Parent IF-PLN-002-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-002-02-01	Parent IF-PLN-002-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-002-02-02	Parent IF-PLN-002-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-002-02-03	Parent IF-PLN-002-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-002-03-01	Parent IF-PLN-002-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-002-03-02	Parent IF-PLN-002-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-002-03-03	Parent IF-PLN-002-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-003-01-01	Parent IF-PLN-003-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-003-01-02	Parent IF-PLN-003-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-003-01-03	Parent IF-PLN-003-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-003-02-01	Parent IF-PLN-003-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-003-02-02	Parent IF-PLN-003-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-003-02-03	Parent IF-PLN-003-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-003-03-01	Parent IF-PLN-003-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-003-03-02	Parent IF-PLN-003-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-003-03-03	Parent IF-PLN-003-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-004-01-01	Parent IF-PLN-004-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-004-01-02	Parent IF-PLN-004-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-004-01-03	Parent IF-PLN-004-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-004-02-01	Parent IF-PLN-004-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-004-02-02	Parent IF-PLN-004-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-004-02-03	Parent IF-PLN-004-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-004-03-01	Parent IF-PLN-004-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-004-03-02	Parent IF-PLN-004-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-004-03-03	Parent IF-PLN-004-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-005-01-01	Parent IF-PLN-005-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-005-01-02	Parent IF-PLN-005-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-005-01-03	Parent IF-PLN-005-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-005-02-01	Parent IF-PLN-005-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-005-02-02	Parent IF-PLN-005-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-005-02-03	Parent IF-PLN-005-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-005-03-01	Parent IF-PLN-005-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-005-03-02	Parent IF-PLN-005-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-005-03-03	Parent IF-PLN-005-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-006-01-01	Parent IF-PLN-006-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-006-01-02	Parent IF-PLN-006-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-006-01-03	Parent IF-PLN-006-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-006-02-01	Parent IF-PLN-006-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-006-02-02	Parent IF-PLN-006-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-006-02-03	Parent IF-PLN-006-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-006-03-01	Parent IF-PLN-006-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-006-03-02	Parent IF-PLN-006-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-006-03-03	Parent IF-PLN-006-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-007-01-01	Parent IF-PLN-007-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-007-01-02	Parent IF-PLN-007-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-007-01-03	Parent IF-PLN-007-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-007-02-01	Parent IF-PLN-007-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-007-02-02	Parent IF-PLN-007-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-007-02-03	Parent IF-PLN-007-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-007-03-01	Parent IF-PLN-007-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-007-03-02	Parent IF-PLN-007-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-007-03-03	Parent IF-PLN-007-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-008-01-01	Parent IF-PLN-008-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-008-01-02	Parent IF-PLN-008-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-008-01-03	Parent IF-PLN-008-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-008-02-01	Parent IF-PLN-008-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-008-02-02	Parent IF-PLN-008-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-008-02-03	Parent IF-PLN-008-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-008-03-01	Parent IF-PLN-008-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-008-03-02	Parent IF-PLN-008-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-008-03-03	Parent IF-PLN-008-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-009-01-01	Parent IF-PLN-009-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-009-01-02	Parent IF-PLN-009-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-009-01-03	Parent IF-PLN-009-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-009-02-01	Parent IF-PLN-009-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-009-02-02	Parent IF-PLN-009-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-009-02-03	Parent IF-PLN-009-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-009-03-01	Parent IF-PLN-009-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-009-03-02	Parent IF-PLN-009-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-009-03-03	Parent IF-PLN-009-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-010-01-01	Parent IF-PLN-010-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-010-01-02	Parent IF-PLN-010-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-010-01-03	Parent IF-PLN-010-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-010-02-01	Parent IF-PLN-010-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-010-02-02	Parent IF-PLN-010-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-010-02-03	Parent IF-PLN-010-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-010-03-01	Parent IF-PLN-010-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-010-03-02	Parent IF-PLN-010-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-010-03-03	Parent IF-PLN-010-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-011-01-01	Parent IF-PLN-011-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-011-01-02	Parent IF-PLN-011-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-011-01-03	Parent IF-PLN-011-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-011-02-01	Parent IF-PLN-011-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-011-02-02	Parent IF-PLN-011-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-011-02-03	Parent IF-PLN-011-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-011-03-01	Parent IF-PLN-011-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-011-03-02	Parent IF-PLN-011-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-011-03-03	Parent IF-PLN-011-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-012-01-01	Parent IF-PLN-012-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-012-01-02	Parent IF-PLN-012-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-PLN-012-01-03	Parent IF-PLN-012-01; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-012-02-01	Parent IF-PLN-012-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-PLN-012-02-02	Parent IF-PLN-012-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-PLN-012-02-03	Parent IF-PLN-012-02; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-PLN-012-03-01	Parent IF-PLN-012-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-012-03-02	Parent IF-PLN-012-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-PLN-012-03-03	Parent IF-PLN-012-03; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-PLAN; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

8.7 Plan review, selection, QA, and transfer (REV) - 81 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-001-01-01	Parent IF-REV-001-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-001-01-02	Parent IF-REV-001-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-REV-001-01-03	Parent IF-REV-001-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-001-02-01	Parent IF-REV-001-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-REV-001-02-02	Parent IF-REV-001-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-REV-001-02-03	Parent IF-REV-001-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-001-03-01	Parent IF-REV-001-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-001-03-02	Parent IF-REV-001-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-REV-001-03-03	Parent IF-REV-001-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-002-01-01	Parent IF-REV-002-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-002-01-02	Parent IF-REV-002-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-002-01-03	Parent IF-REV-002-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-002-02-01	Parent IF-REV-002-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-REV-002-02-02	Parent IF-REV-002-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-REV-002-02-03	Parent IF-REV-002-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-002-03-01	Parent IF-REV-002-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-002-03-02	Parent IF-REV-002-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-REV-002-03-03	Parent IF-REV-002-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-003-01-01	Parent IF-REV-003-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-003-01-02	Parent IF-REV-003-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-003-01-03	Parent IF-REV-003-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-003-02-01	Parent IF-REV-003-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-REV-003-02-02	Parent IF-REV-003-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-REV-003-02-03	Parent IF-REV-003-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-003-03-01	Parent IF-REV-003-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
SIR-REV-003-03-02	Parent IF-REV-003-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-REV-003-03-03	Parent IF-REV-003-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-004-01-01	Parent IF-REV-004-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-004-01-02	Parent IF-REV-004-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-REV-004-01-03	Parent IF-REV-004-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-004-02-01	Parent IF-REV-004-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-REV-004-02-02	Parent IF-REV-004-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-REV-004-02-03	Parent IF-REV-004-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-004-03-01	Parent IF-REV-004-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-004-03-02	Parent IF-REV-004-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-REV-004-03-03	Parent IF-REV-004-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-005-01-01	Parent IF-REV-005-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-005-01-02	Parent IF-REV-005-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-005-01-03	Parent IF-REV-005-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-005-02-01	Parent IF-REV-005-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-REV-005-02-02	Parent IF-REV-005-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-REV-005-02-03	Parent IF-REV-005-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-005-03-01	Parent IF-REV-005-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-005-03-02	Parent IF-REV-005-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-REV-005-03-03	Parent IF-REV-005-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-006-01-01	Parent IF-REV-006-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-006-01-02	Parent IF-REV-006-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-006-01-03	Parent IF-REV-006-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-006-02-01	Parent IF-REV-006-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-REV-006-02-02	Parent IF-REV-006-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-REV-006-02-03	Parent IF-REV-006-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-006-03-01	Parent IF-REV-006-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
SIR-REV-006-03-02	Parent IF-REV-006-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-REV-006-03-03	Parent IF-REV-006-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-007-01-01	Parent IF-REV-007-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-007-01-02	Parent IF-REV-007-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-007-01-03	Parent IF-REV-007-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-007-02-01	Parent IF-REV-007-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-REV-007-02-02	Parent IF-REV-007-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-REV-007-02-03	Parent IF-REV-007-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-007-03-01	Parent IF-REV-007-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-007-03-02	Parent IF-REV-007-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-REV-007-03-03	Parent IF-REV-007-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-008-01-01	Parent IF-REV-008-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-008-01-02	Parent IF-REV-008-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-008-01-03	Parent IF-REV-008-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-008-02-01	Parent IF-REV-008-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-REV-008-02-02	Parent IF-REV-008-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-REV-008-02-03	Parent IF-REV-008-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-008-03-01	Parent IF-REV-008-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-008-03-02	Parent IF-REV-008-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-REV-008-03-03	Parent IF-REV-008-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-009-01-01	Parent IF-REV-009-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-009-01-02	Parent IF-REV-009-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-009-01-03	Parent IF-REV-009-01; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-009-02-01	Parent IF-REV-009-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Gate
SIR-REV-009-02-02	Parent IF-REV-009-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-REV-009-02-03	Parent IF-REV-009-02; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-REV-009-03-01	Parent IF-REV-009-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-REV-009-03-02	Parent IF-REV-009-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-REV-009-03-03	Parent IF-REV-009-03; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-REV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

8.8 Data, interoperability, provenance, and audit (DAT) - 90 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-001-01-01	Parent IF-DAT-001-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-001-01-02	Parent IF-DAT-001-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-DAT-001-01-03	Parent IF-DAT-001-01; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-001-02-01	Parent IF-DAT-001-02; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-DAT-001-02-02	Parent IF-DAT-001-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-DAT-001-02-03	Parent IF-DAT-001-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-001-03-01	Parent IF-DAT-001-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-001-03-02	Parent IF-DAT-001-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-DAT-001-03-03	Parent IF-DAT-001-03; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-002-01-01	Parent IF-DAT-002-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-002-01-02	Parent IF-DAT-002-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-DAT-002-01-03	Parent IF-DAT-002-01; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-002-02-01	Parent IF-DAT-002-02; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-DAT-002-02-02	Parent IF-DAT-002-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-DAT-002-02-03	Parent IF-DAT-002-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-002-03-01	Parent IF-DAT-002-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-002-03-02	Parent IF-DAT-002-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-DAT-002-03-03	Parent IF-DAT-002-03; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-003-01-01	Parent IF-DAT-003-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-003-01-02	Parent IF-DAT-003-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-DAT-003-01-03	Parent IF-DAT-003-01; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

Continued on next page

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-003-02-01	Parent IF-DAT-003-02; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-DAT-003-02-02	Parent IF-DAT-003-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-DAT-003-02-03	Parent IF-DAT-003-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-003-03-01	Parent IF-DAT-003-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-003-03-02	Parent IF-DAT-003-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-DAT-003-03-03	Parent IF-DAT-003-03; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-004-01-01	Parent IF-DAT-004-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-004-01-02	Parent IF-DAT-004-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-DAT-004-01-03	Parent IF-DAT-004-01; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-004-02-01	Parent IF-DAT-004-02; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-DAT-004-02-02	Parent IF-DAT-004-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-DAT-004-02-03	Parent IF-DAT-004-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

Continued on next page

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-004-03-01	Parent IF-DAT-004-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-004-03-02	Parent IF-DAT-004-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-DAT-004-03-03	Parent IF-DAT-004-03; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-005-01-01	Parent IF-DAT-005-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-005-01-02	Parent IF-DAT-005-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-DAT-005-01-03	Parent IF-DAT-005-01; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-005-02-01	Parent IF-DAT-005-02; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-DAT-005-02-02	Parent IF-DAT-005-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-DAT-005-02-03	Parent IF-DAT-005-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-005-03-01	Parent IF-DAT-005-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-005-03-02	Parent IF-DAT-005-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-DAT-005-03-03	Parent IF-DAT-005-03; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-006-01-01	Parent IF-DAT-006-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-006-01-02	Parent IF-DAT-006-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-DAT-006-01-03	Parent IF-DAT-006-01; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-006-02-01	Parent IF-DAT-006-02; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-DAT-006-02-02	Parent IF-DAT-006-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-DAT-006-02-03	Parent IF-DAT-006-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-006-03-01	Parent IF-DAT-006-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-006-03-02	Parent IF-DAT-006-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-DAT-006-03-03	Parent IF-DAT-006-03; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-007-01-01	Parent IF-DAT-007-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-007-01-02	Parent IF-DAT-007-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-DAT-007-01-03	Parent IF-DAT-007-01; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-007-02-01	Parent IF-DAT-007-02; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-DAT-007-02-02	Parent IF-DAT-007-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-DAT-007-02-03	Parent IF-DAT-007-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-007-03-01	Parent IF-DAT-007-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-007-03-02	Parent IF-DAT-007-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-DAT-007-03-03	Parent IF-DAT-007-03; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-008-01-01	Parent IF-DAT-008-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-008-01-02	Parent IF-DAT-008-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-DAT-008-01-03	Parent IF-DAT-008-01; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-008-02-01	Parent IF-DAT-008-02; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-DAT-008-02-02	Parent IF-DAT-008-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-DAT-008-02-03	Parent IF-DAT-008-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-008-03-01	Parent IF-DAT-008-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-008-03-02	Parent IF-DAT-008-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-DAT-008-03-03	Parent IF-DAT-008-03; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-009-01-01	Parent IF-DAT-009-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-009-01-02	Parent IF-DAT-009-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-DAT-009-01-03	Parent IF-DAT-009-01; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-009-02-01	Parent IF-DAT-009-02; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-DAT-009-02-02	Parent IF-DAT-009-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-DAT-009-02-03	Parent IF-DAT-009-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-009-03-01	Parent IF-DAT-009-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-009-03-02	Parent IF-DAT-009-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-DAT-009-03-03	Parent IF-DAT-009-03; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-010-01-01	Parent IF-DAT-010-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-010-01-02	Parent IF-DAT-010-01; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-DAT-010-01-03	Parent IF-DAT-010-01; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-010-02-01	Parent IF-DAT-010-02; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Gate
SIR-DAT-010-02-02	Parent IF-DAT-010-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Gate
SIR-DAT-010-02-03	Parent IF-DAT-010-02; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Gate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-DAT-010-03-01	Parent IF-DAT-010-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-DAT-010-03-02	Parent IF-DAT-010-03; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-DAT-010-03-03	Parent IF-DAT-010-03; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-DATA; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

8.9 AI and inference-model lifecycle (AIM) - 72 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-001-01-01	Parent IF-AIM-001-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-001-01-02	Parent IF-AIM-001-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-001-01-03	Parent IF-AIM-001-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-001-02-01	Parent IF-AIM-001-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-001-02-02	Parent IF-AIM-001-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-TRX; V&V T/D; FMEA S/O/D/ RPN TBD	High
SIR-AIM-001-02-03	Parent IF-AIM-001-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-OBS; V&V T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-001-03-01	Parent IF-AIM-001-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-001-03-02	Parent IF-AIM-001-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-001-03-03	Parent IF-AIM-001-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-002-01-01	Parent IF-AIM-002-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-002-01-02	Parent IF-AIM-002-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-002-01-03	Parent IF-AIM-002-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-002-02-01	Parent IF-AIM-002-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-002-02-02	Parent IF-AIM-002-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-TRX; V&V T/D; FMEA S/O/D/ RPN TBD	High
SIR-AIM-002-02-03	Parent IF-AIM-002-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-OBS; V&V T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-002-03-01	Parent IF-AIM-002-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-002-03-02	Parent IF-AIM-002-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-002-03-03	Parent IF-AIM-002-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-003-01-01	Parent IF-AIM-003-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-003-01-02	Parent IF-AIM-003-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-003-01-03	Parent IF-AIM-003-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-003-02-01	Parent IF-AIM-003-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-003-02-02	Parent IF-AIM-003-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-TRX; V&V T/D; FMEA S/O/D/ RPN TBD	High
SIR-AIM-003-02-03	Parent IF-AIM-003-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-OBS; V&V T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-003-03-01	Parent IF-AIM-003-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-003-03-02	Parent IF-AIM-003-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-003-03-03	Parent IF-AIM-003-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-004-01-01	Parent IF-AIM-004-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-004-01-02	Parent IF-AIM-004-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-004-01-03	Parent IF-AIM-004-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-004-02-01	Parent IF-AIM-004-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-004-02-02	Parent IF-AIM-004-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-TRX; V&V T/D; FMEA S/O/D/ RPN TBD	High
SIR-AIM-004-02-03	Parent IF-AIM-004-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-OBS; V&V T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-004-03-01	Parent IF-AIM-004-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-004-03-02	Parent IF-AIM-004-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-004-03-03	Parent IF-AIM-004-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-005-01-01	Parent IF-AIM-005-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-005-01-02	Parent IF-AIM-005-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-005-01-03	Parent IF-AIM-005-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-005-02-01	Parent IF-AIM-005-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-005-02-02	Parent IF-AIM-005-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-TRX; V&V T/D; FMEA S/O/D/ RPN TBD	High
SIR-AIM-005-02-03	Parent IF-AIM-005-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-OBS; V&V T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-005-03-01	Parent IF-AIM-005-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-005-03-02	Parent IF-AIM-005-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-005-03-03	Parent IF-AIM-005-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-006-01-01	Parent IF-AIM-006-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-006-01-02	Parent IF-AIM-006-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-006-01-03	Parent IF-AIM-006-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-006-02-01	Parent IF-AIM-006-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-006-02-02	Parent IF-AIM-006-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-TRX; V&V T/D; FMEA S/O/D/ RPN TBD	High
SIR-AIM-006-02-03	Parent IF-AIM-006-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-OBS; V&V T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-006-03-01	Parent IF-AIM-006-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-006-03-02	Parent IF-AIM-006-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-006-03-03	Parent IF-AIM-006-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-007-01-01	Parent IF-AIM-007-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-007-01-02	Parent IF-AIM-007-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-007-01-03	Parent IF-AIM-007-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-007-02-01	Parent IF-AIM-007-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-007-02-02	Parent IF-AIM-007-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-TRX; V&V T/D; FMEA S/O/D/ RPN TBD	High
SIR-AIM-007-02-03	Parent IF-AIM-007-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-OBS; V&V T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-007-03-01	Parent IF-AIM-007-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-007-03-02	Parent IF-AIM-007-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-007-03-03	Parent IF-AIM-007-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-008-01-01	Parent IF-AIM-008-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-008-01-02	Parent IF-AIM-008-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-008-01-03	Parent IF-AIM-008-01; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-008-02-01	Parent IF-AIM-008-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-AIM-008-02-02	Parent IF-AIM-008-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-TRX; V&V T/D; FMEA S/O/D/ RPN TBD	High
SIR-AIM-008-02-03	Parent IF-AIM-008-02; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-OBS; V&V T/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-AIM-008-03-01	Parent IF-AIM-008-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-008-03-02	Parent IF-AIM-008-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-AIM-008-03-03	Parent IF-AIM-008-03; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-AI; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

8.10 Human factors, usability, and competency (HFE) - 54 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-001-01-01	Parent IF-HFE-001-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-001-01-02	Parent IF-HFE-001-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-001-01-03	Parent IF-HFE-001-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-001-02-01	Parent IF-HFE-001-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-001-02-02	Parent IF-HFE-001-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-001-02-03	Parent IF-HFE-001-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-001-03-01	Parent IF-HFE-001-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-001-03-02	Parent IF-HFE-001-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-001-03-03	Parent IF-HFE-001-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-002-01-01	Parent IF-HFE-002-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-002-01-02	Parent IF-HFE-002-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-002-01-03	Parent IF-HFE-002-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-002-02-01	Parent IF-HFE-002-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-002-02-02	Parent IF-HFE-002-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-002-02-03	Parent IF-HFE-002-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-002-03-01	Parent IF-HFE-002-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-002-03-02	Parent IF-HFE-002-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-002-03-03	Parent IF-HFE-002-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-003-01-01	Parent IF-HFE-003-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-003-01-02	Parent IF-HFE-003-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-003-01-03	Parent IF-HFE-003-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-003-02-01	Parent IF-HFE-003-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-003-02-02	Parent IF-HFE-003-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-003-02-03	Parent IF-HFE-003-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-003-03-01	Parent IF-HFE-003-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-003-03-02	Parent IF-HFE-003-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-003-03-03	Parent IF-HFE-003-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-004-01-01	Parent IF-HFE-004-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-004-01-02	Parent IF-HFE-004-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-004-01-03	Parent IF-HFE-004-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-004-02-01	Parent IF-HFE-004-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-004-02-02	Parent IF-HFE-004-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-004-02-03	Parent IF-HFE-004-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-004-03-01	Parent IF-HFE-004-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-004-03-02	Parent IF-HFE-004-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-004-03-03	Parent IF-HFE-004-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-005-01-01	Parent IF-HFE-005-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-005-01-02	Parent IF-HFE-005-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-005-01-03	Parent IF-HFE-005-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-005-02-01	Parent IF-HFE-005-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-005-02-02	Parent IF-HFE-005-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-005-02-03	Parent IF-HFE-005-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-005-03-01	Parent IF-HFE-005-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-005-03-02	Parent IF-HFE-005-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-005-03-03	Parent IF-HFE-005-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-006-01-01	Parent IF-HFE-006-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-006-01-02	Parent IF-HFE-006-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-SCH; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-006-01-03	Parent IF-HFE-006-01; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

Continued on next page

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-006-02-01	Parent IF-HFE-006-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-006-02-02	Parent IF-HFE-006-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-006-02-03	Parent IF-HFE-006-02; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-HFE-006-03-01	Parent IF-HFE-006-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-HFE-006-03-02	Parent IF-HFE-006-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-HFE-006-03-03	Parent IF-HFE-006-03; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-UX; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

8.11 Privacy, cybersecurity, and secure development (SEC) - 72 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-001-01-01	Parent IF-SEC-001-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-001-01-02	Parent IF-SEC-001-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-001-01-03	Parent IF-SEC-001-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-001-02-01	Parent IF-SEC-001-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-001-02-02	Parent IF-SEC-001-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-SEC-001-02-03	Parent IF-SEC-001-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-001-03-01	Parent IF-SEC-001-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-001-03-02	Parent IF-SEC-001-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SEC-001-03-03	Parent IF-SEC-001-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-002-01-01	Parent IF-SEC-002-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-ICD; V&V I/A; FMEA S/O/D/RPN TBD	High
SIR-SEC-002-01-02	Parent IF-SEC-002-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-SCH; V&V I/T; FMEA S/O/D/RPN TBD	High
SIR-SEC-002-01-03	Parent IF-SEC-002-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-002-02-01	Parent IF-SEC-002-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-002-02-02	Parent IF-SEC-002-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-SEC-002-02-03	Parent IF-SEC-002-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-002-03-01	Parent IF-SEC-002-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-002-03-02	Parent IF-SEC-002-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-002-03-03	Parent IF-SEC-002-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-003-01-01	Parent IF-SEC-003-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-003-01-02	Parent IF-SEC-003-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-003-01-03	Parent IF-SEC-003-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-003-02-01	Parent IF-SEC-003-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-003-02-02	Parent IF-SEC-003-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-SEC-003-02-03	Parent IF-SEC-003-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-003-03-01	Parent IF-SEC-003-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-003-03-02	Parent IF-SEC-003-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-003-03-03	Parent IF-SEC-003-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-004-01-01	Parent IF-SEC-004-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-ICD; V&V I/A; FMEA S/O/D/RPN TBD	High
SIR-SEC-004-01-02	Parent IF-SEC-004-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-SCH; V&V I/T; FMEA S/O/D/RPN TBD	High
SIR-SEC-004-01-03	Parent IF-SEC-004-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-004-02-01	Parent IF-SEC-004-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-004-02-02	Parent IF-SEC-004-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-SEC-004-02-03	Parent IF-SEC-004-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-004-03-01	Parent IF-SEC-004-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-004-03-02	Parent IF-SEC-004-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-004-03-03	Parent IF-SEC-004-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-005-01-01	Parent IF-SEC-005-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-005-01-02	Parent IF-SEC-005-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-005-01-03	Parent IF-SEC-005-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-005-02-01	Parent IF-SEC-005-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-005-02-02	Parent IF-SEC-005-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-SEC-005-02-03	Parent IF-SEC-005-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-005-03-01	Parent IF-SEC-005-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
SIR-SEC-005-03-02	Parent IF-SEC-005-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SEC-005-03-03	Parent IF-SEC-005-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-006-01-01	Parent IF-SEC-006-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-006-01-02	Parent IF-SEC-006-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-006-01-03	Parent IF-SEC-006-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-006-02-01	Parent IF-SEC-006-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-006-02-02	Parent IF-SEC-006-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-SEC-006-02-03	Parent IF-SEC-006-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-006-03-01	Parent IF-SEC-006-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-006-03-02	Parent IF-SEC-006-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SEC-006-03-03	Parent IF-SEC-006-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-007-01-01	Parent IF-SEC-007-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-007-01-02	Parent IF-SEC-007-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-007-01-03	Parent IF-SEC-007-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-007-02-01	Parent IF-SEC-007-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-007-02-02	Parent IF-SEC-007-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-SEC-007-02-03	Parent IF-SEC-007-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-007-03-01	Parent IF-SEC-007-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V I/A; FMEA S/O/D/RPN TBD	High
SIR-SEC-007-03-02	Parent IF-SEC-007-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SEC-007-03-03	Parent IF-SEC-007-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-008-01-01	Parent IF-SEC-008-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-008-01-02	Parent IF-SEC-008-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-008-01-03	Parent IF-SEC-008-01; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-008-02-01	Parent IF-SEC-008-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-SEC-008-02-02	Parent IF-SEC-008-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-SEC-008-02-03	Parent IF-SEC-008-02; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-SEC-008-03-01	Parent IF-SEC-008-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-SEC-008-03-02	Parent IF-SEC-008-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-SEC-008-03-03	Parent IF-SEC-008-03; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-SEC; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

8.12 Clinical operations, monitoring, and resilience (OPS) - 63 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-001-01-01	Parent IF-OPS-001-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-001-01-02	Parent IF-OPS-001-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-001-01-03	Parent IF-OPS-001-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-001-02-01	Parent IF-OPS-001-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-001-02-02	Parent IF-OPS-001-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-001-02-03	Parent IF-OPS-001-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-001-03-01	Parent IF-OPS-001-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-001-03-02	Parent IF-OPS-001-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-001-03-03	Parent IF-OPS-001-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-002-01-01	Parent IF-OPS-002-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-002-01-02	Parent IF-OPS-002-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-002-01-03	Parent IF-OPS-002-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-002-02-01	Parent IF-OPS-002-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-002-02-02	Parent IF-OPS-002-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-002-02-03	Parent IF-OPS-002-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-002-03-01	Parent IF-OPS-002-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-002-03-02	Parent IF-OPS-002-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-002-03-03	Parent IF-OPS-002-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-003-01-01	Parent IF-OPS-003-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-003-01-02	Parent IF-OPS-003-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-003-01-03	Parent IF-OPS-003-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-003-02-01	Parent IF-OPS-003-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-003-02-02	Parent IF-OPS-003-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-003-02-03	Parent IF-OPS-003-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-003-03-01	Parent IF-OPS-003-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-003-03-02	Parent IF-OPS-003-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-003-03-03	Parent IF-OPS-003-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-004-01-01	Parent IF-OPS-004-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-004-01-02	Parent IF-OPS-004-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-004-01-03	Parent IF-OPS-004-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-004-02-01	Parent IF-OPS-004-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-004-02-02	Parent IF-OPS-004-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-004-02-03	Parent IF-OPS-004-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-004-03-01	Parent IF-OPS-004-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-004-03-02	Parent IF-OPS-004-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-004-03-03	Parent IF-OPS-004-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-005-01-01	Parent IF-OPS-005-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-005-01-02	Parent IF-OPS-005-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-005-01-03	Parent IF-OPS-005-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-005-02-01	Parent IF-OPS-005-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-005-02-02	Parent IF-OPS-005-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-005-02-03	Parent IF-OPS-005-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-005-03-01	Parent IF-OPS-005-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-005-03-02	Parent IF-OPS-005-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-005-03-03	Parent IF-OPS-005-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-006-01-01	Parent IF-OPS-006-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-006-01-02	Parent IF-OPS-006-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-006-01-03	Parent IF-OPS-006-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-006-02-01	Parent IF-OPS-006-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-006-02-02	Parent IF-OPS-006-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-006-02-03	Parent IF-OPS-006-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-006-03-01	Parent IF-OPS-006-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-006-03-02	Parent IF-OPS-006-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-006-03-03	Parent IF-OPS-006-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-007-01-01	Parent IF-OPS-007-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-007-01-02	Parent IF-OPS-007-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-007-01-03	Parent IF-OPS-007-01; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-007-02-01	Parent IF-OPS-007-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VAL; V&V I/T; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-007-02-02	Parent IF-OPS-007-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-007-02-03	Parent IF-OPS-007-02; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	Moderate

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-OPS-007-03-01	Parent IF-OPS-007-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate
SIR-OPS-007-03-02	Parent IF-OPS-007-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	Moderate
SIR-OPS-007-03-03	Parent IF-OPS-007-03; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-OPS; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	Moderate

8.13 Verification, commissioning, release, and change control (VAL) - 81 records

Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-001-01-01	Parent IF-VAL-001-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-001-01-02	Parent IF-VAL-001-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-001-01-03	Parent IF-VAL-001-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or re-release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-001-02-01	Parent IF-VAL-001-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-001-02-02	Parent IF-VAL-001-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-VAL-001-02-03	Parent IF-VAL-001-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-001-03-01	Parent IF-VAL-001-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-001-03-02	Parent IF-VAL-001-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-001-03-03	Parent IF-VAL-001-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-002-01-01	Parent IF-VAL-002-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-002-01-02	Parent IF-VAL-002-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-002-01-03	Parent IF-VAL-002-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-002-02-01	Parent IF-VAL-002-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-002-02-02	Parent IF-VAL-002-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-VAL-002-02-03	Parent IF-VAL-002-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-002-03-01	Parent IF-VAL-002-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-002-03-02	Parent IF-VAL-002-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-002-03-03	Parent IF-VAL-002-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-003-01-01	Parent IF-VAL-003-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-003-01-02	Parent IF-VAL-003-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-003-01-03	Parent IF-VAL-003-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-003-02-01	Parent IF-VAL-003-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-003-02-02	Parent IF-VAL-003-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-VAL-003-02-03	Parent IF-VAL-003-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-003-03-01	Parent IF-VAL-003-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-003-03-02	Parent IF-VAL-003-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V T/A; FMEA S/O/D/RPN TBD	High
SIR-VAL-003-03-03	Parent IF-VAL-003-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-004-01-01	Parent IF-VAL-004-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-004-01-02	Parent IF-VAL-004-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-004-01-03	Parent IF-VAL-004-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-004-02-01	Parent IF-VAL-004-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-004-02-02	Parent IF-VAL-004-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-VAL-004-02-03	Parent IF-VAL-004-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-004-03-01	Parent IF-VAL-004-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-004-03-02	Parent IF-VAL-004-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-004-03-03	Parent IF-VAL-004-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-005-01-01	Parent IF-VAL-005-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-005-01-02	Parent IF-VAL-005-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-005-01-03	Parent IF-VAL-005-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-005-02-01	Parent IF-VAL-005-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-005-02-02	Parent IF-VAL-005-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-VAL-005-02-03	Parent IF-VAL-005-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-005-03-01	Parent IF-VAL-005-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-005-03-02	Parent IF-VAL-005-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-005-03-03	Parent IF-VAL-005-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-006-01-01	Parent IF-VAL-006-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-006-01-02	Parent IF-VAL-006-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-006-01-03	Parent IF-VAL-006-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-006-02-01	Parent IF-VAL-006-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-006-02-02	Parent IF-VAL-006-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-VAL-006-02-03	Parent IF-VAL-006-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-006-03-01	Parent IF-VAL-006-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-006-03-02	Parent IF-VAL-006-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-006-03-03	Parent IF-VAL-006-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-007-01-01	Parent IF-VAL-007-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-007-01-02	Parent IF-VAL-007-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-007-01-03	Parent IF-VAL-007-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-007-02-01	Parent IF-VAL-007-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-007-02-02	Parent IF-VAL-007-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-VAL-007-02-03	Parent IF-VAL-007-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-007-03-01	Parent IF-VAL-007-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-007-03-02	Parent IF-VAL-007-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-007-03-03	Parent IF-VAL-007-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-008-01-01	Parent IF-VAL-008-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-008-01-02	Parent IF-VAL-008-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-008-01-03	Parent IF-VAL-008-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-008-02-01	Parent IF-VAL-008-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-008-02-02	Parent IF-VAL-008-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-VAL-008-02-03	Parent IF-VAL-008-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-008-03-01	Parent IF-VAL-008-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-008-03-02	Parent IF-VAL-008-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-008-03-03	Parent IF-VAL-008-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-009-01-01	Parent IF-VAL-009-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind the parent scope to....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-ICD; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-ICD; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-009-01-02	Parent IF-VAL-009-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and invalid examples, field type....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-SCH; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-SCH; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-009-01-03	Parent IF-VAL-009-01; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: place each boundary record, schema or object profile, example set, dependency and compatibility matrix, limitation, owner approval, and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-009-02-01	Parent IF-VAL-009-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request or object, the gate....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VAL; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VAL; V&V I/T; FMEA S/O/D/ RPN TBD	High
SIR-VAL-009-02-02	Parent IF-VAL-009-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic state transition, and is....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-TRX; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-TRX; V&V T/D; FMEA S/O/D/RPN TBD	High
SIR-VAL-009-02-03	Parent IF-VAL-009-02; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input and output identities and....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-OBS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-OBS; V&V T/A; FMEA S/O/D/RPN TBD	High

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Entity	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
SIR-VAL-009-03-01	Parent IF-VAL-009-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family sub-profile into objective inspection, analysis, demonstration....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V I/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-009-03-02	Parent IF-VAL-009-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved compatibility matrix. Signed results....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-VVP; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-VVP; V&V T/A; FMEA S/O/D/ RPN TBD	High
SIR-VAL-009-03-03	Parent IF-VAL-009-03; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete, a required result failed....	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Implement package RP-RLS; test producer, consumer, integrated route, wrong-context, incompatibility, timeout, partial, recovery, and misuse behavior as applicable.	T-VV; RP-RLS; V&V I/A; FMEA S/O/D/ RPN TBD	High

9 Functional component risk records

9.1 Functional component - 41 records

Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-GOV-01	Program component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Owns intended use, scope, residual-risk acceptance, evidence policy, clinical authority, and release decisions.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-GOV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-QMS-01	Program component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls documents, changes, deviations, suppliers, incidents, CAPA, approvals, and quality records.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-QMS-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-REQ-01	Program component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Maintains bidirectional links from ConOps through requirements, hazards, components, tests, results, anomalies, and releases.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-REQ-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-ARCH-01	Program component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls system boundaries, service contracts, data ownership, interface-control documents, and component compatibility.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give C-ARCH-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-IAM-01	Control component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides authentication, role and context authorization, separation of duties, approval identity, and access review evidence.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-IAM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
C-SAFE-01	Control component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Evaluates authorization, context, scope, preconditions, hard stops, approval validity, and export eligibility independently of generative models.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-SAFE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-STATE-01	Control component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls draft, review, approval, QA, transfer, degraded, invalidated, and superseded states with atomic transitions.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-STATE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-UX-01	Experience component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Presents context, provenance, warnings, previews, comparisons, approvals, and human-controlled actions.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-UX-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-UX; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-HFE-01	Experience component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Defines critical tasks, formative and summative validation, role curricula, competency, renewal, and use-safety evidence.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give C-HFE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-UX; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-VOICE-01	Experience component; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides bounded speech capture, transcription, safety-critical token flagging, correction, and confirmed transcript output.	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Give C-VOICE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-NLI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-INTENT-01	Experience component; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Converts text or confirmed speech into versioned typed intent and validates units, negation, laterality, objects, and required fields.	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Give C-INTENT-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-NLI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-WF-01	Orchestration component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Coordinates governed tasks, human gates, clinical services, external TPS interactions, QA routing, cancellation, and recovery.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give C-WF-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-EVID-01	Clinical service component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Stores and retrieves institution-approved, versioned, applicable evidence without unrestricted clinical-mode internet retrieval.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-EVID-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-EVD; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-RULE-01	Clinical service component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Evaluates rule applicability, units, authority order, conflicts, lifecycle state, and no-applicable-rule outcomes.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-RULE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-EVD; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-CASE-01	Clinical service component; PAT	Wrong, stale, or incomplete patient, course, study, plan, or action context is accepted. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Binds patient, course, studies, frames, prescriptions, anatomy, prior treatment, task, mode, and destination into an immutable case context.	Work may be applied to the wrong patient or object, or valid downstream approvals may become unsafe.	Bind authenticated actor and immutable case context, confirm two patient identifiers for consequential actions, reject mismatches, and invalidate stale approvals. Give C-CASE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-IMG; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-REG-01	Clinical service component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Invokes commissioned registration and accumulation methods and retains transforms, visualization, uncertainty, approval, and lineage.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give C-REG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-IMG; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
C-SEG-01	Clinical service component; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls contour-model applicability, draft generation, uncertainty, validation, editing, review, approval, and nomenclature.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Give C-SEG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-IMG; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
C-PLAN-01	Clinical service component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Builds structured planning intent and coordinates commissioned optimizers, dose services, robustness, diversity, and candidate lineage.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give C-PLAN-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-PLAN; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-DOSE-01	Clinical service component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides a controlled adapter to commissioned planning, optimization, robustness, and dose engines without generative dose calculation.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give C-DOSE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-PLAN; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
C-REV-01	Clinical service component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Presents consistent metrics, tradeoffs, spatial dose, robustness, uncertainty, provenance, comments, and role-specific decisions.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give C-REV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-REV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
C-QA-01	Clinical service component; ASSUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Routes independent calculation, patient-specific QA, transfer QA, technique checks, and physics review while preserving independence.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Give C-QA-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-REV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-DICOM-01	Interface component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides approved DICOM query, retrieve, storage, import, export, and routing for imaging and radiotherapy objects at external system....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-DICOM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
C-DICOM-02	Interface component; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Validates identities, UIDs, references, frames, geometry, units, required attributes, completeness, and source-to-destination fidelity.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Give C-DICOM-02 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
C-TPS-01	Interface component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Encapsulates each commissioned external TPS, its DICOM/DICOM-RT conformance statement, supported workflows, optional approved API, limitations, and site configuration.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-TPS-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-FHIR-01	Interface component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Implements approved profiles or guides for workflow and non-DICOM clinical exchanges where deployment scope requires them.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-FHIR-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-TERM-01	Interface component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls codes, structure names, units, dose basis, coordinate systems, mappings, conversions, and rounding rules.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give C-TERM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
C-OBJ-01	Data component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Stores immutable object versions, lifecycle state, ownership, integrity values, approvals, and source linkage.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-OBJ-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-PROV-01	Data component; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Traces derived objects to inputs, transformations, software, models, evidence, configurations, users, approvals, and downstream impact.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Give C-PROV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-AUD-01	Data component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Records ordered commands, context, evidence, versions, outputs, warnings, edits, approvals, QA, transfers, failures, overrides, and reconciliation.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-AUD-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-CFG-01	Data component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Locks approved scope, modes, rules, thresholds, services, models, machines, interfaces, operating responses, and release manifests.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-CFG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-MGW-01	AI lifecycle component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Resolves approved locked models, validates intended use and input contracts, executes inference, and enforces abstention and routing.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-MGW-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-AI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-MREG-01	AI lifecycle component; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls model versions, provenance, intended use, artifacts, signatures, configurations, limitations, dependencies, and software bills of materials.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Give C-MREG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-AI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-MVAL-01	AI lifecycle component; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Manages evaluation datasets, subgroup and calibration measures, failure detection, edit burden, thresholds, drift, and change regression.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Give C-MVAL-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-AI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-SEC-01	Platform component; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides identity federation, encryption, segmentation, secrets, egress control, artifact verification, detection, and privacy safeguards.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Give C-SEC-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-MON-01	Platform component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Calculates safety, model, evidence, workflow, reliability, subgroup, and security signals and invokes approved actions.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-MON-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-JOB-01	Platform component; GEN	The required behavior or component responsibility is absent, incomplete, inconsistent, bypassed, or not auditable. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls pending, running, complete, failed, canceled, timed-out, partial, commit, cleanup, and reconciliation states.	The parent requirement may not be achieved and a dependent clinical or operational control may fail without timely detection.	Define an accountable contract and owner, implement deterministic preconditions and explicit outcomes, retain evidence, test positive and failure paths, monitor performance, and gate release. Give C-JOB-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-RES-01	Platform component; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides validated backups, staged restoration, downtime and read-only modes, known-good rollback, and reconciliation.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Give C-RES-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate
C-INC-01	Platform component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Supports reporting, triage, containment, patient-impact assessment, investigation, evidence preservation, trends, CAPA, and closure.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give C-INC-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-VV-01	Assurance component; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Executes inspection, analysis, test, demonstration, HFE, site acceptance, commissioning, regression, shadow, and end-to-end protocols.	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Give C-VV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
C-TEST-01	Assurance component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls representative, boundary, wrong-context, failure, adversarial, recovery, interoperability, and clinical challenge cases with expected results.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give C-TEST-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
C-SBX-01	Assurance component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Separates research, training, commissioning, shadow, clinical draft, approved, degraded, and read-only execution and destinations.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give C-SBX-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

10 Interface component risk records

10.1 Interface component - 57 records

Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-BND-01	Control component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Creates the canonical identity for every human, service, file, message, DICOM, DICOMweb, IHE, and vendor-API producer-consumer route and....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give IC-BND-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-ICD-01	Control component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Versions and approves the interface-control record, diagrams, responsibility boundaries, authority limits, normal and degraded behavior, and references to....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-ICD-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-CAP-01	Control component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Resolves the approved producer, consumer, vendor, version, site, machine, modality, technique, SOP Class, role, transfer syntax, profile, API....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-CAP-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-OWN-01	Control component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Assigns accountable producer, consumer, clinical, systems, data, security, operations, supplier, support, and independent-verification authorities for each route and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-OWN-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
IC-CFG-01	Control component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Publishes signed effective route, endpoint, trust, schema, terminology, timeout, retry, monitoring, feature, scope, and dependency configuration and prevents....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give IC-CFG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-RLS-01	Control component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Allows activation or continued use only when required approvals, compatibility, risk controls, tests, commissioning, limitations, support, monitoring, rollback....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-RLS-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-CHG-01	Control component; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Detects contract or endpoint changes, performs impact analysis, defines migration and regression scope, coordinates staged activation, and restores....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Give IC-CHG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/RPN TBD	High
IC-SCH-01	Contract component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Stores machine-validatable message, API, event, file, DICOM object, DICOMweb resource, and IHE content profiles with cardinality, conditionality, constraints....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give IC-SCH-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/RPN TBD	Gate
IC-SEM-01	Contract component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls clinical meaning, terminology, nomenclature, units, dose basis, coordinate systems, rounding, state meaning, authority, provenance, and cross-system mappings....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give IC-SEM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-OBJ-01	Contract component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Defines required identifiers, references, lifecycle fields, approval state, provenance, integrity, status, and family-specific content for each exchanged object....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-OBJ-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-EXM-01	Contract component; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Maintains controlled valid, invalid, boundary, wrong-context, stale-version, private-attribute, partial, and recovery examples linked to the exact contract and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Give IC-EXM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate
IC-VER-01	Contract component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Negotiates or rejects contract, profile, model, configuration, endpoint, and dependency versions against the approved matrix and prohibits unapproved....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-VER-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-GTW-01	Runtime component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Terminates the first trusted boundary, authenticates the route, applies allowlists and size or rate controls, establishes correlation, and....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give IC-GTW-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-CTX-01	Runtime component; PAT	Wrong, stale, or incomplete patient, course, study, plan, or action context is accepted. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Validates user, patient, course, study, task, object, plan, frame, operating mode, destination, source, freshness, and approval context and....	Work may be applied to the wrong patient or object, or valid downstream approvals may become unsafe.	Bind authenticated actor and immutable case context, confirm two patient identifiers for consequential actions, reject mismatches, and invalidate stale approvals. Give IC-CTX-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-IMG; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
IC-REF-01	Runtime component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Validates identifiers, UIDs, reference graphs, frames, transforms, dimensions, orientation, spacing, structure linkage, dose quantity and basis, units, precision....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give IC-REF-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-POL-01	Runtime component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Evaluates human and service identity, role, purpose, case access, operating mode, intended-use scope, separation of duties, approval authority....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-POL-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
IC-SAF-01	Runtime component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Enforces deterministic preconditions, allowlisted transitions, hard stops, draft and approved separation, invalidation, export eligibility, and fail-closed behavior independently....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-SAF-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
IC-INT-01	Runtime component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Checks hashes or signatures where applicable, object completeness, replay and staleness, configuration and model identity, source authenticity, and....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give IC-INT-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-QRT-01	Runtime component; GEN	The required behavior or component responsibility is absent, incomplete, inconsistent, bypassed, or not auditable. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Isolates malformed, incompatible, orphaned, partial, conflicting, suspicious, or unacknowledged content with reason, ownership, retention, investigation, and controlled disposition.	The parent requirement may not be achieved and a dependent clinical or operational control may fail without timely detection.	Define an accountable contract and owner, implement deterministic preconditions and explicit outcomes, retain evidence, test positive and failure paths, monitor performance, and gate release. Give IC-QRT-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate
IC-ORC-01	Transaction component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Coordinates producers, validators, human gates, clinical services, external systems, QA, acknowledgment, cancellation, recovery, and release states through a....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-ORC-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-TXN-01	Transaction component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Applies one approved state mutation or an explicit compensation, rechecks authorization at commit, and prevents a partial technical....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-TXN-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-IDM-01	Transaction component; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Assigns request keys and sequence rules, detects replay, duplicate, stale, out-of-order, and conflicting work, and makes safe retry....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Give IC-IDM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate
IC-ACK-01	Transaction component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Returns one correlated accepted, rejected, canceled, partial, failed, or degraded outcome and distinguishes transport, storage, semantic import, review....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-ACK-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-CMP-01	Transaction component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Executes the approved cleanup, compensating action, prior-version restoration, approval invalidation, and reconciliation plan without bypassing a hard stop....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-CMP-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-TMO-01	Transaction component; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Enforces bounded timeouts, backoff, retry limits, cancellation, late-result handling, dead-letter behavior, and safe user-visible status for synchronous and....	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Give IC-TMO-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate
IC-AUD-01	Evidence component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Records correlated requests, decisions, versions, input and output identities, actors, timing, state changes, outcomes, retries, approvals, transfers, exceptions....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-AUD-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-PRV-01	Evidence component; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Links exchanged and derived objects to sources, transforms, schemas, services, models, configurations, evidence, users, approvals, integrity values, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Give IC-PRV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-MON-01	Operations component; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Calculates availability, latency, queue, validation, rejection, duplicate, partial, reconciliation, drift, security, and safety signals and evaluates controlled thresholds.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Give IC-MON-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-REC-01	Operations component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Compares producer, consumer, repository, TPS, OIS, and destination identities, content, references, states, acknowledgments, and integrity and creates controlled....	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give IC-REC-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-ALT-01	Operations component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Routes safety, clinical, interoperability, security, availability, and supplier events to accountable teams with severity, context, evidence preservation, containment....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give IC-ALT-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-BDR-01	Operations component; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides validated backup, restore, read-only or degraded operation, known-good configuration recovery, route reactivation, and post-restoration reconciliation.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Give IC-BDR-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate
IC-TIM-01	Operations component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Maintains trusted time, clock-health evidence, correlation identifiers, ordering metadata, and explicit handling when timing assurance is unavailable or....	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give IC-TIM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-DCM-01	External component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Implements commissioned association negotiation and approved SCU or SCP services, SOP Classes, transfer syntaxes, statuses, timeouts, and audit....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-DCM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-DWP-01	External component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Implements approved user-agent or origin-server QIDO-RS, WADO-RS, STOW-RS, or other scoped DICOMweb behavior with authentication, media negotiation, status....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-DWP-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
IC-RTP-01	External component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Applies site-approved RT object profiles and validates prescription, structure, plan, ion plan, dose, registration, treatment-record, geometry, unit, machine....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give IC-RTP-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-DVA-01	External component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Performs paired conformance analysis, capability comparison, object validation, private-attribute policy, semantic round-trip comparison, and source-to-destination fidelity assessment.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give IC-DVA-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
IC-TPS-01	External component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Encapsulates each vendor, product version, site, machine, modality, technique, route, workflow, limitation, acknowledgment, and support model while preserving....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-TPS-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
IC-VAP-01	External component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Exposes only approved vendor operations with explicit authentication, authorization, schema, version, timeout, error, audit, change, fallback, and reconciliation....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-VAP-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-IHE-01	External component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Implements approved IHE-RO actor, transaction, content-module, option, and version behavior and reconciles profile workflow state with authoritative DICOM....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-IHE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-UIB-01	Adapter component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Connects the clinical workspace to context, preview, comparison, warning, confirmation, cancellation, annotation, approval, and outcome services while preserving....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-UIB-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-UX; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-NLG-01	Adapter component; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Connects typed text and push-to-talk speech to confirmed transcript, typed intent, terminology, units, ambiguity handling, clarification, read-back, abstention....	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Give IC-NLG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-NLI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-EVG-01	Adapter component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Connects the closed approved evidence service and computable-rule resolver to source authority, applicability, conflict, exception, citation, lifecycle, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-EVG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-EVD; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-IMG-01	Adapter component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Connects case imaging, registration, accumulated dose, segmentation or contours, uncertainty, edit lineage, review, and approval services through versioned....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give IC-IMG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-IMG; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
IC-PLN-01	Adapter component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Connects structured planning intent to commissioned optimization, dose, robustness, machine, technique, candidate, and calculation services without generative dose....	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give IC-PLN-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-PLAN; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-REV-01	Adapter component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Connects candidates to comparison, selection, professional approval, physics review, independent verification, patient-specific QA, transfer gating, and destination acknowledgment.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-REV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-REV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
IC-MDG-01	Adapter component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Connects only approved signed model packages to contract-validated inputs, inference, uncertainty, abstention, draft outputs, lineage, monitoring, and model-change....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-MDG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-AI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-DAT-01	Adapter component; DATA	Clinical data, configuration, provenance, dependency, lifecycle state, or audit evidence is missing, altered, stale, or misattributed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Connects services to immutable versioned objects, terminology, identity, dependency lineage, integrity, audit, retention, lifecycle state, and authoritative-source rules.	Users may act on the wrong version and the team may be unable to reconstruct, invalidate, investigate, or safely recover the workflow.	Use immutable versioned objects, integrity values, dependency graphs, synchronized tamper-evident audit, retention, reconciliation, controlled configuration, and approval invalidation. Give IC-DAT-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-IAM-01	Security component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Federates identity and maps actor or service, role, purpose, case, session, signature, approval meaning, revocation, and separation-of-duties state....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-IAM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
IC-SEC-01	Security component; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides mutual endpoint trust, approved encryption, network segmentation, certificate and secret lifecycle, rate and egress control, logging, detection....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Give IC-SEC-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-PTM-01	Security component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Enforces purpose-bound minimum-necessary content, field and object allowlists, de-identification where approved, destination and residency rules, retention, and privacy-event....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-PTM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-TRC-01	Assurance component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Maintains bidirectional links from ConOps, HLR, HLR, SIR, family profile, hazard, component, contract, test, result, anomaly, monitoring, change....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-TRC-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-TST-01	Assurance component; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Executes schema, semantic, role, state, security, positive, negative, boundary, wrong-context, stale-version, timeout, duplicate, cancellation, partial, recovery, and misuse....	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Give IC-TST-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-ITL-01	Assurance component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides controlled producer, consumer, simulator, capture, reference-comparison, failure-injection, load, multi-vendor, version, route, and semantic round-trip environments.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give IC-ITL-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-COM-01	Assurance component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Binds the actual institution, vendor, product version, endpoint, machine, modality, technique, disease site, workflow, configuration, users, and acceptance....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-COM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-HFE-01	Assurance component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Evaluates representative users and critical tasks including context confirmation, speech correction, warning, interruption, comparison, approval, cancellation, recovery, and....	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-HFE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-UX; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
IC-AIE-01	Assurance component; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Tests the combined model and interface route for input contract, intended-use scope, subgroup behavior, uncertainty, abstention, failure, lineage....	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Give IC-AIE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-AI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
IC-RPT-01	Assurance component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Produces reproducible signed coverage, actual-result, anomaly, deviation, residual-risk, limitation, approval, commissioning, release, monitoring, and revalidation reports.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give IC-RPT-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

11 Category component risk records

11.1 Functional category component - 30 records

Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
FC-REQ-01	Universal component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Maintains bidirectional trace from ConOps and source HLRs through categorized requirements, hazards, tests, results, anomalies, and releases.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-REQ-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
FC-ARCH-01	Universal component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls functional boundaries, service contracts, data ownership, interface dependencies, and compatible component versions.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give FC-ARCH-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
FC-CFG-01	Universal component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Locks approved functional scope, workflows, parameters, models, evidence, machines, interfaces, and release manifests.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-CFG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
FC-PROV-01	Universal component; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Links each input, action, transformation, result, approval, and transfer to the exact actor, object, service, evidence, model, and....	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Give FC-PROV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
FC-VV-01	Universal component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Executes positive, negative, boundary, recovery, interoperability, and end-to-end tests with retained objective evidence.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give FC-VV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
FC-IAM-01	Control component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Authenticates users and services and enforces role, context, approval identity, and separation of duties.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-IAM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
FC-SAFE-01	Control component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Evaluates scope, context, preconditions, hard stops, human authority, approval validity, and export eligibility independently of generative models.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-SAFE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
FC-STATE-01	Control component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls draft, review, approval, QA, transfer, invalidated, degraded, and superseded object and workflow states.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-STATE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
FC-UX-01	Experience component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Presents context, sources, assumptions, warnings, previews, comparisons, edits, approvals, and human-controlled actions.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-UX-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-UX; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
FC-VOICE-01	Experience component; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides bounded speech capture, transcription, critical-token flagging, correction, read-back, and confirmed transcript output.	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Give FC-VOICE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-NLI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
FC-INTENT-01	Experience component; NLI	Speech or text is misrecognized, ambiguous, incompletely typed, or assigned the wrong unit, object, laterality, negation, or priority. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Converts text or confirmed speech into inspectable typed intent and validates objects, units, laterality, negation, and required fields.	The system may execute or propose an unintended clinical change while the user believes another action was requested.	Require confirmed transcript, typed intent, deterministic validation, ambiguity resolution, structured read-back, preview, explicit confirmation, cancellation, and full interaction audit. Give FC-INTENT-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-NLI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/RPN TBD	High
FC-WF-01	Orchestration component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Coordinates governed clinical tasks, human gates, commissioned services, existing TPS interactions, QA routing, cancellation, and recovery.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give FC-WF-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/RPN TBD	High
FC-EVID-01	Clinical component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Stores and retrieves institution-approved, versioned, applicable literature and local policies without unrestricted clinical-mode internet retrieval.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-EVID-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-EVD; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
FC-RULE-01	Clinical component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Evaluates evidence applicability, units, authority, conflicts, lifecycle state, and no-applicable-rule outcomes.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-RULE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-EVD; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
FC-CASE-01	Clinical component; PAT	Wrong, stale, or incomplete patient, course, study, plan, or action context is accepted. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Binds patient, course, studies, frames, prescription, anatomy, prior treatment, task, mode, and destination.	Work may be applied to the wrong patient or object, or valid downstream approvals may become unsafe.	Bind authenticated actor and immutable case context, confirm two patient identifiers for consequential actions, reject mismatches, and invalidate stale approvals. Give FC-CASE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-IMG; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
FC-REG-01	Clinical component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Invokes commissioned registration and accumulation methods and retains transforms, uncertainty, approval, and lineage.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give FC-REG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-IMG; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
FC-SEG-01	Clinical component; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls contour-model applicability, draft generation, uncertainty, editing, review, approval, and nomenclature.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Give FC-SEG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-IMG; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
FC-PLAN-01	Clinical component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Builds structured planning intent and coordinates candidate generation, diversity, optimization, dose, and robustness services.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give FC-PLAN-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-PLAN; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
FC-DOSE-01	Clinical component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Connects only to commissioned optimization, robustness, and dose engines; it does not permit generative dose calculation.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give FC-DOSE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-PLAN; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
FC-REV-01	Clinical component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, un-commissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Presents consistent metrics, spatial dose, trade-offs, robustness, uncertainty, provenance, comments, and role-specific decisions.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give FC-REV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-REV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
FC-QA-01	Clinical component; AS-SUR	Verification, validation, commissioning, regression, anomaly disposition, or release evidence is incomplete or not representative. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Routes independent calculation, patient-specific QA, transfer verification, technique checks, and physics review while preserving independence.	A latent defect or unsupported route may enter or remain in clinical service.	Maintain bidirectional trace, risk-based tests and challenge cases, independent review, site commissioning, controlled anomaly disposition, release gates, monitoring, change regression, and revalidation. Give FC-QA-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-REV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
FC-DICOM-01	Interface component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides approved query, retrieve, store, import, export, validation, and reconciliation for images and radiotherapy objects.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-DICOM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
FC-TPS-01	Interface component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Encapsulates each commissioned TPS, its DICOM conformance, supported workflows, optional approved API, limitations, and site configuration.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-TPS-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-INT; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
FC-TERM-01	Data component; DOSE	A prescription, objective, optimization, robustness, MU, or dose result is incorrect, uncommissioned, or applied outside approved scope. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls codes, structure names, units, dose basis, coordinate systems, mappings, conversions, and rounding.	A candidate may have inadequate target coverage, excessive normal-tissue dose, or misleading clinical acceptability.	Use only commissioned TPS, optimizer, robustness, and dose services; lock scope and versions; retain calculation lineage; require professional review and independent QA. Give FC-TERM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Gate
FC-OBJ-01	Data component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Stores immutable object versions, lifecycle state, ownership, integrity values, approvals, and source links.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-OBJ-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
FC-MGW-01	AI component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Resolves approved locked models, validates intended use and input contracts, executes inference, and enforces abstention and routing.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-MGW-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-AI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
FC-MREG-01	AI component; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls model versions, provenance, intended use, artifacts, signatures, configurations, limitations, and dependencies.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Give FC-MREG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-AI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
FC-MVAL-01	AI component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Manages validation datasets, subgroup and calibration measures, edit burden, thresholds, drift, and regression evidence.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give FC-MVAL-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-AI; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
FC-JOB-01	Platform component; GEN	The required behavior or component responsibility is absent, incomplete, inconsistent, bypassed, or not auditable. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls pending, running, complete, failed, canceled, timed-out, partial, commit, cleanup, and reconciliation states.	The parent requirement may not be achieved and a dependent clinical or operational control may fail without timely detection.	Define an accountable contract and owner, implement deterministic preconditions and explicit outcomes, retain evidence, test positive and failure paths, monitor performance, and gate release. Give FC-JOB-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate
FC-AUD-01	Data component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Records ordered commands, context, evidence, outputs, warnings, edits, approvals, QA, transfers, failures, and reconciliation.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give FC-AUD-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

11.2 Non-functional category component - 22 records

Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NFC-REQ-01	Universal component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Maintains non-functional requirements, hazards, controls, verification, anomalies, residual-risk decisions, and release trace.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give NFC-REQ-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NFC-ARCH-01	Universal component; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Defines safety, usability, privacy, security, integrity, observability, and fail-safe contracts at component and interface boundaries.	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Give NFC-ARCH-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/RPN TBD	High
NFC-CFG-01	Universal component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Locks quality-attribute thresholds, safety policies, warning tiers, trust rules, security baselines, and approved exceptions.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give NFC-CFG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/RPN TBD	High
NFC-MON-01	Universal component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Collects versioned safety, usability, security, integrity, and control-effectiveness measures with approved response thresholds.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give NFC-MON-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NFC-VV-01	Universal component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Runs safety, misuse, human-factors, security, resilience, integrity, and negative tests with retained acceptance evidence.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give NFC-VV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
NFC-SAFE-01	Safety component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Implements fail-closed preconditions, authority limits, hard stops, approval invalidation, and safe export rules.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give NFC-SAFE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
NFC-STATE-01	Safety component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls valid states and atomically invalidates affected approvals, QA, and transfer eligibility when dependencies change.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give NFC-STATE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NFC-IAM-01	Safety/security component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Enforces unique identity, current role and context, least privilege, clinical approval identity, and incompatible-role controls.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give NFC-IAM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
NFC-CTX-01	Safety component; GEN	The required behavior or component responsibility is absent, incomplete, inconsistent, bypassed, or not auditable. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Creates immutable action context, validates patient and object identity, and rejects stale, incomplete, or mismatched context.	The parent requirement may not be achieved and a dependent clinical or operational control may fail without timely detection.	Define an accountable contract and owner, implement deterministic preconditions and explicit outcomes, retain evidence, test positive and failure paths, monitor performance, and gate release. Give NFC-CTX-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate
NFC-HFE-01	Human factors component; HFE	A user misreads context, provenance, status, warning, confirmation, comparison, or recovery information, or training is inadequate. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls critical-task analysis, formative and summative evaluation, foreseeable misuse, training, competency, and remediation.	Use error may cause wrong-patient action, missed hazard, inappropriate plan selection, or ineffective recovery.	Use persistent context and source labels, accessible multimodal warnings, explicit confirmations, role training and competency, formative evaluation, and representative-user summative validation. Give NFC-HFE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-UX; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NFC-UX-01	Human factors component; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides persistent context, semantic source labels, accessible multimodal status, confirmations, and usable recovery paths.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Give NFC-UX-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-UX; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate
NFC-ALERT-01	Human factors component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls risk tier, presentation, acknowledgment, escalation, suppression, burden measures, and corrective review.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give NFC-ALERT-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-UX; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
NFC-SEC-01	Security component; ANAT	Contours, registration, transforms, frames, geometry, nomenclature, or accumulated dose are incorrect or mismatched. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Provides segmented trust boundaries, least-privilege services, secure configuration, endpoint control, detection, and enforcement.	Dose may be evaluated or optimized against the wrong anatomy or spatial relationship.	Validate identity, UID/reference graph, frame, geometry, units, contour status, transform approval, uncertainty, and edit lineage; require visual and professional review. Give NFC-SEC-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NFC-PRIV-01	Privacy component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Classifies protected data, enforces permitted purpose and destination, and limits disclosure to approved minimum-necessary use.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give NFC-PRIV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
NFC-CRY-01	Security component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls approved encryption, keys, secrets, certificates, rotation, revocation, status, and recovery.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give NFC-CRY-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
NFC-EGRESS-01	Security component; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Allowlists connectors, tenants, destinations, data classes, and purposes and blocks incompatible protected-data transfer.	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Give NFC-EGRESS-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NFC-INPUT-01	Security component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Separates external content from policy and tool authority and sanitizes or rejects embedded authority-changing instructions.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give NFC-INPUT-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
NFC-ART-01	Security component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Validates manifests, signatures, provenance, integrity, compatibility, approval, and revocation before clinical load or use.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give NFC-ART-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
NFC-SDL-01	Security component; MODEL	A model is out of scope, altered, drifted, poorly calibrated, supplied incompatible input, or fails to abstain when uncertain. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls threat modeling, secure coding, review, vulnerability management, penetration testing, patching, SBOM, and supplier risk.	An unreliable contour, interpretation, or planning suggestion may appear plausible and influence clinical work.	Lock signed model and dependency versions, validate intended use and input contract, expose uncertainty and limitations, keep output draft, require review, monitor performance and drift, and support rollback. Give NFC-SDL-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
NFC-AUD-01	Assurance component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Retains synchronized, integrity-protected evidence of context, controls, warnings, access, transfer, exceptions, incidents, and response.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give NFC-AUD-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
NFC-DATA-01	Assurance component; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Validates completeness, identity, consistency, version, integrity, and provenance of data used by safety or security decisions.	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Give NFC-DATA-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
NFC-INC-01	Assurance component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Coordinates triage, containment, patient-impact assessment, evidence preservation, communication, recovery, CAPA, and closure.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give NFC-INC-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

11.3 Operational category component - 22 records

Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OC-REQ-01	Universal component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Maintains operational needs, procedures, risks, controls, exercises, results, anomalies, approvals, and release trace.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-REQ-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-GOV-01	Universal component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Owns intended use, scope, clinical authority, residual-risk acceptance, evidence policy, and accountable decisions.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-GOV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-QMS-01	Universal component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls documents, changes, deviations, incidents, CAPA, approvals, records, and quality review.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-QMS-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OC-CFG-01	Universal component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Binds approved scope, modes, controls, thresholds, services, models, interfaces, procedures, and release manifests.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-CFG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SYS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-AUD-01	Universal component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Retains decisions, events, exercises, exceptions, results, approvals, actions, and immutable operational chronology.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-AUD-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-DATA; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-VV-01	Universal component; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Executes inspections, exercises, commissioning, recovery, shadow, site-acceptance, and end-to-end operational protocols.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Give OC-VV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OC-SCOPE-01	Governance component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls approved institutions, users, populations, sites, modalities, machines, techniques, tasks, and exclusions.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-SCOPE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-MODE-01	Governance component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Segregates research, training, commissioning, shadow, clinical-draft, clinical-approved, degraded, and read-only modes and destinations.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-MODE-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-ACCESS-01	Governance component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Administers role matrices, competency dependencies, exceptions, separation of duties, and periodic access review.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-ACCESS-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-SEC; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OC-REL-01	Lifecycle component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Confirms approved baseline, residual risk, verification, commissioning, training, readiness, and authorization before clinical activation.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-REL-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-CHG-01	Lifecycle component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Classifies changes, assesses impact, defines regression and migration, approves activation, and preserves known-good rollback.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-CHG-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-MON-01	Operations component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Monitors service health, workflow, model, evidence, interface, safety, quality, and subgroup indicators under controlled definitions.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give OC-MON-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OC-ALERT-01	Operations component; INT	A schema, version, DICOM/DICOM-RT object, UID/reference, unit, status, acknowledgment, or vendor semantic is incompatible, partial, or silently transformed. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Evaluates thresholds, opens actionable records, routes escalation, records acknowledgment, and tracks disposition.	Wrong or incomplete clinical objects may enter the TPS, repository, OIS, review, or transfer workflow without detection.	Use paired conformance analysis, allowlisted routes, semantic validation, atomic transfer, explicit outcomes, quarantine, round-trip reconciliation, site commissioning, and change regression. Give OC-ALERT-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-INC-01	Operations component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Supports reporting, triage, containment, patient-impact assessment, evidence preservation, investigation, communication, and closure.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give OC-INC-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-CAPA-01	Operations component; GEN	The required behavior or component responsibility is absent, incomplete, inconsistent, bypassed, or not auditable. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls root-cause analysis, assigned actions, due dates, effectiveness checks, trend review, and quality closure.	The parent requirement may not be achieved and a dependent clinical or operational control may fail without timely detection.	Define an accountable contract and owner, implement deterministic preconditions and explicit outcomes, retain evidence, test positive and failure paths, monitor performance, and gate release. Give OC-CAPA-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OC-BKP-01	Resilience component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Protects approved objects and configurations and validates encrypted backup, retention, restoration authority, and restore integrity.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-BKP-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-DOWN-01	Resilience component; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Defines safe access, external work practices, communication, restrictions, reconciliation, and controlled restoration.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Give OC-DOWN-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate
OC-REC-01	Resilience component; OPS	A dependency fails, degrades, times out, recovers incorrectly, or exceeds a threshold without effective detection and response. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Exercises recovery objectives, known-good restoration, deviations, corrective actions, and continued adequacy.	Clinical work may be delayed, silently altered, partially completed, or continued through an unsafe workaround.	Expose health state, bound timeout and retry, fail safely, alert and escalate, quarantine partial work, validate backup and recovery, reconcile after restoration, and exercise downtime procedures. Give OC-REC-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-OPS; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	Moderate

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OC-COMM-01	Assurance component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls local configuration, machine and TPS interfaces, reference cases, clinical workflow tests, acceptance, and approval.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-COMM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-TEST-01	Assurance component; CYBER	Identity, endpoint trust, confidentiality, integrity, availability, or minimum-necessary controls are bypassed or compromised. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Controls representative, boundary, failure, misuse, recovery, interoperability, and clinical cases with expected results.	Protected information may be disclosed or altered, service authority may be escalated, or clinical availability and integrity may be impaired.	Apply least privilege, MFA, segmentation, encryption, managed secrets, allowlisted egress, artifact verification, secure development, monitoring, containment, recovery, and incident response. Give OC-TEST-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High
OC-ANOM-01	Assurance component; AUTH	An unauthorized user or automated service performs, represents, bypasses, or preserves an approval, release, QA waiver, or clinical decision. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Classifies discrepancies, blocks inappropriate acceptance, records investigation, and requires approved disposition and retest.	An unreviewed or unsuitable plan may be promoted, transferred, or represented as clinically approved.	Enforce unique identity, current role and context, separation of duties, deterministic state transitions, hard stops, approval invalidation, and human-only approval and release. Give OC-ANOM-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-VV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/ RPN TBD	High

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Component	Trace / hazard	Failure / risk	Potential effect	Mitigation	Evidence / owner	Band
OC-REV-01	Lifecycle component; EVID	Evidence or a computable rule is outdated, inapplicable, lower authority, conflicted, misquoted, or used outside its approved lifecycle. Risk increases when this obligation is absent, wrong, stale, bypassed, or incomplete: Reviews field signals, scope, residual risk, controls, evidence, models, changes, and continued authorization at defined intervals.	Planning objectives or evaluations may be presented as evidence-based when they are not applicable to the patient or institution.	Use a closed approved repository, versioned authority and applicability rules, exact citations, conflict and exception handling, effective dates, and no-applicable-rule behavior. Give OC-REV-01 a versioned contract, accountable owner, controlled configuration, health and safety monitoring, explicit failure behavior, recovery or rollback, and independent acceptance evidence.	Lead T-GOV; contract, tests, monitor, anomaly and release evidence; FMEA S/O/D/RPN TBD	High

12 Risk-control completion and residual-risk rule

A mitigation is not effective merely because it is specified or implemented. Effectiveness requires traceable verification under representative and failure conditions, site commissioning where applicable, monitoring with controlled thresholds, anomaly disposition, and confirmation that the control does not introduce unacceptable new risk. Control independence, common-cause failure, automation bias, alert burden, cybersecurity, downtime, and recovery shall be included in the evaluation.

No clinical release may rely on an unapproved numerical risk score, an automatically generated risk acceptance, or a trade-off score as a substitute for professional judgment and governance. Residual risk and benefit-risk decisions shall identify the approved criteria, evidence, uncertainty, affected population and scope, decision authority, limitations, monitoring, and revalidation triggers.

13 Structural risk-register audit

Entity set	Expected	Unique IDs	Risk records	Missing	Audit
System high-level requirements	111	111	111	0	Pass
Detailed sub-requirements	333	333	333	0	Pass
High-level interface requirements	333	333	333	0	Pass
Sub-interface requirements	999	999	999	0	Pass
Functional components	41	41	41	0	Pass
Interface components	57	57	57	0	Pass
F/NF/O category components	74	74	74	0	Pass
TOTAL	1,948	1,948	1,948	0	Pass

Structural pass establishes complete record coverage only. It does not establish hazard complete-

ness, FMEA consensus, control effectiveness, risk acceptability, benefit-risk acceptability, regulatory compliance, commissioning, or clinical release.

Bottom line. Every unique controlled requirement and component now has an explicit preliminary failure-risk statement, mitigation strategy, evidence expectation, owner, and review priority, with clinical risk scoring and acceptance reserved for the approved multidisciplinary process.